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Mouse Thalamic Neuron Types and Classification

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Classification systems and naming conflicts

Mouse thalamic neurons are named in several overlapping ways: by nucleus, projection target, transmitter, morphology, developmental origin, and transcriptomic type. These schemes often agree at coarse scale but conflict at fine scale because many molecular types span nuclei, some nuclei contain gradients, and connectivity does not always follow nuclear borders.

The mouse thalamus has traditionally been divided into anatomical nuclei using cytoarchitecture, stains, and mesoscale connectivity. That scheme is still useful, but it is incomplete: some nuclear borders are disputed, some nuclei contain subdomains with different connectivity and function, and some gene-expression patterns cut across classic boundaries. Spatial transcriptomics in THALMANAC shows all of these cases: some transcriptomic types map well to one nucleus, some split a nucleus into parts, and many span multiple nuclei. (Turner et al., 2025)

Modern work uses several partly overlapping classification systems:

- **Nuclear anatomy**: ~40 adult mouse thalamic labels are commonly used, but this granularity can miss finer topographic structure. (Sun et al., 2026) (Sun et al., 2025)
- **Projection-defined types**: retrograde labeling and RNA-seq identified thalamic neurons by their main targets and found repeated molecular profiles across sensory, motor, and cognitive pathways rather than one scheme per modality. (Phillips et al., 2019) (Phillips et al., 2017)
- **Core/matrix and first-order/higher-order schemes**: these are widely used, but boundary definitions are not consistent across theories and do not capture all nuclei. Phillips and colleagues argued that a single mediolateral transcriptional axis is a better summary than a simple core/matrix dichotomy. (Phillips et al., 2018) (Phillips et al., 2017) Roy et al. note that older sensory-based and core/matrix schemes overlap only partly and are not comprehensive. (Roy et al., 2022)
- **Transcriptomic taxonomy**: the Allen/ABC atlas organizes cells into hierarchical classes, subclasses, supertypes, and clusters. In the thalamic region analyzed by THALMANAC, thalamic neurons fell into 5 classes, 58 subclasses, 163 supertypes, and 563 clusters in taxonomy version CCN20230722; numbers will change as the taxonomy evolves. A single glutamatergic class covers much of thalamus proper,

whereas inhibitory and habenular classes cut across broader regions. (Turner et al., 2025) (Yao et al., 2023)
- **Developmental lineage**: embryonic single-cell studies support at least three major lineage streams, including Shh-responsive progenitors that form many lateral sensory/motor/anterior glutamatergic nuclei, a Shh-independent lineage that contributes to more dorsomedial nuclei such as PV and MD, and a rostral thalamic lineage that forms GABAergic neurons in LGv/IGL. (Govek et al., 2022)

A practical conflict is that "cell type" can mean different things depending on the assay. Roy et al. recommend keeping apart cell types defined by a single feature from broader "subpopulations" and functionally defined "subnetworks." This is important in thalamus because morphology, physiology, gene expression, and connectivity often align only partly. (Roy et al., 2022) Another conflict is that clustering methods applied to the same spatial thalamus dataset can give discordant boundaries, especially near transitions between nuclei, showing that even molecular parcellation is not unique. (Sun et al., 2026) (Sun et al., 2025)

Evidence

(Turner et al., 2025)

Exploring the correspondence between gene expression and thalamic nuclei using the THALMANAC resource

"The thalamus connects the sensory organs and major subcortical brain regions with the neocortex. The thalamus has long been divided into multiple discrete nuclei, based on cytoarchitecture, histochemical stains, and mesoscale connectivity. However, thalamic nuclei do not completely describe thalamic organization. For example, some boundaries between thalamic nuclei are disputed, whereas other nuclei are known to contain subdomains with distinct connectivity and function. Moreover, the correspondence between cellular gene expression and other properties of thalamic projection neurons remains to be established. Spatial analysis of single cell gene expression provides a basis for reevaluating thalamic organization. We present the THALMANAC, (THALamusMERFISH ANalysis and ACcess) a Findable, Accessible, Interoperable, Reusable and Reproducible (FAIRR) resource for exploring and analyzing single-cell transcriptomic variation in the thalamus. The THALMANAC provides streamlined access to thalamic gene expression data registered to the common coordinate framework and tools for quantitative analysis and visualization of these data, all encapsulated in a reproducible, cloud computing platform. Using this resource, we find that gene expression generally supports the parcellation of thalamus into distinct nuclei. Some nuclei, such as the anteromedial nucleus, are additionally composed of discrete subdomains, while other nuclei share patterns of gene expression or are arrayed on a spatial gradient of gene expression. The THALMANAC establishes spatial transcriptomic data as a foundation for delineating thalamic organization. Graphical Abstract"

"This fine-scale structure could correspond to discrete, molecularly defined subdivisions of VM that have not yet been annotated, or to topographic organization of thalamocortical and corticothalamic connectivity.

A major question is how gene expression relates to existing parcellations and thalamic organization in general. Single cell RNA sequencing (scRNA-seq) provides detailed measurements of transcription in large populations of individual neurons (Tasic et al. 2016), which can be used to define transcriptomic cell types ('t-types'). The combination of scRNA-seq with spatial transcriptomics has been used to localize t-types across the whole mouse brain (Langlieb et al. 2023;Yao et al. 2023;Zhang et al. 2023). With t-types mapped into the CCFv3 coordinate space (Wang et al. 2020), we can evaluate how their spatial distributions correspond to anatomical parcellations (Yao et al. 2023).

Here, building on the Allen Brain Cell (ABC) Atlas scRNA-seq and spatial transcriptomics data, we developed a resource, the THALamus MERFISH ANalysis and ACcess (THALMANAC), which provides access to spatial gene expression data in the mouse thalamus; targeted analysis and data visualization of gene expression patterns; a platform for reproducible computations; and a no-code visualization app for data exploration. We demonstrate the utility of the THALMANAC by addressing how thalamic structural organization relates to gene expression. Aligning spatial transcriptomic data and nuclei boundaries reveals a wide range of relationships between t-types and thalamic nuclei: some t-types correspond to a single thalamic nucleus; others subdivide nuclei into multiple parts; many t-types span multiple thalamic nuclei.

Analysis of expression by itself supports these same conclusions: some nuclei boundaries are associated with sharp changes in gene expression; other nuclei contain gene expression gradients; and, in many cases, gene expression patterns span multiple nuclei.

Understanding how information is routed through the thalamus requires integrating insights gained from gene expression with other modalities, including connectivity and neurophysiology."

"The THALMANAC resource provides access to thalamic spatial gene expression data, visualization of the relationships between gene expression, t-types, and CCFv3 anatomical parcellations, and quantitative analysis of targeted quantitative analyses, all

encapsulated in a cloud platform that enables reproducible computations and customized exploration of the data.

The THALMANAC is built on a thalamus-specific subset of spatial transcriptomics data from multiplexed error-robust fluorescence in situ hybridization (MERFISH) from the ABC Atlas. The ABC Atlas dataset measures 500 transcripts, chosen to distinguish t-types across the whole mouse brain, in spatially resolved, single cells. MERFISH cells were registered to the CCFv3 coordinate space and mapped to the scRNA-seq data by assigning each MERFISH cell to the label of the most similar t-type cluster. We identified neurons that belong to the thalamus (including the zona incerta and habenula) according to their t-type label (Figure S1A) and CCFv3 parcellation label (Figure S1B). The thalamic dataset analyzed here includes 80,170 MERFISH cells across 14 coronal sections collected from a single male mouse (Methods).

The THALMANAC includes three avenues for engaging with the t-type and gene expression data and accompanying analyses (Figure 1). First, a Streamlit web app (RRID:SCR_024354) provides nocode exploration, including interactive visualizations and on-the-fly di"erential gene expression analysis for thalamic nuclei (Interactive Streamlit App: thalmanac.allenneuraldynamics.org). Second, Jupyter notebooks (RRID:SCR_018315) provide step-by-step demonstrations of the Python code needed to reproduce all results presented here (github.com/AllenNeuralDynamics/thalmanac). Both tools allow, users to customize the thalamic cells that are selected from the full ABC Atlas."

"In the ABC Atlas, transcriptomic variation is used to assign cells to a hierarchy of discrete t-types (from broad to narrow categories: class, subclass, supertype, cluster). Thalamic neurons belong to 5 classes, 58 subclasses, 163 supertypes, and 563 clusters (Table S1; Figure S2) (taxonomy version CCN20230722) (Yao et al. 2023), but these numbers will change as the taxonomy evolves. Whereas the cluster level is based on unsupervised clustering, higher levels of the taxonomy are additionally informed by developmental origin via shared transcription factor expression, anatomical location, and classical (pre-scRNAseq) neuron types (DeFelipe et al. 2013) defined by key molecular markers (e.g. neuropeptides, calcium-binding proteins).

Transcriptomic classes typically correspond to major brain structures. Glutamatergic neurons in thalamus proper form a single class (class 18 TH Glut), whereas habenula neurons belong to a distinct class (class 17 MH-LH Glut) (Figure S2B). The other three classes are not unique to the thalamus: class 12 HY GABA is found in the reticular thalamus, zona incerta and ventral lateral geniculate nucleus (CCFv3 annotations; RT, ZI, and LGv, respectively, see Table S1) but also in the hypothalamus; class 19 MB Glut is found in thalamic nucleus SPA but also in the midbrain; class 20 MB GABA is found in thalamic nuclei LGv and SPF and the midbrain. Transcriptomic subclasses span multiple thalamic nuclei(Figure S2C), whereas transcriptomic supertypes often define individual nuclei (e.g. AM), and several nuclei contain multiple supertypes (e.g. ZI) (Figure S2D). Most nuclei contain multiple transcriptomic clusters (Figure 2A, Figure S2E).

We explored correspondence between clusters and thalamic nuclei."

"A comprehensive understanding of how the thalamus routes information to the cortex will require multiple types of data and a community-wide e"ort. Our work contributes to this program by curating and analyzing the correspondence between state-of-the-art spatially resolved, single-cell transcriptomic data, and widely used parcellations of thalamus into discrete nuclei. In addition, we make our data and analyses available in an accessible, documented, and reproducible bioinformatics resource in the cloud. Using the AM nucleus and the somatosensory nuclei as examples, we illustrate how this interactive resource can be used to identify novel patterns of gene expression and how these patterns correspond to prior parcellations of the thalamus.

The THALMANAC resource streamlines researchers' access to thalamic spatial transcriptomic data. We anticipate that scientists will extend our analysis based on the ABC atlas data, or apply similar analyses to additional spatial transcriptomics datasets registered to the CCF coordinate system (Langlieb et al. 2023;Zhang et al. 2023). The tool should also be used as a reference to validate recent studies focused on specific nuclei, such as the spatially segregated projections from PF to distinct dorsal striatum regions (Mandelbaum et al. 2019) and spatially segregated inputs to PO (Casas-Torremocha et al. 2022), the molecular organization of the PVT (Gao et al. 2023), and the core/shell organization of neurons in RT (Li et al. 2020). Our interactive Streamlit app can be used to identify marker genes for post-hoc localization of neurophysiological and neuroanatomical experiments, or to select gene panels for multiplexed mFISH experiments. The THALMANAC is already being used in the e"ort to create and characterize enhancerbased genetic tools (Ben-Simon et al. 2025) to target thalamic cell types and regions.

The THALMANAC shows that at the scale of thalamic nuclei, the molecular organization of the thalamus resembles existing parcellations. "

"However, both discrete cell type composition and gene expression gradients vary at finer spatial scales within and across nuclei. Given the evidence for circuit and functional variation at the same length scales, additional data need to be integrated with molecular characterization to understand thalamic organization. We see two promising paths forward for data integration. First, existing data registered to the CCFv3 3D coordinate space, such as the databases of single-neuron morphological reconstruction (Winnubst et al. 2019) or neurophysiology data (Chen et al. 2024), could be linked to cell types and gene expression described in the THALMANAC. Second, targeted multimodal datasets, such as single-neuron morphological reconstructions combined with posthoc mFISH for transcriptomic cell typing would link cell types and gene expression to projection targets at the single cell level.

Beyond data integration, we also recognize the need for thalamus-focused, community annotation, similar to previous e"orts to

annotate the fly connectome (Schlegel et al. 2024). We envision a refined parcellation of the thalamus where scientists with specific domain knowledge would combine molecular information from the THALMANAC with neuroanatomical and functional data to reflect progress in understanding of thalamic organization. Diagram illustrating the three workflows that can be launched from the Code Ocean Collection accompanying this paper. The Code Ocean platform provides built-in support for code versioning (via Git), environment replicability (via Docker), data and analysis provenance (via tracked data assets and linked pipeline runs), cloud compute resources (via AWS EC2 instances). Users can launch an interactive, no-code Streamlit application to explore spatial distributions of transcriptomic cell types and gene expression from their browser, as well as visualize locations of annotated cell types for each thalamic nucleus and highlight a specific nucleus or group of related thalamic nuclei. Tutorial Jupyter notebooks and Python code replicate all analyses and visualizations in a cloud workstation. These visualizations can be customized and extended. Users can download an AnnData-formatted .h5ad"

(Sun et al., 2026)

Beyond benchmarking: an expert-guided consensus approach to spatially aware clustering

"The adult mouse brain thalamus has been parcellated into approximately 40 anatomical labels, called thalamic nuclei (Fig. 4a), on the basis of cytoarchitecture (cell shape, size and density) and histochemical stains. However, this level of granularity insufficiently captures the finer topographical patterns of neuronal connectivity and function within the thalamus 55 .

SAC methods can identify novel, functionally relevant parcellations. However, the results of eight SAC methods on the mouse thalamus MERFISH dataset were highly discordant (Supplementary Figs. 23 and 24). Without further refinement, this lack of agreement would only introduce confusion, rather than improve our understanding of molecular thalamic organization. Therefore, we leveraged consensus clustering and entropy analyses to interpret the outputs of SAC methods on this dataset.

We compared SAC results with each other and to reference anatomical labels from the Allen Mouse Brain Common Coordinate Framework (CCFv3) 56 using pairwise ARIs (Fig. 4b). Again, SAC methods agreed more with each other than with the anatomical reference labels (Fig. 4a,b), reiterating the limitations of using anatomical annotations as GT for SAC evaluation.

Expert reviewers compared clusterings with reference anatomical labels and their thalamic biology expertise. The CME (Fig. 4c) revealed regions of high entropy (low cross-method agreement) near the boundaries between anatomical labels. Two key insights emerged. First, boundaries between thalamic nuclei often correspond to transitions in gene expression, and the entropy plot visually highlights how small differences in method or parameterization lead to variation in SAC calls at these transitions. Second, the entropy-indicated border identifies regions where the reference anatomical labels could be refined. While reference anatomical labels were derived from spatial registration to a whole-brain coordinate space (CCFv3), this process imperfectly results in misalignments, most notably at borders between anatomical regions. The ventral posteromedial nucleus (VPM) shows key marker gene expression aligning better with entropy-indicated borders than reference labels (Fig. 4c,d, arrowheads).

Next, we generated consensus clusterings for the thalamus (Fig. 4e-h)."

(Sun et al., 2025)

Beyond benchmarking: an expert-guided consensus approach to spatially aware clustering

"In the adult mouse brain, researchers have parcellated the thalamus region into approximately 40 anatomical labels, known as thalamic nuclei (Figure 4A), based on cytoarchitecture (cell shape, size, and density) and histochemical stains. However, this level of granularity is insu cient to capture the finer topographical patterns of neuronal connectivity and function that are known to exist within the thalamus [55].

SAC methods have the potential to identify novel, functionally relevant parcellations based on the information-rich and reproducible data of single-cell SRT. However, the results of eight SAC methods on the mouse thalamus MERFISH dataset showed a high level of discordance (Supplementary Figure S16).

Without further refinement, this lack of agreement would only introduce confusion, rather than improve our understanding of molecular thalamic organization. Therefore, we leveraged our consensus clustering and SAC entropy analyses to interpret and apply the outputs of SAC methods on the same SRT dataset.

We compared the SAC results with each other and to the reference anatomical labels from the Allen Mouse Brain Common Coordinate Framework (CCFv3) [56] using pairwise ARIs (Figure 4B). Consistent with our findings in the Visium LIBD DLPFC dataset, we found that SAC methods tended to agree more with each other than with the anatomical reference labels (Figure 4A, B). This reiterates the limitations and risks of using anatomical annotations as GT for SAC evaluation.

Expert reviewers compared the resulting clusterings to both the reference anatomical labels and their domain knowledge of thalamic biology. The CME plot (Figure 4C) revealed regions of the thalamus that exhibited good agreement between SAC methods, as well as regions where the agreement was poor. High entropy (low cross-method agreement) frequently occurred near the boundaries between di erentanatomical labels. Two key insights emerged from this analysis. First, boundaries between thalamic nuclei often correspond to transitions in gene expression profile, and the entropy plot visually highlights how small di erencesin

method choice or parameter setting can lead to variation in SAC calls at these transitions. Second, the entropy-indicated border helped us identify regions where the reference anatomical labels could be refined."

(Phillips et al., 2019)

A repeated molecular architecture across thalamic pathways

"The thalamus is the central communication hub of the forebrain and provides the cerebral cortex with inputs from sensory organs, subcortical systems and the cortex itself. Multiple thalamic regions send convergent information to each cortical region, but the organizational logic of thalamic projections has remained elusive. Through comprehensive transcriptional analyses of retrogradely labeled thalamic neurons in adult mice, we identify three major profiles of thalamic pathways. These profiles exist along a continuum that is repeated across all major projection systems, such as those for vision, motor control and cognition. The largest component of gene expression variation in the mouse thalamus is topographically organized, with features conserved in humans. Transcriptional differences between these thalamic neuronal identities are tied to cellular features that are critical for function, such as axonal morphology and membrane properties. Molecular profiling therefore reveals covariation in the properties of thalamic pathways serving all major input modalities and output targets, thus establishing a molecular framework for understanding the thalamus. This work describes comprehensive transcriptomic sequencing from murine thalamic pathways. By integrating this molecular information with anatomical and functional features, this study reveals a repeated architecture across thalamocortical systems."

(Phillips et al., 2017)

A topographic axis of transcriptional identity in thalamus

"A fundamental goal in neuroscience is to uncover common principles by which different modalities of information are processed. In the mammalian brain, thalamus acts as the essential hub for forebrain circuits handling inputs from sensory, motor, limbic, and cognitive pathways. Whether thalamus imposes common transformations on each of these modalities is unknown. Molecular characterization offers a principled approach to revealing the organization of thalamus. Using near-comprehensive and projection-specific transcriptomic sequencing, we found that almost all thalamic nuclei fit into one of three profiles. These profiles lie on a single axis of genetic variance which is aligned with the mediolateral spatial axis of thalamus. Genes defining this axis of variance include receptors and ion channels, providing a systematic diversification of input/output transformations across the topography of thalamus. Single cell transcriptional profiling revealed graded heterogeneity within individual thalamic nuclei, demonstrating that a spectrum of cell types and potentially diverse input/output transforms exist within a given thalamic nucleus. Together, our data argue for an archetypal organization of pathways serving diverse input modalities, and provides a comprehensive organizational scheme for thalamus."

"We produced a near-comprehensive transcriptomic atlas of murine thalamus via microdissection of retrogradely labeled cells and subsequent RNAseq analysis (22 nuclei, 120 samples). Anterograde tracing of inputs to thalamus was used when identification of nuclear boundaries was ambiguous (Figure 1A,B for an example dissection approach). Labeled neurons from a given thalamic nucleus were pooled and profiled, an approach chosen because nuclear identity and projection target are major definers of cell type. See supplementary tables 1 and 2 for details on labelling strategies.

To identify the relationship between the transcriptomes of thalamic nuclei in an unsupervised manner, we selected the top 500 most differentially expressed genes between thalamic nuclei (via an ANOVA-like test, see methods). Using this differentially expressed gene set, we grouped the thalamic nuclei via hierarchical clustering (1 -Spearman's correlation as distance metric), identifying five major subdivisions of nuclei (Figure 1C). We termed these profiles, not clusters, as we do not mean to imply discreteness between the classifications (see below). These profiles are not explained by the modality of information processed, since subnuclei of motor, somatosensory and visual complexes all split across different profiles. For example, the blue profile (Figure 1C) included both the cerebellar-recipient motor thalamus and the retina-recipient visual nucleus (lateral geniculate, LGd), whilst the red profile contained the basal ganglia recipient motor thalamus and the superior colliculus-recipient visual nucleus (lateral posterior, LP). Interestingly, the anterior dorsal nucleus (AD) and nucleus reuniens (RE) had distinct profiles and each formed profiles of their own. When schematic sections of thalamus were colored by these profiles, a mediolateral separation was observed, with the 'blue' nuclei being most lateral and 'yellow' nuclei most medial (Figure 1D). Thus, the largest fraction of linear gene expression variance in thalamus is not explained by modality, but rather a topographic axis spanning the structure's mediolateral extent."

"A. Overview of clusters (identified using the Seurat package, see methods) within each modality visualized using tSNE, cells colored by cluster identity. B. Violin plots for top marker genes for each cluster (Likelihood Ratio Test, p-value<10^-5, logFC>0.5, top 4-5 signal-to-noise ratio). C. Projection of single-cell data onto bulk principal component 1 (as in figure 2 Thus far we analyzed thalamic nuclei by pooling projection neurons from their anatomical position. The resulting transcriptional profiles could represent homogenous populations, discrete subtypes, or cells with graded differences. To probe these possibilities, we profiled the transcriptomes of individual neurons from motor, somatosensory, auditory, and prefrontal projection classes (supplementary table 3). Analysis of the single-cell RNAseq dataset resulted in multiple clusters for each projection class which, when projected onto the first principal component of our bulk RNAseq dataset, spanned the same axis of variance identified from bulk data (Figure 4 A,B,C, supplementary figure 4). Markers for this axis were spatially separated at the single-cell level, with clear intermediate cells present (Figure 4D, supplementary figure 5,6,7). However, some individual single-cell clusters had similar PC1 positions (e.g. clusters 1 and 2 of the motor projections neurons, or clusters 1 and 2 of the somatosensory nuclei, supplementary figure 4). This suggests that

distinct clusters within a single modality could exist in similar anatomical positions. We examined this possibility by performing RNAscope for genes which distinguished these clusters (Pvalb and Tnnt1) . Using motor thalamus as an example, Pvalb and Tnnt1 expression were both found within the confines of a single thalamic nucleus, VL. While neurons selectively expressing each gene could be found, intermediate cells were also apparent. Pvalb-selective, intermediate, and Tnnt1-selective cells were distributed along the mediolateral axis (Figure 4E). "

"Therefore spatially organized genetic gradients can exist even within individual thalamic nuclei.

In this study we produced a near-comprehensive transcriptomic atlas of thalamus via RNAseq in order to investigate its organizational structure. Though input modality has historically defined thalamic nuclei, several alternative, overlapping organizational classifications have been proposed 1,6 . Early work perturbing thalamus showed that nuclei of the same modality can have strikingly different contributions to behavior, leading to a division between arousal-linked intralaminar nuclei and the rest of thalamus [7][8][9] . More recently, theories developed in the sensory thalamus have subdivided thalamic nuclei into two groups, based upon anatomical or neurochemical features [10][11][12] . However, precise delineation of groups is not consistent between theories, and no theory is comprehensive. If and how these later theories apply beyond sensory thalamus, for example, is unknown.

Here, we reconcile and extend these theories by placing almost all thalamic nuclei into one of three profiles, lying on a single topographic axis identified via unsupervised analysis. This axis extends from the midline/intralaminar groups through to a profile that includes the classic 'relay' sensory nuclei. Though prior theories did not incorporate electrophysiological properties, we found that genes directly affecting these properties were highly differentially expressed between groups, implying conserved, systematic variation of response properties across thalamus.

Understanding how the pattern of intra-thalamic variability identified here is used to support different circuit functions will be an important challenge 13 .

Prior theories of thalamic organization have placed thalamic cells into discrete nuclei, and thalamic nuclei into discrete groups. However, it remains unclear if subtypes of cells exist within nuclei and how discrete vs. continuous the cells type differences are. Recent studies in other non-thalamic areas have argued that gradients of projection neuron identities can exist 14,15 . Our single-cell transcriptomic data revealed heterogeneity within thalamic nuclei and suggest that thalamic genetic profiles exist on a spectrum, with mixtures of genetic profiles often existing in single neurons. Intermediate cell types exist both within and between nuclei, and are topographically localised."

(Phillips et al., 2018)

A single spectrum of neuronal identities across thalamus

"A. Schematic of experimental pipeline to obtain transcriptomic atlas of the thalamus. In this example, motor thalamic neurons were retrogradely labelled from their primary projection target (motor cortex), manually dissected and sorted. Viruses expressing green and blue fluorescent proteins (GFP, BFP), respectively, were injected to the deep cerebellar nuclei (DCN) and substantia nigra pars reticulata (SNr) to label motor nuclear subdivisions (ventral lateral (VL) and ventral anterior (VA), respectively) previously identified 8,9 . B. Example labelling from scheme shown in A. Coronal sections. Scale bars = 200 μm. C. Hierarchical clustering of thalamic nuclei using Spearman's correlation of top 500 most differentially expressed genes (Extended Data Fig. 1B) across all 22 nuclei. Major clusters defined as the top 5 branches of cluster dendrogram. D. Topographic localization of gene expression profiles in thalamus. Coronal thalamic section schematics with nuclei colored as in Fig. 1C. Unsampled nuclei are left uncolored. (see Fig. 1B). To understand the pattern of gene expression differences between the thalamic projection profiles, we performed principal component analysis (PCA) on our data. Principal component 1 (PC1, 38% explained variance) separated nuclei into the same major profiles identified via hierarchical clustering (Fig. 2A, Extended Data Fig. 3A). Again, position on this first dimension strongly correlated with mediolateral position, demonstrating topogenetic architecture in thalamus (Fig. 2B, Extended Data Fig. 3B). Based on their relative order on this first component, we named the three major profiles primary, secondary, and tertiary. The progressive difference from primary to tertiary nuclei was also evident in the number of genes differentially expressed between the groups, with the primary and tertiary nuclei being most distinct, and the other two comparisons being less so (Fig. 2C). This primary axis was prominently enriched in genes encoding neurotransmitter receptors, ion channels, and signaling molecules (Fig. 2D and E)."

"A. Overview of single-cell clusters within each projection system visualized via tSNE. Cells colored by cluster identity. B. Violin plots for cluster marker genes (inclusion criteria were Likelihood Ratio Test Pvalue < 10 -5 and log2 fold change > 0.5 for each cluster). C. Projection of single-cell data onto pooled-cell PC1 from Fig. 2A. Each dot is a single cell colored as in A. D. Topographic distribution of marker genes within 6 major thalamic modalities. Multi-FISH with probes for Calb2 (pink), Tnnt1 (red) and Necab1 (green). See Extended Data Fig. 7 and 8 for expanded views and quantification. Scale bars = 200 μm. E. Multi-FISH within VL thalamus comparing genes marking clusters in single-cell data.

Left panel shows field of view (coronal section, scale bar = 100 μm). Right three panels (scale bars = 50μm) show expansion of the three boxed areas, moving left to right.

Middle box shows intermediate cells expressing both single cell cluster markers. (red = Tnnt1, green = Pvalb, blue = DAPI). Thus far we analyzed thalamic nuclei by pooling projection neurons from specific anatomical positions. The resulting transcriptional profiles

could represent homogenous populations, discrete subtypes, or cells with graded differences. To probe these possibilities, we profiled the transcriptomes of individual neurons from motor, somatosensory, visual, auditory, and prefrontal projection classes (Extended Data Table 3). Analysis of this single-cell RNAseq dataset resulted in multiple clusters for each projection class, and cluster markers included many genes that also distinguished nuclei (Fig. 4B). Single-cell clusters also separated along the first principal component derived from our pooled-cell RNAseq dataset (Fig. 4C, Extended Data Fig. 6). "

"Markers for the single-cell clusters were spatially separated at the single-cell level, but with intermediate cells especially prominent near the anatomical boundaries of thalamic nuclei (Fig. 4D, Extended Data Fig. 7). This is consistent with spatially organized heterogeneity or a gradient-like organization rather than intermingled, discrete populations 15,16 . Given the strong relationship between PC1 and anatomical nuclei position (Fig. 2C and Extended Data Fig. 3), the presence of single-cell clusters occupying similar PC1 positions (e.g. clusters 1 and 2 of the motor projections neurons, or clusters 1 and 2 of the somatosensory nuclei, Extended Data Fig. 6) suggested that distinct neuron types could also coexist within anatomical boundaries of nuclei. We examined this possibility by performing multi-color fluorescent in situ hybridization (multi-FISH) for genes which distinguished amongst the major profiles (e.g. Pvalb and Tnnt1). Taking motor thalamus as an example, Pvalb and Tnnt1 expressing cells were found within the anatomical boundaries of a single thalamic nucleus (VL; Fig. 4E). Some individual VL cells expressed both Pvalb and Tnnt1; Pvalb-selective, intermediate, and Tnnt1-selective cells were distributed along the mediolateral axis (Fig. 4E). Therefore, spatially organized transcriptomic differences can exist even within individual thalamic nuclei. A common organizing principle of thalamus divides nuclei into discrete core or matrix subtypes that span modalities 1,7 . To date the strongest evidence for cross-modal organization is that genetic differences can be larger within a sensory modality than between sensory modalities 17 (Extended Data Fig. 9). However, previous studies have focused either on select sensory systems and/or have had limited scope with respect to the molecules investigated 18 . Here, using the full transcriptomes of nearly all thalamic pathways, we confirm cross-modal organization but replace the core/matrix dichotomy with a spectrum of profiles that span a single axis of genetic variance."

(Roy et al., 2022)

Thalamic subnetworks as units of function

"To appreciate the levels of organization discussed in this Review, we need to explicitly define the three key terms that we use throughout this paper. We refer to the term 'cell type' as a feature that is identified via a singular approach. Within individual thalamic nuclei, cell types have traditionally been defined on the basis of morphology, neurochemistry or physiology. For example, morphology can identify neurons based on differences in soma size and dendritic branching patterns 1,3 , whereas neurochemistry indicates that cell types can be either excitatory or inhibitory (for example, by the expression of enzymes for the synthesis and packaging of glutamate or GABA) 3 . Physiology, on the other hand, can identify neurons using distinct sensory response patterns (for example, in the LGN) 2 . The advent of modern genetic profiling has enabled the identification of 'subpopulations': groups of cells that share a particular pattern of gene expression, morphology, physiology and fine-grained connectivity. In this Review, we define one or more subpopulations serving a particular function as a 'subnetwork' (Fig. 2). Because each neuron or subpopulation often has multiple inputs and outputs, using the same subpopulation in different functional subnetworks is likely to be a common and efficient approach for functional circuit assembly. For instance, subpopulation A may form a subnetwork with subpopulation B for a given function, but forms a subnetwork with subpopulation X to serve a different function. Admittedly, this definition is forward-looking and will require future experiments to continue to validate and refine it.

One example of a putative thalamic subnetwork came from examining neural activity patterns within the TRN 15 , which, up to that point, had been considered a functional monolith 16 . A combination of electrophysiological recordings and connectivity-based optical tagging revealed the existence of subpopulations that share activity patterns depending on the principal thalamic structures to which they project 15 . More recently, state-of-the-art single-cell RNA sequencing (scRNA-seq) has indicated that these 'TRN subnetworks' could be further specified using gene expression gradients 17 subpopulations in several brain regions, such as"

"brainstem, histaminergic afferents from the hypothalamus and noradrenergic afferents from the parabrachial region have heterogeneous terminals in the thalamus 1,2 .

Physiology. Building on these findings, researchers have consistently observed electrophysiological differences in thalamic nuclei.

Whereas thalamic principal cells are thought to have comparable intrinsic properties across sensory nuclei, there is substantial heterogeneity in their action potential parameters [39][40][41][42][43] . Recently, whole-cell recordings from three motor-related thalamic nuclei (central medial (CM), ventral anterior (VA) and ventral lateral (VL) nuclei) showed that electrophysiological membrane and synaptic properties varied along a gradient from CM to VA to VL nuclei 44 . Whereas previous studies revealed electrophysiological heterogeneity of thalamic neurons, this study 44 found that single-cell physiological properties across thalamic nuclei exist along a continuum rather than forming unique, discrete profiles (Fig. 3b). However, the functional implications of this observation have not been revealed. This study also linked these physiological observations to graded transcriptional profiles (Fig. 3c) and morphological differences, suggesting that cellular morphology, terminal arborization, input type and electrophysiological properties are effective approaches for characterizing thalamic cellular diversity. Regarding the fundamental intrinsic feature of thalamic principal neurons 45-47 their two distinct firing modes (tonic versus bursting)-heterogeneity in these modes has been associated with somatodendritic morphological differences 27,48,49 .

Neurotransmitters and receptors. Finally, differences in the localization of neurotransmitters and their receptors [50][51][52][53][54]

[55][56][57] led to the idea that variation in gene expression profiles may serve as an organizing principle across thalamic nuclei. One group identified four transcription factor genes that showed differential expression within excitatory thalamic nuclei 58 that these candidate genes might reflect distinct subpopulations that are shared across excitatory or inhibitory nuclei, many groups attempted to identify thalamic nucleus-and subpopulation-specific molecular markers ["

"Although studies that have identified subnetworks within the thalamus up to now have used rodents, there is substantial evidence to suggest that heterogeneity among thalamic neurons is conserved from rodents to primates. Here, we briefly discuss several primate studies that support the idea of diverse thalamic subpopulations or show that some cellular properties of primate thalamic neurons are comparable to those in rodents. Neurotransmitter expression patterns are usually conserved across species; for instance, nicotinic binding sites are enriched in the LGN, the MD nucleus and the rhomboid nucleus in mice and monkeys 52 . Similarly, a study examined the expression of genes in the monkey thalamus that were enriched in different mouse thalamic nuclei and found highly consistent patterns 63 . Several genes were expressed in excitatory nuclei in monkeys but not in rodents. A study using high-density oligonucleotide arrays to identify thalamic nuclei-specific gene expression in adult monkeys 120 showed that over 550 genes were selectively expressed in the anterior thalamus or the CM, MD or ventral posterior nucleus. This study identified a marker for excitatory nuclei (coding for transcription factor 7-like 2), a marker for excitatory nuclei but not the CM nucleus or the PF (coding for Purkinje cell protein 4), a CM-PF-specific marker (coding for cerebellin 1 precursor) and an

LGN marker (Spp1). These markers have comparable expression patterns in mice 69 . A study that showed distinct subpopulations labeled by PV or somatostatin (SST) in the mouse TRN showed that these two subpopulations are also found in human TRN samples 87 . Additionally, in primates, SST-expressing neurons exhibit differential distribution in the visual sector of the TRN 121 . As in rodents, the LGN in monkeys has distinct subpopulations based on single-cell electrophysiological properties, spatial localization and inputs from distinct RGC subtypes 122,123 .

Tracing studies have also revealed subpopulations within thalamic nuclei in monkeys 109,124,125 . For example, the PF in monkeys contains projection-specific subpopulations 90 , including neurons that project to striatal regions and neurons that project"

(Yao et al., 2023)

A high-resolution transcriptomic and spatial atlas of cell types in the whole mouse brain

"The mammalian brain is composed of millions to billions of cells that are organized into numerous cell types with specific spatial distribution patterns and structural and functional properties. An essential step towards understanding brain function is to obtain a parts list, i.e., a catalog of cell types, of the brain. Here, we report a comprehensive and high-resolution transcriptomic and spatial cell type atlas for the whole adult mouse brain. The cell type atlas was created based on the combination of two single-cell-level, whole-brain-scale datasets: a single- cell RNA-sequencing (scRNA-seq) dataset of ~7 million cells profiled, and a spatially resolved transcriptomic dataset of ~4.3 million cells using MERFISH. The atlas is hierarchically organized into five nested levels of classification: 7 divisions, 32 classes, 306 subclasses, 1,045 supertypes and 5,200 clusters. We systematically analyzed the neuronal, non-neuronal, and immature neuronal cell types across the brain and identified a high degree of correspondence between transcriptomic identity and spatial specificity for each cell type. The results reveal unique features of cell type organization in different brain regions, in particular, a dichotomy between the dorsal and ventral parts of the brain: the dorsal part contains relatively fewer yet highly divergent neuronal types, whereas the ventral part contains more numerous neuronal types that are more closely related to each other. We also systematically characterized cell-type specific expression of neurotransmitters, neuropeptides, and transcription factors. The study uncovered extraordinary diversity and heterogeneity in neurotransmitter and neuropeptide expression and co-expression patterns in different cell types across the brain, suggesting they mediate a myriad of modes of intercellular communications. Finally, we found that transcription factors are major determinants of cell type classification in the adult mouse brain and identified a combinatorial transcription factor code that defines cell types across all parts of the brain. The whole-mouse-brain transcriptomic and spatial cell type atlas establishes a benchmark reference atlas and a foundational resource for deep and integrative investigations of cell type and circuit function, development, and evolution of the mammalian brain."

"In this study, we created a comprehensive, high-resolution transcriptomic cell type atlas for the whole adult mouse brain based on the combination of two whole-brain-scale datasets: a scRNAseq dataset of ~7 million cells and a MERFISH dataset of a similar scale (~4.3 million cells from the AIBS MERFISH brain). We used ~4.1 million high-quality single-cell transcriptomes after stringent QC to create a transcriptomic cell type taxonomy. We used the MERFISH data, which were generated using marker genes derived from the whole-brain transcriptomic taxonomy, to annotate the spatial location of each subclass and each cluster in the taxonomy. We then built a hierarchically organized transcriptomic and spatial cell type atlas with five nested levels: 7 divisions, 32 classes, 306 subclasses, 1,045 supertypes and 5,200 clusters (Figure 1). The neuronal cell type composition in each major brain structure were systematically analyzed (Figure 2) and distinct features in different brain structures identified (Figure 6). We discovered many sets of neuronal types with varying degrees of similarity with each other, including transitional neuronal types across regions as well as highly distinct neuronal types. We also systematically analyzed all divisions of non-neuronal cell types as well as immature neuronal types present in the adult brain, and identified their unique spatial distribution and spatial interaction patterns (Figure 4). Finally, we systematically characterized cell-type specific neurotransmitters, neuropeptides, and transcription factors, and discovered unique characteristics for each as discussed below. This large-scale study allowed us to delineate several principles regarding cell type organization across the whole mouse brain. It provides a benchmark reference cell type atlas as a resource for the community that will enable many more discoveries in the future.

One of the most striking findings from our study is the high degree of correspondence between transcriptomic identity and spatial specificity. Every subclass (and all supertypes and many clusters within each) has a unique and specific spatial localization pattern within the brain. Furthermore, the relative relatedness between transcriptomic types (as revealed in 2D and 3D UMAPs) is strongly correlated with the spatial relationship between them. Transcriptomically related cell types"

(Govek et al., 2022)

Developmental trajectories of thalamic progenitors revealed by single-cell transcriptome profiling and Shh perturbation

"Our study provides a comprehensive transcriptome-wide analysis of thalamic progenitors and their trajectories into thalamic nuclei during embryonic stages of brain development. We demonstrate that molecular signatures of thalamic neuronal subtypes can be readily distinguished as early as E12.5, prior to their aggregation into histologically distinct thalamic nuclei. 26 These data lend support to the outside-in model of thalamic neurogenesis, whereby early-born neurons contribute to lateral thalamic nuclei and later-born neurons contribute to medial thalamic nuclei. 36,58 Our work further extends this model with molecular insights into the mechanisms by which diverse thalamic nuclei acquire their identities. By following single-cell trajectories over developmental time, our data are consistent with a model in which thalamic nuclei arise from three distinct cellular lineages. In this model, Shh-responsive cTh.Pro1 progenitors give rise to glutamatergic neurons (cTh.N1) within sensory (DLG, VPM/VPL, vMG), motor (VA/VL, VM), and anterior (AD, AV, AM) nuclei that occupy predominantly ventrolateral regions of the thalamus. Shh-responsive rTh.Pro progenitors give rise to GABAergic neurons (rTh.N) in VLG and IGL thalamic nuclei, in agreement with previous findings. 12,13 In contrast, Shh-independent cTh.Pro2 progenitors give rise to neurons (cTh.N2) within nuclei (e.g., PV, MD, CM) that form at more dorsomedial positions of the thalamus.

The bifurcation of transcriptomic trajectories according to thalamic progenitor identity is in general agreement with results from in vivo clonal analyses. 36,59 These studies revealed that individual thalamic progenitors give rise to many neurons that populate multiple thalamic nuclei. They also demonstrated that the clonal relationship between thalamic nuclei is determined primarily by the rostro-caudal and dorsoventral positions of thalamic progenitors."

Major neuronal classes with soma in the mouse thalamic region

At broad scale, neurons with somata in the mouse thalamic region fall into glutamatergic projection neurons and several GABAergic populations. The main inhibitory groups are the thalamic reticular nucleus, ventral/lateral geniculate-related nuclei, sparse local interneurons in dLGN, and some midbrain-derived inhibitory cells in nearby regions.

The broadest division is by fast transmitter.

- **Glutamatergic projection neurons**
  - These are the dominant neurons of thalamus proper and are marked by **Slc17a6/VGLUT2** expression. In the ABC atlas they largely form one thalamic glutamatergic class. (Turner et al., 2025) (Yao et al., 2023)
  - They usually send long-range axons to cortex and/or subcortical targets and are the main source of thalamocortical output. (Clascá et al., 2012) (Phillips et al., 2019)
- **GABAergic neurons of thalamic reticular nucleus (TRN/RT)**
  - TRN is the major source of thalamic inhibition and contains molecularly graded inhibitory neurons rather than a single uniform type. (Li et al., 2020)
- **GABAergic neurons in LGv/IGL and related ventrolateral thalamic regions**
  - Developmental data place GABAergic neurons of **ventral lateral geniculate (VLG/LGv)** and **intergeniculate leaflet (IGL)** in a rostral thalamic lineage distinct from most glutamatergic nuclei. (Govek et al., 2022)
  - In ABC-based analyses, GABAergic classes are also found in **LGv** and **SPF**, and class 12 HY GABA spans RT, zona incerta, and LGv. (Turner et al., 2025)
- **Sparse local GABAergic interneurons in dLGN**
  - Zhou et al. note that the central thalamus is almost entirely excitatory except for a few local GABAergic interneurons in **LGd/dLGN**. (Zhou et al., 2023)
- **Special cases in neighboring thalamic-region structures**

- Some datasets include **habenula** and **zona incerta** with thalamic-region analyses, but these are not thalamus proper and should be kept separate. (Turner et al., 2025)
- The supramammillary nucleus can include a dual-transmitter population co-expressing **Slc32a1** and **Slc17a6**, but SUM is hypothalamic rather than thalamic. (Mickelsen et al., 2020)

In humans, medial thalamus contains more inhibitory neurons than classic mouse dorsal thalamus, including **SOX14**-positive midbrain-derived interneurons and additional inhibitory populations, underscoring that some inhibitory diversity is species- and region-dependent. (Schulmann et al., 2024) (Trujillo et al., 2026)

Paper	Neuron Classification Scheme	Anatomical Scope	Cell Type Identity	Neurotransmitter Profile	Molecular Markers	Neuropeptide Profile	Anatomical Location And Distribution	Connectivity Pattern	Functional Role	Tran ipton Evide e An Hete gene

Paper	Neuron Classification Scheme	Anatomical Scope	Cell Type Identity	Neurotransmitter Profile	Molecular Markers	Neuropeptide Profile	Anatomical Location And Distribution	Connectivity Pattern	Functional Role	Tranipton Evide e An Hete gene
Exploring the correspondence between gene expression and thalamic nuclei using the THALMANAC resource	Thalamic neurons are grouped using transcriptomic t-types, organized hierarchically into class, subclass, supertype, and cluster; higher levels also incorporate developmental origin, location, and classical markers.	The analysis focuses on a thalamus-specific MERFISH dataset, including thalamus proper, zona incerta, and habenula; specific nuclei also include TRN, dLGN, LGv, SPF, and related neighboring regions.	Major neuron types are organized into transcriptomic classes / subclasses, including thalamic glutamatergic neurons, GABAergic reticular/ hypothalamic populations, and habenula-specific classes; subclasses often span multiple nuclei.	Not reported in the provided snippets; neurotransmitter phenotype was not described as a primary analysis variable.	They are used as key molecular markers to define classical neuron types and distinguish thalamic cell populations.	It is a neuron-type-specific annotation of neuropeptides or peptide-related genes, indicating characteristic peptide markers that may further subdivide the population.	Neuron types show nucleus-specific localization, spatial gradients, and some laminar/shell-core or regional enrichment patterns within the thalamus.	Not directly addressed; the paper focuses on gene expression and thalamic nuclei, not detailed connectivity patterns.	N/A	Supp ts thala c nuc parc ation but revea hete enec t-typ subd sions and gene expr ion grad ts withi and acro nucl

Paper	Neuron Classification Scheme	Anatomical Scope	Cell Type Identity	Neurotransmitter Profile	Molecular Markers	Neuropeptide Profile	Anatomical Location And Distribution	Connectivity Pattern	Functional Role	Tran ipton Evide e An Hete gene
A high-resolution transcriptomic and spatial atlas of cell types in the whole mouse brain	A hierarchical transcriptomic-and-spatial taxonomy; transcription factors are major determinants, with cells grouped into divisions, classes, subclasses, supertypes, and clusters.	Whole adult mouse brain; no specific thalamic-region inclusion/exclusion is stated in the snippets.	Major neuron subclasses show specific spatial localization and transcriptomic identity, including glutamatergic, GABAergic, local interneuron, and dual-transmitter populations.	Cell types show diverse neurotransmitter expression and co-expression patterns; the paper characterizes neurotransmitters and identifies a primary fast transmitter phenotype for each neuron type.	Cell types are identified and distinguished by marker genes, and transcription factors form a combinatorial code defining them.	The paper systematically characterized cell-type specific neuropeptide expression and co-expression patterns, revealing extraordinary diversity across brain cell types.	Neuron types show nucleus-specific localization, spatial gradients, laminar or shell-core organization, and regionally enriched distribution patterns.	N/A	N/A	A whol brain scRN seq and MER H atl revea d ma disti hete eneo cell type inclu g trans onal type with trans ptom ident tight mato ng spati spec ity.

Paper	Neuron Classification Scheme	Anatomical Scope	Cell Type Identity	Neurotransmitter Profile	Molecular Markers	Neuropeptide Profile	Anatomical Location And Distribution	Connectivity Pattern	Functional Role	Tran ipton Evide e An Hete gene
Unveiling the diversity of thalamocortical neuron subtypes	No specific neuron classification scheme is described; the abstract only notes subtypes are reviewed by morphology, connectivity, target specificity, gene expression, and development.	Focuses on thalamic projection neurons within and across anatomically defined nuclei; does not specify particular nuclei included or excluded.	Distinct subtypes of thalamic projection neurons, varying in morphology, connectivity, target specificity, gene expression, and development.	Not discussed in the abstract.	Neuron type-specific gene expression is reviewed as a distinguishing feature of thalamocortical neuron subtypes.	N/A	It emphasizes that thalamic projection neuron subtypes vary both within and across anatomically defined nuclei, rather than being uniformly distributed.	Distinct thalamic projection neuron subtypes differ in morphology, input/output connectivity, and target specificity.	The abstract does not specify a particular functional role; it says TC neurons are important for understanding perception, movement, cognition, and attention.	Ment ns recen findi s on neur type- spec gene expr ion, b provi s no trans ptom data s, clust s, or acce on num s.

Paper	Neuron Classification Scheme	Anatomical Scope	Cell Type Identity	Neurotransmitter Profile	Molecular Markers	Neuropeptide Profile	Anatomical Location And Distribution	Connectivity Pattern	Functional Role	Tran ipton Evide e An Hete gene
A repeated molecular architecture across thalamic pathways	Transcriptomic profiles define three major thalamic pathway identities across modalities; no specific naming framework or conflict-resolution scheme is described.	The study examined comprehensive transcriptomic sequencing from murine thalamic pathways across major projection systems; specific nuclei are not detailed in the provided text.	Three major profiles of thalamic pathways, varying along a continuum across major projection systems.	N/A	A repeated molecular architecture across thalamic pathways, with three major transcriptional profiles distinguishing thalamic neuronal identities.	Neuropeptide-related genes are reported for each neuron type, including characteristic peptide markers that can further subdivide populations.	Topographically organized across the thalamus, with features conserved in humans, forming repeated profiles across major projection systems.	A repeated architecture across thalamocortical systems, with convergent inputs to cortical regions and three major thalamic pathway profiles repeated across projection systems.	Repeated molecular profiles across thalamic pathways linked to cellular features critical for function, establishing a molecular framework for thalamocortical output.	Comehen e trans ption analy s of retro adely label thala c neur s ident ed three majo path y profi with repe d cont um-li orga ation acro proje on syste s.

Paper	Neuron Classification Scheme	Anatomical Scope	Cell Type Identity	Neurotransmitter Profile	Molecular Markers	Neuropeptide Profile	Anatomical Location And Distribution	Connectivity Pattern	Functional Role	Tran ipton Evide e An Hete gene
Distinct subnetworks of the thalamic reticular nucleus	Uses a transcriptomic gradient with mutually exclusive Spp1 and Ecel1 extremes, forming molecularly defined TRN–thalamus subnetworks.	The study examined the mouse thalamic reticular nucleus and its connections to functionally distinct first-order and higher-order thalamic nuclei.	Two TRN neuron subpopulations defined by a transcriptomic gradient, with mutually exclusive Spp1 and Ecel1 expression.	TRN neurons are GABAergic inhibitory neurons; the snippet does not mention glutamatergic or mixed transmitter identities.	Mutually exclusive markers Spp1 and Ecel1 define the extremes of the TRN transcriptomic gradient.	N/A	TRN neuron types show core- and shell-like anatomical organization at opposite extremes of a transcriptomic gradient.	TRN neuron subpopulations show differential connectivity with first-order and higher-order thalamic nuclei, forming molecularly defined TRN–thalamus subnetworks.	Differentially regulates sleep through distinct TRN–thalamus subnetworks.	TRN cellu heter eneit is defin by a trans ptom grad t of t nega ely corre ed gene expr ion profi , with mutu y exclu ve Spp1 and Ecel expr ion a its extre es.

Paper	Neuron Classification Scheme	Anatomical Scope	Cell Type Identity	Neurotransmitter Profile	Molecular Markers	Neuropeptide Profile	Anatomical Location And Distribution	Connectivity Pattern	Functional Role	Tran ipton Evide e An Hete gene
Developmental trajectories of thalamic progenitors revealed by single-cell transcriptome profiling and Shh perturbation	Thalamic neurons are classified mainly by developmental lineage, transmitter identity, anatomical nucleus, and transcriptomic cluster; transcriptomic identity distinguishes subtypes early.	Thalamic progenitors and their trajectories into thalamic nuclei; includes sensory, motor, anterior, VLG, IGL, PV, MD, and CM nuclei.	Three major thalamic neuron classes are described: glutamatergic neurons (cTh.N1/cTh.N2) and GABAergic neurons (rTh.N).	Glutamatergic cTh.N1 neurons arise from Shh-responsive cTh.Pro1 progenitors, GABAergic rTh.N neurons from rTh.Pro progenitors; cTh.N2 identity is not specified.	Distinct thalamic neuronal subtypes are molecularly distinguishable as early as E12.5, before nuclei become histologically distinct.	No specific neuropeptide profile is reported in the provided text.	Thalamic neuron types show nucleus-specific localization with ventrolateral or dorsomedial regional enrichment, consistent with an outside-in spatial gradient.	N/A	N/A	Singl cell traje ries revea three thala c prog tor linea s, disti uishi cTh.I 1, rTh.P and cTh.I 2 an their desc dant neur al nucl

Paper	Neuron Classification Scheme	Anatomical Scope	Cell Type Identity	Neurotransmitter Profile	Molecular Markers	Neuropeptide Profile	Anatomical Location And Distribution	Connectivity Pattern	Functional Role	Tranipton Evide e An Hete gene
Brain-wide Correspondence Between Neuronal Epigenomics and Long-Distance Projections	Thalamic neurons are grouped by molecular and projection-defined clusters; spatial nuclei and projection identity both map to these clusters, with projection-specific enrichment overriding broad anatomical grouping.	Thalamic neurons were analyzed broadly across the thalamus; no finer anatomical inclusion/exclusion list is provided.	Distinct thalamic cell types are organized into projection-specific clusters, with cortical versus subcortical targets enriched in different cell-type subsets.	N/A	No specific molecular markers are stated in the provided snippets.	N/A	Thalamic cell clusters were spatially organized by anatomical region, with nucleus-specific localization and projection-specific enrichment patterns.	Connectivity patterns distinguish neuron types and align with projection-specific epigenomic clusters, including distinct thalamic target and pathway enrichments.	N/A	Sing cell RNA- seq show thala c proje on neur s are hete enec with each proje on enric d in spec , part shar clust s.

Paper	Neuron Classification Scheme	Anatomical Scope	Cell Type Identity	Neurotransmitter Profile	Molecular Markers	Neuropeptide Profile	Anatomical Location And Distribution	Connectivity Pattern	Functional Role	Tran ipton Evide e An Hete gene
A conserved cell-type gradient across the human mediodorsal and paraventricular thalamus	Neuron types were classified by transcriptomic clusters, then mapped to anatomical nuclei and developmental/transmitter markers; conflicting schemes were resolved by nearest-centroid annotation and integration with reference datasets.	Human MD and adjacent midline nuclei were examined, including PVT and PT, with surrounding nuclei such as intralaminar, motor/somatosensory, and lateral dorsal thalamus.	Excitatory SLC17A6/VGluT2+ projection neurons, inhibitory SOX14+ neurons, and several non-neuronal populations; inhibitory neurons were nearly 40% in human MD.	ExN_H2 and ExN_H3 were glutamatergic excitatory neuron types, while inhibitory neurons were GABAergic; some cells co-expressed multiple markers.	Excitatory neurons expressed SLC17A6/VGluT2, inhibitory neurons expressed SOX14, and marker genes defined opposite ends of the MD–PVT gradient.	Not reported in the provided snippets; no specific neuropeptide profile is described.	Neuronal types show nucleus-specific localization and a spatial gradient from MDmc to PVT/PT, with enrichments in MDpc/MDdc and adjacent nuclei.	A major MD–PVT excitatory gradient maps to primary versus tertiary thalamus and likely reflects distinct connectivity and functional properties.	The gradient likely reflects distinct connectivity and functional properties of these neurons; primary cells may expand in humans, but specific roles are not detailed.	Sing cell a spati data revea d ma thala c cel class and a excit ory-neur trans ption grad t wit disti MDm MDp and PVT/ ends

Paper	Neuron Classification Scheme	Anatomical Scope	Cell Type Identity	Neurotransmitter Profile	Molecular Markers	Neuropeptide Profile	Anatomical Location And Distribution	Connectivity Pattern	Functional Role	Tran ipton Evide e An Hete gene
Selective vulnerability and resilience of thalamic neuronal populations in Frontotemporal Dementia	Thalamic neurons were grouped into four populations using a transcriptomic classification based on developmental lineage and transmitter identity: InNeu SOX14+, InNeu SOX14-, ExNeu RNF220+, and ExNeu RNF220-.	Human medial thalamus; integration showed populations across several nuclei in the human medial thalamus.	Four major thalamic neuronal populations: two excitatory and two inhibitory subtypes, annotated as InNeu SOX14+, InNeu SOX14-, ExNeu RNF220+, and ExNeu RNF220-.	Thalamic neurons were classified as GABAergic or glutamatergic: inhibitory populations expressed GAD2 and MEIS2, while excitatory populations expressed SLC17A6.	Thalamic neuronal populations were distinguished by GAD2, MEIS2, SOX14, NR2F2, ERBB4, LAMP5, BCL11B, SLC17A6, and RNF220; inhibitory subtypes also showed LHX6, SST, NPY, CXCL14, VIP.	The provided text does not report a neuropeptide profile for the thalamic neuron types.	InNeu SOX14+ and ExNeu RNF220+ populations were present across several nuclei in the human medial thalamus.	N/A	Thalamic excitatory populations are thalamocortical projection neurons; inhibitory populations likely provide local gating/modulatory control.	Four majo thala c neur al popu ions were iden ed: t excit ory a two inhib ry subt es, w hete enec linea spec signa res.

Evidence

(Turner et al., 2025)

Exploring the correspondence between gene expression and thalamic nuclei using the THALMANAC resource

"The thalamus connects the sensory organs and major subcortical brain regions with the neocortex. The thalamus has long been divided into multiple discrete nuclei, based on cytoarchitecture, histochemical stains, and mesoscale connectivity. However, thalamic nuclei do not completely describe thalamic organization. For example, some boundaries between thalamic nuclei are

disputed, whereas other nuclei are known to contain subdomains with distinct connectivity and function. Moreover, the correspondence between cellular gene expression and other properties of thalamic projection neurons remains to be established. Spatial analysis of single cell gene expression provides a basis for reevaluating thalamic organization. We present the THALMANAC, (THALamusMERFISH ANalysis and ACcess) a Findable, Accessible, Interoperable, Reusable and Reproducible (FAIRR) resource for exploring and analyzing single-cell transcriptomic variation in the thalamus. The THALMANAC provides streamlined access to thalamic gene expression data registered to the common coordinate framework and tools for quantitative analysis and visualization of these data, all encapsulated in a reproducible, cloud computing platform. Using this resource, we find that gene expression generally supports the parcellation of thalamus into distinct nuclei. Some nuclei, such as the anteromedial nucleus, are additionally composed of discrete subdomains, while other nuclei share patterns of gene expression or are arrayed on a spatial gradient of gene expression. The THALMANAC establishes spatial transcriptomic data as a foundation for delineating thalamic organization. Graphical Abstract"

"This fine-scale structure could correspond to discrete, molecularly defined subdivisions of VM that have not yet been annotated, or to topographic organization of thalamocortical and corticothalamic connectivity.

A major question is how gene expression relates to existing parcellations and thalamic organization in general. Single cell RNA sequencing (scRNA-seq) provides detailed measurements of transcription in large populations of individual neurons (Tasic et al. 2016), which can be used to define transcriptomic cell types ('t-types'). The combination of scRNA-seq with spatial transcriptomics has been used to localize t-types across the whole mouse brain (Langlieb et al. 2023;Yao et al. 2023;Zhang et al. 2023). With t-types mapped into the CCFv3 coordinate space (Wang et al. 2020), we can evaluate how their spatial distributions correspond to anatomical parcellations (Yao et al. 2023).

Here, building on the Allen Brain Cell (ABC) Atlas scRNA-seq and spatial transcriptomics data, we developed a resource, the THALamus MERFISH ANalysis and ACcess (THALMANAC), which provides access to spatial gene expression data in the mouse thalamus; targeted analysis and data visualization of gene expression patterns; a platform for reproducible computations; and a no-code visualization app for data exploration. We demonstrate the utility of the THALMANAC by addressing how thalamic structural organization relates to gene expression. Aligning spatial transcriptomic data and nuclei boundaries reveals a wide range of relationships between t-types and thalamic nuclei: some t-types correspond to a single thalamic nucleus; others subdivide nuclei into multiple parts; many t-types span multiple thalamic nuclei.

Analysis of expression by itself supports these same conclusions: some nuclei boundaries are associated with sharp changes in gene expression; other nuclei contain gene expression gradients; and, in many cases, gene expression patterns span multiple nuclei.

Understanding how information is routed through the thalamus requires integrating insights gained from gene expression with other modalities, including connectivity and neurophysiology."

"The THALMANAC resource provides access to thalamic spatial gene expression data, visualization of the relationships between gene expression, t-types, and CCFv3 anatomical parcellations, and quantitative analysis of targeted quantitative analyses, all encapsulated in a cloud platform that enables reproducible computations and customized exploration of the data.

The THALMANAC is built on a thalamus-specific subset of spatial transcriptomics data from multiplexed error-robust fluorescence in situ hybridization (MERFISH) from the ABC Atlas. The ABC Atlas dataset measures 500 transcripts, chosen to distinguish t-types across the whole mouse brain, in spatially resolved, single cells. MERFISH cells were registered to the CCFv3 coordinate space and mapped to the scRNA-seq data by assigning each MERFISH cell to the label of the most similar t-type cluster. We identified neurons that belong to the thalamus (including the zona incerta and habenula) according to their t-type label (Figure S1A) and CCFv3 parcellation label (Figure S1B). The thalamic dataset analyzed here includes 80,170 MERFISH cells across 14 coronal sections collected from a single male mouse (Methods).

The THALMANAC includes three avenues for engaging with the t-type and gene expression data and accompanying analyses (Figure 1). First, a Streamlit web app (RRID:SCR_024354) provides nocode exploration, including interactive visualizations and on-the-fly di"erential gene expression analysis for thalamic nuclei (Interactive Streamlit App: thalmanac.allenneuraldynamics.org). Second, Jupyter notebooks (RRID:SCR_018315) provide step-by-step demonstrations of the Python code needed to reproduce all results presented here (github.com/AllenNeuralDynamics/thalmanac). Both tools allow, users to customize the thalamic cells that are selected from the full ABC Atlas."

"In the ABC Atlas, transcriptomic variation is used to assign cells to a hierarchy of discrete t-types (from broad to narrow categories: class, subclass, supertype, cluster). Thalamic neurons belong to 5 classes, 58 subclasses, 163 supertypes, and 563 clusters (Table S1; Figure S2) (taxonomy version CCN20230722) (Yao et al. 2023), but these numbers will change as the taxonomy evolves. Whereas the cluster level is based on unsupervised clustering, higher levels of the taxonomy are additionally informed by developmental origin via shared transcription factor expression, anatomical location, and classical (pre-scRNAseq) neuron types (DeFelipe et al. 2013) defined by key molecular markers (e.g. neuropeptides, calcium-binding proteins).

Transcriptomic classes typically correspond to major brain structures. Glutamatergic neurons in thalamus proper form a single class (class 18 TH Glut), whereas habenula neurons belong to a distinct class (class 17 MH-LH Glut) (Figure S2B). The other three

classes are not unique to the thalamus: class 12 HY GABA is found in the reticular thalamus, zona incerta and ventral lateral geniculate nucleus (CCFv3 annotations; RT, ZI, and LGv, respectively, see Table S1) but also in the hypothalamus; class 19 MB Glut is found in thalamic nucleus SPA but also in the midbrain; class 20 MB GABA is found in thalamic nuclei LGv and SPF and the midbrain. Transcriptomic subclasses span multiple thalamic nuclei(Figure S2C), whereas transcriptomic supertypes often define individual nuclei (e.g. AM), and several nuclei contain multiple supertypes (e.g. ZI) (Figure S2D). Most nuclei contain multiple transcriptomic clusters (Figure 2A, Figure S2E).

We explored correspondence between clusters and thalamic nuclei."

"A comprehensive understanding of how the thalamus routes information to the cortex will require multiple types of data and a community-wide e"ort. Our work contributes to this program by curating and analyzing the correspondence between state-of-the-art spatially resolved, single-cell transcriptomic data, and widely used parcellations of thalamus into discrete nuclei. In addition, we make our data and analyses available in an accessible, documented, and reproducible bioinformatics resource in the cloud. Using the AM nucleus and the somatosensory nuclei as examples, we illustrate how this interactive resource can be used to identify novel patterns of gene expression and how these patterns correspond to prior parcellations of the thalamus.

The THALMANAC resource streamlines researchers' access to thalamic spatial transcriptomic data. We anticipate that scientists will extend our analysis based on the ABC atlas data, or apply similar analyses to additional spatial transcriptomics datasets registered to the CCF coordinate system (Langlieb et al. 2023;Zhang et al. 2023). The tool should also be used as a reference to validate recent studies focused on specific nuclei, such as the spatially segregated projections from PF to distinct dorsal striatum regions (Mandelbaum et al. 2019) and spatially segregated inputs to PO (Casas-Torremocha et al. 2022), the molecular organization of the PVT (Gao et al. 2023), and the core/shell organization of neurons in RT (Li et al. 2020). Our interactive Streamlit app can be used to identify marker genes for post-hoc localization of neurophysiological and neuroanatomical experiments, or to select gene panels for multiplexed mFISH experiments. The THALMANAC is already being used in the e"ort to create and characterize enhancerbased genetic tools (Ben-Simon et al. 2025) to target thalamic cell types and regions.

The THALMANAC shows that at the scale of thalamic nuclei, the molecular organization of the thalamus resembles existing parcellations. "

"However, both discrete cell type composition and gene expression gradients vary at finer spatial scales within and across nuclei. Given the evidence for circuit and functional variation at the same length scales, additional data need to be integrated with molecular characterization to understand thalamic organization. We see two promising paths forward for data integration. First, existing data registered to the CCFv3 3D coordinate space, such as the databases of single-neuron morphological reconstruction (Winnubst et al. 2019) or neurophysiology data (Chen et al. 2024), could be linked to cell types and gene expression described in the THALMANAC. Second, targeted multimodal datasets, such as single-neuron morphological reconstructions combined with posthoc mFISH for transcriptomic cell typing would link cell types and gene expression to projection targets at the single cell level.

Beyond data integration, we also recognize the need for thalamus-focused, community annotation, similar to previous e"orts to annotate the fly connectome (Schlegel et al. 2024). We envision a refined parcellation of the thalamus where scientists with specific domain knowledge would combine molecular information from the THALMANAC with neuroanatomical and functional data to reflect progress in understanding of thalamic organization. Diagram illustrating the three workflows that can be launched from the Code Ocean Collection accompanying this paper. The Code Ocean platform provides built-in support for code versioning (via Git), environment replicability (via Docker), data and analysis provenance (via tracked data assets and linked pipeline runs), cloud compute resources (via AWS EC2 instances). Users can launch an interactive, no-code Streamlit application to explore spatial distributions of transcriptomic cell types and gene expression from their browser, as well as visualize locations of annotated cell types for each thalamic nucleus and highlight a specific nucleus or group of related thalamic nuclei. Tutorial Jupyter notebooks and Python code replicate all analyses and visualizations in a cloud workstation. These visualizations can be customized and extended. Users can download an AnnData-formatted .h5ad"

(Yao et al., 2023)

A high-resolution transcriptomic and spatial atlas of cell types in the whole mouse brain

"The mammalian brain is composed of millions to billions of cells that are organized into numerous cell types with specific spatial distribution patterns and structural and functional properties. An essential step towards understanding brain function is to obtain a parts list, i.e., a catalog of cell types, of the brain. Here, we report a comprehensive and high-resolution transcriptomic and spatial cell type atlas for the whole adult mouse brain. The cell type atlas was created based on the combination of two single-cell-level, whole-brain-scale datasets: a single- cell RNA-sequencing (scRNA-seq) dataset of ~ 7million cells profiled, and a spatially resolved transcriptomic dataset of ~ 4.3million cells using MERFISH. The atlas is hierarchically organized into five nested levels of classification: 7 divisions, 32 classes, 306 subclasses, 1,045 supertypes and 5,200 clusters. We systematically analyzed the neuronal, non-neuronal, and immature neuronal cell types across the brain and identified a high degree of correspondence between transcriptomic identity and spatial specificity for each cell type. The results reveal unique features of cell type organization in different brain regions, in particular, a dichotomy between the dorsal and ventral parts of the brain: the dorsal part contains relatively fewer yet highly divergent neuronal types, whereas the ventral part contains more numerous neuronal types that are more closely related to each other. We also systematically characterized cell-type specific expression of neurotransmitters, neuropeptides, and

transcription factors. The study uncovered extraordinary diversity and heterogeneity in neurotransmitter and neuropeptide expression and co-expression patterns in different cell types across the brain, suggesting they mediate a myriad of modes of intercellular communications. Finally, we found that transcription factors are major determinants of cell type classification in the adult mouse brain and identified a combinatorial transcription factor code that defines cell types across all parts of the brain. The whole-mouse-brain transcriptomic and spatial cell type atlas establishes a benchmark reference atlas and a foundational resource for deep and integrative investigations of cell type and circuit function, development, and evolution of the mammalian brain."

"In this study, we created a comprehensive, high-resolution transcriptomic cell type atlas for the whole adult mouse brain based on the combination of two whole-brain-scale datasets: a scRNAseq dataset of ~7 million cells and a MERFISH dataset of a similar scale (~4.3 million cells from the AIBS MERFISH brain). We used ~4.1 million high-quality single-cell transcriptomes after stringent QC to create a transcriptomic cell type taxonomy. We used the MERFISH data, which were generated using marker genes derived from the whole-brain transcriptomic taxonomy, to annotate the spatial location of each subclass and each cluster in the taxonomy. We then built a hierarchically organized transcriptomic and spatial cell type atlas with five nested levels: 7 divisions, 32 classes, 306 subclasses, 1,045 supertypes and 5,200 clusters (Figure 1). The neuronal cell type composition in each major brain structure were systematically analyzed (Figure 2) and distinct features in different brain structures identified (Figure 6). We discovered many sets of neuronal types with varying degrees of similarity with each other, including transitional neuronal types across regions as well as highly distinct neuronal types. We also systematically analyzed all divisions of non-neuronal cell types as well as immature neuronal types present in the adult brain, and identified their unique spatial distribution and spatial interaction patterns (Figure 4). Finally, we systematically characterized cell-type specific neurotransmitters, neuropeptides, and transcription factors, and discovered unique characteristics for each as discussed below. This large-scale study allowed us to delineate several principles regarding cell type organization across the whole mouse brain. It provides a benchmark reference cell type atlas as a resource for the community that will enable many more discoveries in the future.

One of the most striking findings from our study is the high degree of correspondence between transcriptomic identity and spatial specificity. Every subclass (and all supertypes and many clusters within each) has a unique and specific spatial localization pattern within the brain. Furthermore, the relative relatedness between transcriptomic types (as revealed in 2D and 3D UMAPs) is strongly correlated with the spatial relationship between them. Transcriptomically related cell types"

(Clascá et al., 2012)

Unveiling the diversity of thalamocortical neuron subtypes

"Our current understanding of thalamocortical (TC) circuits is largely based on studies investigating so-called 'specific' thalamic nuclei, which receive and transmit sensory-triggered input to specific cortical target areas. TC neurons in these nuclei have a striking point-to-point topography and a stereotyped laminar pattern of termination in the cortex, which has made them ideal models to study the organization, plasticity, and development of TC circuits. However, despite their experimental importance, neurons within these nuclei only represent a fraction of all thalamic neurons and do not reflect the diversity of the TC neuron population. Here we review the distinct subtypes of projection neurons that populate the thalamus, both within and across anatomically-defined nuclei, with regard to differences in their morphology, input/output connectivity and target specificity, as well as more recent findings on their neuron type-specific gene expression and development. We argue that a detailed understanding of the biology of TC neurons is critical to understand the role of the thalamus in normal and pathological perception, voluntary movement, cognition and attention."

(Phillips et al., 2019)

A repeated molecular architecture across thalamic pathways

"The thalamus is the central communication hub of the forebrain and provides the cerebral cortex with inputs from sensory organs, subcortical systems and the cortex itself. Multiple thalamic regions send convergent information to each cortical region, but the organizational logic of thalamic projections has remained elusive. Through comprehensive transcriptional analyses of retrogradely labeled thalamic neurons in adult mice, we identify three major profiles of thalamic pathways. These profiles exist along a continuum that is repeated across all major projection systems, such as those for vision, motor control and cognition. The largest component of gene expression variation in the mouse thalamus is topographically organized, with features conserved in humans. Transcriptional differences between these thalamic neuronal identities are tied to cellular features that are critical for function, such as axonal morphology and membrane properties. Molecular profiling therefore reveals covariation in the properties of thalamic pathways serving all major input modalities and output targets, thus establishing a molecular framework for understanding the thalamus. This work describes comprehensive transcriptomic sequencing from murine thalamic pathways. By integrating this molecular information with anatomical and functional features, this study reveals a repeated architecture across thalamocortical systems."

(Li et al., 2020)

Distinct subnetworks of the thalamic reticular nucleus

"The thalamic reticular nucleus (TRN), the major source of thalamic inhibition, regulates thalamocortical interactions that are critical for sensory processing, attention and cognition1–5. TRN dysfunction has been linked to sensory abnormality, attention deficit and sleep disturbance across multiple neurodevelopmental disorders6–9. However, little is known about the organizational principles that underlie its divergent functions. Here we performed an integrative study linking single-cell molecular and electrophysiological

features of the mouse TRN to connectivity and systems-level function. We found that cellular heterogeneity in the TRN is characterized by a transcriptomic gradient of two negatively correlated gene-expression profiles, each containing hundreds of genes. Neurons in the extremes of this transcriptomic gradient express mutually exclusive markers, exhibit core or shell-like anatomical structure and have distinct electrophysiological properties. The two TRN subpopulations make differential connections with the functionally distinct first-order and higher-order thalamic nuclei to form molecularly defined TRN–thalamus subnetworks. Selective perturbation of the two subnetworks in vivo revealed their differential role in regulating sleep. In sum, our study provides a comprehensive atlas of TRN neurons at single-cell resolution and links molecularly defined subnetworks to the functional organization of thalamocortical circuits. A study integrating single-cell RNA-sequencing and electrophysiology data shows that in mouse, the cellular repertoire of the thalamic reticular nucleus is characterized by a transcriptomic gradient defined at its extremes by mutually exclusive expression of Spp1 and Ecel1, providing insights into the organizational principles underlying the divergent functions of this brain region."

(Govek et al., 2022)

Developmental trajectories of thalamic progenitors revealed by single-cell transcriptome profiling and Shh perturbation

"Our study provides a comprehensive transcriptome-wide analysis of thalamic progenitors and their trajectories into thalamic nuclei during embryonic stages of brain development. We demonstrate that molecular signatures of thalamic neuronal subtypes can be readily distinguished as early as E12.5, prior to their aggregation into histologically distinct thalamic nuclei. 26 These data lend support to the outside-in model of thalamic neurogenesis, whereby early-born neurons contribute to lateral thalamic nuclei and later-born neurons contribute to medial thalamic nuclei. 36,58 Our work further extends this model with molecular insights into the mechanisms by which diverse thalamic nuclei acquire their identities. By following single-cell trajectories over developmental time, our data are consistent with a model in which thalamic nuclei arise from three distinct cellular lineages. In this model, Shh-responsive cTh.Pro1 progenitors give rise to glutamatergic neurons (cTh.N1) within sensory (DLG, VPM/VPL, vMG), motor (VA/VL, VM), and anterior (AD, AV, AM) nuclei that occupy predominantly ventrolateral regions of the thalamus. Shh-responsive rTh.Pro progenitors give rise to GABAergic neurons (rTh.N) in VLG and IGL thalamic nuclei, in agreement with previous findings. 12,13 In contrast, Shh-independent cTh.Pro2 progenitors give rise to neurons (cTh.N2) within nuclei (e.g., PV, MD, CM) that form at more dorsomedial positions of the thalamus.

The bifurcation of transcriptomic trajectories according to thalamic progenitor identity is in general agreement with results from in vivo clonal analyses. 36,59 These studies revealed that individual thalamic progenitors give rise to many neurons that populate multiple thalamic nuclei. They also demonstrated that the clonal relationship between thalamic nuclei is determined primarily by the rostro-caudal and dorsoventral positions of thalamic progenitors."

(Zhou et al., 2023)

Brain-wide Correspondence Between Neuronal Epigenomics and Long-Distance Projections

"In any given brain, each neuron contributes uniquely to brain function. Nevertheless, neurons can be grouped into types based on similarities and differences across multiple dimensions, including epigenetic state, gene expression, anatomy, and physiology. Single-cell genomic technologies have been particularly impactful for cell type classification due to their high throughput (millions of cells assayed) and dimensionality (thousands of genes and even more genetic loci) leading to the identification of large numbers of transcriptomic and epigenomic clusters corresponding to possible cell types across the entire mouse brain.

A prominent and distinguishing anatomical feature of many brain neuron types is their long-distance axonal projections. Long-distance projections can be directly related to single neuron gene expression or epigenomes by use of powerful linking technologies, including BARseq 1,2 , Retro-seq 3,4 , and Epi-Retro-Seq 5 . Previous studies have used Retro-Seq and Epi-Retro-Seq to link mouse neocortical [3][4][5] , hypothalamic 6 and thalamic projection cell types 7 to their genetic and epigenetic clusters, revealing complex but predictable relationships. For example, cortical neurons projecting solely to intra-telencephalic (IT) targets fall into different clusters than those that project to extra-telencephalic (ET) targets. On the other hand, cortical layer 2/3 (L2/3) IT neuron types projecting to different cortical areas typically co-cluster despite having quantifiable and predictable genetic and epigenetic differences across the populations 4,5 . In the face of this complexity, how can single-cell genetic and epigenetic assays be used to inform the structure and function of brain cell types and how can neuronal structure predict genetics, epigenetics, and function? Further, can the principles learned from more limited previous studies be extended to the entire brain? Are there different principles linking projection status to epigenetics for different brain areas?

To address these questions we employed Epi-Retro-Seq to assay 33,034 neurons from 225 source target combinations across the entire mouse brain."

"The thalamus is a primary hub in sensory and cortical information processing and also projects to subcortical structures. Similar to hypothalamus, thalamus consists of a large number of nuclei that are organized into multiple functional groups. The main, central regions of the thalamus are composed of exclusively excitatory regions (except for a few local GABAergic interneurons in the dorsal lateral geniculate nucleus (LGd)) that are reciprocally connected with cortical areas 20 . Other more ventral and lateral regions of the thalamus (such as LGv and RT) contain GABAergic inhibitory neurons that are either reciprocally connected with thalamic excitatory neurons or project to subcortical structures such as the basal ganglia and brainstem 20 . In contrast to the hypothalamus, the thalamus had a lower degree of cell type complexity as shown by the smaller number of cell clusters identified through gene

expression analysis 16 . Despite both the thalamus and hypothalamus showing a high level of heterogeneity in their anatomical nuclei and projections, the differences in their cell type complexity prompted us to investigate whether the relationships between cell types, their projections, and their spatial locations in the thalamus differ from those observed in the hypothalamus, as discussed above.

We analyzed thalamic neurons that project to twelve different targets, including nine cortical areas (PFC, MOp, SSp, ACA, AI, AUDp, RSP, PTLp, and VISp), SC, VTA, and P. To gain a comprehensive understanding of these neurons, we combined Epi-Retro-Seq data with unbiased snmC-seq and single-cell RNA-seq data from the thalamus. "

"Through this integration, we identified a total of 58 thalamic neuronal cell clusters (Fig. 4a), of which 33 clusters were enriched for Epi-Retro-Seq neurons (Fig. 4b). It is worth noting that neurons dissected from different anatomical regions within the thalamus were located in distinct sets of clusters 17 (Fig. 4c), as expected from prior descriptions based on analysis of scRNA-seq data 7 , suggesting that these molecularly defined cell clusters also have a spatial organization.

In Figure 4b, we assessed the degree of enrichment of each projection in each cluster. Similar to what was observed in the hypothalamus, each population of thalamic projection neurons exhibited enrichment in distinct subsets of cell clusters, with each cluster showing enrichment for a specific set of projections, sometimes only one. Notably, the clusters enriched for TH⁻ SC, TH⁻ VTA, TH⁻ P, and TH⁻ cortex were mostly mutually exclusive. Regarding cortical projections, TH⁻ PTLp and TH⁻ VISp neurons exhibited enrichment in a largely overlapping set of clusters, but with varying degrees of enrichment. TH⁻ MOp and TH⁻ SSp neurons also shared most of their enriched clusters, which differed from those enriched for TH⁻ PTLp and TH⁻ VISp. These results support the notion of a separation of thalamic cell types between the visual and motor pathways in the thalamus and highlight the heterogeneity of cell types within each pathway. Notably, TH⁻ RSP neurons showed no overlap in enriched clusters with any other cortical projections, and were uniquely enriched in clusters 13, 26 and 47. These clusters were annotated by their gene expression patterns as belonging to the anteroventral (AV) nucleus (clusters 13 and 26), and anterodorsal (AD) nucleus (cluster 47), which is consistent with TH⁻ RSP projections originating from anterior thalamic nuclei 21 . In summary, TH neurons projecting to cortex vs. subcortical targets were enriched in distinct sets of clusters. "

"The enriched cell clusters for cortical projections were further segregated by different thalamic pathways, with multiple enriched cell clusters observed for each pathway or projection. These findings highlight the cell type specificity as well as heterogeneity at the level of TH projections.

Such cell type specificity and heterogeneity of TH projection neurons were also reported in transcriptomic analysis of single TH projection neurons. Phillips et al. conducted a single-cell RNA-seq study on thalamic neurons projecting to the prefrontal, motor, somatosensory, auditory, and visual cortices 7 . We will refer to their dataset as Retro-Seq of these thalamocortical (TC) projections. In their study, clustering analysis of Retro-Seq revealed that neurons of each projection were enriched in a specific set of clusters, indicating specific and shared cell types between different TC projections, as well as cell type heterogeneity within each TC projection.

To compare Retro-Seq and Epi-Retro-Seq data for these TC projections, we mapped the Retro-Seq data onto our 58 integrated TH clusters (Extended Data Fig. 8a, Methods). The t-SNE of Retro-Seq and Epi-Retro-Seq neurons from comparisons of the most closely corresponding projections showed that they occupied similar spaces and were enriched in a common set of clusters (Extended Data Fig. 8a,b). However, in addition to the shared projection-enriched clusters, certain clusters were found to be enriched for each projection only in either Retro-Seq or Epi-Retro-Seq. These differences are most likely due to the use of different injection coordinates for each cortical target (Supplementary Table 4), resulting in overlapping yet different populations of retrogradely labeled TH neurons being analyzed in the two data sets. The degree of alignment between Epi-Retro-Seq and Retro-Seq was quantified by the overlap score and cosine distance (Extended Data Fig. 8c, Methods), which revealed that Retro-Seq neurons of each projection were more similar to Epi-Retro-Seq neurons for the corresponding projections than for any other projections."

(Mickelsen et al., 2020)

Cellular taxonomy and spatial organization of the murine ventral posterior hypothalamus

"Another notable exception is a putative dual identity cluster 8 subcluster, which exhibits high co-expression of both Slc32a1 and Slc17a6, and corresponds well to markers of the SUM (Figure 4-figure supplement 1) and, possibly, to a known dual phenotype, hippocampal-projecting population previously identified in the SUM (Pedersen et al., 2017;Boulland et al., 2009;Soussi et al., 2010;Hashimotodani et al., 2018).

Within each fast neurotransmitter identity, discrete classifications of VPH neurons were arrived at based on a suite of other discriminatory markers with few classifications being made on the basis of single markers alone.Most neuronal clusters are also defined by a combination of neuropeptide transcripts, which are associated with both well-known and novel populations.The co-expression of both fast neurotransmitter and multiple neuropeptide markers in single neurons is suggestive of a capacity for co-transmission (van den Pol, 2012).Other important cluster-specific markers include some combination of transcripts that encode transcription factors, calcium binding proteins, synaptic and extracellular matrix proteins, receptors and other gene categories.That these combinations of markers specify the identity of distinct populations of VPH neurons likely reflects a convergence of their unique molecular specification during development, their neurochemistry and specific synaptic connectivity within functional

circuits.

While several of the diverse neuronal clusters that emerged from our scRNA-seq analysis were readily identifiable based on well-known signatures (e.g.Kiss1+/Tac2+/Pdyn+ KNDy Arc neurons, Agrp+/Npy+ Arc neurons, Hdc+ TMN HA neurons, etc.), most were novel and differentially expressed genes required both cross-referencing with the ABA (Lein et al., 2007) and extensive multiplexed FISH to effectively trace them to anatomical regions within the VPH.We found that a number of novel neuronal cell populations exhibited striking compartmentalization within distinct VPH nuclei (Figure 10b)."

(Schulmann et al., 2024)

A conserved cell-type gradient across the human mediodorsal and paraventricular thalamus

"MD was dissected from coronal slabs based on anatomical landmarks, followed by nuclei isolation and snRNA-seq via the 10X Chromium platform (Fig. 1a). Data underwent extensive quality control and filtering, clustering, and cell-type annotation based on marker genes and data from the Human Brain Cell Atlas 15 . Despite differences in tissue processing between the brain banks, both datasets showed the same major cell classes (Fig. 1b, Supplementary fig. 1a-d, see Methods for details). We identified excitatory neurons, marked by expression of SLC17A6 (VGluT2), inhibitory neurons, which comprised nearly 40% of neurons in our samples and expressed SOX14 -consistent with midbrain origin -and several non-neuronal populations (Fig. 1c; Supplementary data table 1).

To investigate thalamic projection neuron subpopulations in MD, we re-clustered the excitatory neurons, integrating HBCC and MSSM data (Fig. 1d, Supplementary fig. 1e-f). We found that most excitatory neurons in MD were part of a transcriptional gradient captured by the first principal component (PC1, Fig. 1e). Genes aligned with PC1 overlapped with and correlated in their loadings with those of a previously described trans-thalamic axis separating primary, secondary and tertiary thalamic cells in mice 8 (Supplementary fig. 1g-h). Thus, projection neurons in human MD and midline thalamus exhibit a gene expression gradient that closely resembles a cell-type gradient with distinct functional and morphological properties in mice. Human MD has traditionally been divided into a magnocellular (MDmc, medial), parvocellular (MDpc, lateral-ventral) and multiform/densocellular (MDdc, lateral-dorsal) part 16 . We hypothesized that these cytoarchitectonic subdivisions could correspond to transcriptionally defined neuronal populations. "

"To map cell populations identified by snRNA-seq onto anatomical space, we applied spatial transcriptomics to two coronal thalamic sections, then deconvoluted the cell types, using our snRNA-seq data as a reference 17 . This revealed prominent spatial separation of neuronal populations. Some excitatory neuron clusters preferentially mapped to MDmc, MDpc, or adjacent nuclei, while inhibitory interneurons appeared to be particularly enriched in MDpc and MDdc (Fig. 2a; Supplementary fig. 2a).

Similarly, projection of the excitatory neuron PC1 from Fig. 1e onto the spatial transcriptomics data showed that the primary end of the gradient mapped to MD, especially MDmc, while the tertiary end mapped to PVT and PT (Supplementary fig. 2b). Among the surrounding nuclei, intralaminar nuclei were on the tertiary end, while motor/somatosensory and lateral dorsal thalamus were on the primary end, consistent with their profiles in mouse thalamus 8 . RNA-FISH confirmed this spatial distribution of marker genes associated with opposite ends of the gradient and showed that numerous cells co-expressed multiple markers (Fig. 2b, Supplementary fig. 3). These analyses suggest that the gene expression gradient of excitatory neurons corresponds to a spatial gradient of neurons between MD and PT/PVT. Having identified a cell-type gradient in human MD and PVT with major similarities to the thalamic cell-type gradient in mice, we sought to better understand the commonalities and differences in MD cellular composition between humans and mice. Since previous snRNA-seq from retrogradely labeled cells did not include enough neurons in MD 8 , we collected additional snRNA-seq from mouse MD based on anatomical dissections. As expected, this identified the same major cell classes as human MD, except that inhibitory neurons were not observed in mouse MD (Supplementary fig. 4). To directly compare thalamic projection neurons between species, we integrated human and mouse excitatory neurons (Fig. 3a, Supplementary fig."

"We tested for association with cell type specificity scores derived from our data using the top 10% most specific gene sets, as previously described 20 . Neuronal populations in the thalamus were significantly associated with genetic risk for severe mental illness: ExN_H2 (MDpc-enriched) was most strongly associated with genetic risk for schizophrenia and bipolar disorder, while thalamic inhibitory neurons and ExN_H3 (PVTenriched) were most strongly associated with genetic risk for major depressive disorder. GWAS for Alzheimer's disease and height, included for control purposes, implicated microglia and vascular cells, respectively, as expected. a. UMAP plot illustrating cross-species integration of human and mouse excitatory neurons in medial thalamus, colored by dataset: human data from HBCC and MSSM, and mouse data from this study based on anatomical dissections and a previous study using retrograde labeling from prefrontal cortex 8 . b. UMAP plot of integrated mouse and human data, with mouse cells labeled by their nuclei of origin inferred from integration with public datasets and Allen Brain Atlas ISH data (see also Supplementary fig. 4 We produced a cell type atlas of adult human MD and adjacent midline nuclei, spatially mapped the major neuronal populations, and compared mouse and human neuronal identities. Recent studies examined cell populations across the entire human brain 15 , fetal human thalamus 11 , and multiple structures within the mouse thalamus 8,13,14,21-23 and habenula 18,24 , but were not focused on the MD, and cross-species comparisons have been limited to visual thalamus 21 , making our study the first in-depth analysis of cell types in human MD.

We identified a major transcriptional gradient between MD and PVT, which closely aligns with a previously-described gradient from primary to tertiary thalamus and likely reflects distinct connectivity and functional properties of these neurons 8 . This gene

expression gradient mapped onto anatomical space, and respective clusters were enriched in MDmc, MDpc, and adjacent nuclei. Whilst MD has historically been considered 'higher-order', we identify major primary cell populations within both mouse and human MD. Cross-species analyses show that primary cells in MD are substantially expanded in humans relative to mice."

"Human Brain Cell Atlas integration: Data from the medial thalamus and epithalamus regions of interest of the Human Brain Cell Atlas 15 were downloaded and integrated with the HBCC and MSSM data, respectively, using anchors defined via canonical correlation analysis (CCA) in Seurat (Supplementary fig. 1c). Nearest centroid classification of HBCC and MSSM cells by supercluster label was performed to map corresponding cell types (Supplementary fig. 1d). Brain bank integration: Excitatory neurons from HBCC and MSSM were integrated using CCAbased anchors in Seurat. SNN clustering with a resolution of 0.5 on the top 5 PCs was used to define the clusters. To alleviate clustering being driven by individual samples, the four most closely related clusters were merged, resulting in 10 clusters (Supplementary fig. 1e-f).

Mouse anatomical and retrograde labeling data integration: Mouse thalamic neurons obtained in this study were integrated with those based on retrograde labeling from prefrontal cortex in a previous study 8 using CCA-based anchors in Seurat. SNN clustering with a resolution of 0.8 on the top 10 PCs was used, resulting in 19 clusters (Supplementary fig. 4b). PVT data integration: Five populations previously annotated and validated in the mouse PVT 13 were integrated with the mouse thalamic neuron clusters using anchors based on reciprocal principal component analysis (RPCA) in Seurat. Nearest centroid classification was used to map the PVT clusters onto corresponding excitatory neuron clusters (Supplementary fig. 4c). Habenula data integration: Previously annotated and validated neuron populations from the mouse habenula 18 were re-identified and integrated with the mouse thalamic neurons using RPCA-based anchors. Nearest centroid classification was used to map the re-identified habenula populations onto the excitatory neuron clusters (Supplementary fig. 4d). All integrations were performed on the top 2,000 most variable genes across datasets (SelectIntegrationFeatures in Seurat). Nearest neighbor classifiers were implemented via the lolR library (lol.classify.nearestCentroid)"

(Trujillo et al., 2026)

Selective vulnerability and resilience of thalamic neuronal populations in Frontotemporal Dementia

"To define the neuronal populations underlying selective thalamic vulnerability, we subclustered thalamic neurons and identified four major neuronal populations composed of two excitatory and two inhibitory neuronal subtypes (Figure 2A-C). Cell-type annotation was guided by previously reported human thalamic transcriptional signatures (Bakken et al., 2021;Marsan et al., 2023;Mathys et al., 2024;Siletti et al., 2023), and refined using the large scale and cross-cohort structure of our integrated dataset. These neuronal populations were distinguished by robust transcriptional signatures, including pan-inhibitory expression of GAD2 and MEIS2 together with differential expression of SOX14, NR2F2, ERBB4, LAMP5, and BCL11B across inhibitory neuronal populations. While InNeu SOX14+ neurons correspond to midbrain-derived interneurons, the InNeu SOX14population is composed of ganglionic eminence-derived subtypes, encompassing medial ganglionic eminence (MGE; marked by LHX6, SST, and NPY) and caudal ganglionic eminence (CGE; marked by CXCL14, VIP, and LAMP5) lineages (Supplementary Figure 2A). In contrast, excitatory neuronal populations were defined by SLC17A6 expression, consistent with thalamic projection neuron identity, and differential RNF220 expression (Figure 2A,B). We therefore annotated these populations as InNeu SOX14+, InNeu SOX14-, ExNeu RNF220+, and ExNeu RNF220-(Figure 2C). Integration with human spatial transcriptomic datasets generated by Schulmann et al. (Schulmann et al., 2024) (Schulmann et al. under revision) showed that InNeu SOX14+ and ExNeu RNF220+ populations were present across several nuclei in the human medial thalamus (Figure 2D)."

Canonical glutamatergic thalamic projection neurons

Most neurons with soma in mouse thalamus proper are glutamatergic projection neurons, but they do not form a single uniform type. Across nuclei and even within single nuclei, they vary along a strong mediolateral transcriptional axis that covaries with ion channels, receptors, physiology, morphology, and projection patterns.

Type/family: glutamatergic thalamic projection neurons (principal thalamocortical and other long-range thalamic projection cells).

Structure: These neurons are classically described as relay-like projection cells with soma-derived axons and limited local collaterals; bushy and radiate dendritic morphologies have both been observed in thalamic projection neurons, including midline groups. Morphological surveys also support broad core-like versus matrix-like projection patterns, with lower within-type variability in thalamic core neurons and more variability in matrix neurons. (Lara-Vasquez et al., 2016) (Peng et al., 2021) (Clascá et al., 2012)

Function: They relay and transform sensory, motor, limbic, and cognitive signals to cortex and other targets. A notable result from projection-specific transcriptomics is that their largest axis of variation is not sensory modality but a repeated molecular spectrum across pathways; genes on this axis are enriched for receptors, ion channels, and signaling molecules, implying systematic differences in input/output transformation.

(Phillips et al., 2019) (Phillips et al., 2018)

Location: Throughout thalamus proper, including sensory relay nuclei, motor thalamus, anterior nuclei, mediodorsal and midline nuclei. In the ABC atlas they mostly fall in a single thalamic glutamatergic class. (Turner et al., 2025)

Connectivity: Projection-defined sequencing found distinct but repeated profiles across motor, somatosensory, visual, auditory, and prefrontal-projecting neurons. Epi-Retro-Seq likewise found that thalamic neurons projecting to cortex versus subcortical targets occupy distinct cluster sets, and that different cortical projection streams such as VIS/PTLp, MOp/SSp, and RSP are enriched in different combinations of clusters. (Phillips et al., 2019) (Zhou et al., 2023)

Neurotransmitter: glutamate; major marker **Slc17a6/VGLUT2**. (Yao et al., 2023) (Schulmann et al., 2024)

Neuropeptides and markers: marker sets vary by nucleus and position on the trans-thalamic axis. Widely used markers include **Pvalb**, **Calb1/Calb2**, **Necab1**, and **Tnnt1**; different positions along the axis also differ in transcription factors, receptor genes, and ion channels. (Phillips et al., 2018) (Phillips et al., 2017) (Yao et al., 2023)

Known heterogeneity: Heterogeneity is both discrete and continuous. Single-cell data showed multiple clusters within each projection class, but these clusters distribute along the same principal axis seen in bulk data; intermediate cells are common, especially near nuclear borders. Within single nuclei such as VL, **Pvalb**-selective, **Tnnt1**-selective, and intermediate neurons can coexist in a mediolateral gradient. THALMANAC extends this view by showing that some transcriptomic types are nucleus-specific, some subdivide a nucleus, and many span multiple nuclei. (Phillips et al., 2018) (Phillips et al., 2017) (Turner et al., 2025)

Transcriptomics and related datasets:

- Projection-specific bulk and single-cell atlas across 22 nuclei and multiple cortical projection systems: often called **ThalamoSeq/Retro-Seq** in later papers; accession number is not given in the provided quotes. (Phillips et al., 2018) (Phillips et al., 2019)
- Whole-brain **ABC Atlas** scRNA-seq plus **MERFISH** spatial data, used by **THALMANAC**; 80,170 thalamic MERFISH cells from 14 coronal sections, taxonomy version **CCN20230722**. Resource identifiers given in the quote include Streamlit app **RRID:SCR_024354** and Jupyter notebooks **RRID:SCR_018315**. (Turner et al., 2025)
- Whole-brain cell atlas: ~7 million scRNA-seq cells and ~4.3 million MERFISH cells, including thalamus. (Yao et al., 2023)
- Integrated projection/epigenomic analysis of 33,034 Epi-Retro-Seq neurons from 225 source target combinations, with 58 thalamic neuronal clusters total and 33 enriched for Epi-Retro-Seq neurons. (Zhou et al., 2023)

Evidence

(Lara-Vasquez et al., 2016)

Midline thalamic neurons are differentially engaged during hippocampus network oscillations

> *"Calretinin and calbindin expression in midline thalamic neurons. We sampled the action potential firing of individual neurons from thalamic nuclei across the midline (Table 1). After recording, neurons were individually labeled with the juxtacellular filling method 38 and classified by their expression of neurochemical markers (Supplementary Table 1). In order to map the location of recorded neurons on the thalamus, we projected all labelled neurons on the neuroanatomical atlas of the mouse brain 39 . Neurons were recovered from most midline thalamic nuclei (Fig. 1a). The somatodendritic distributions of midline thalamic neurons were relatively uniform, with no obvious distinct morphological types. However, we could identify cells conforming to the distinct branching patterns of dendrites described in thalamic relay cells, such as "bushy" and "radiate" neurons 1,40 . Indeed, some cells had more angular and multipolar cell bodies, with dendritic trees branching dichotomously, characteristic of "radiate" cells (Fig. 1c). Conversely, other cells had elongated somata, with dendrites longer and branching less frequently, particularly in regions proximal to the soma, and followed a "bushy" organization (Fig. 1c). We were not able to fill the entire extension of axonal arborizations, even in cases where intense somatic labeling was achieved. In fact, most labelled cells did not show evidence of axonal labelling. When*

labelled, axons arose in all cases from the soma and rarely branched locally (Supplementary Figure 1). Given the above mentioned dendritic branching patterns, as well as axonal branching patterns, with little or no local collaterals, we considered our sample to be entirely comprised of thalamic projection neurons, which are known to be glutamatergic and typically reach the cortical mantle 1 ."

"By recording and labeling individual neurons in anesthetized mice, we show here evidence for functionally distinct neuronal populations in the midline thalamus. Indeed, our anatomical and physiological characterization leads us to propose that at least two different populations of midline thalamic projection cell deliver differential synaptic output to cortical targets according to ongoing brain states as defined by hippocampal activity. Our data show that rather than anatomical location, a major defining feature in the physiological properties of midline thalamic neurons is the expression of CR, a calcium-binding protein. Hence, distinct thalamic neuronal populations, identified by the somatic expression of CR, are differentially regulated by hippocampal pathways during network rhythms associated to behavioural contexts, likely supporting different stages of memory processing. Midline thalamic nuclei give rise to large glutamatergic axon terminals that densely distribute within frontal and temporal cortical lobes 27,45 . It has been reported that a dense fiber plexus located in the entorhinal cortex contains both CR and vesicular glutamate transporter 2 in the same axons. Complementary retrograde tracing experiments found labeled cell bodies in the nucleus reuniens co-expressing CR 46 . Our results show that CR+ cells are widely expressed across the midline thalamus, and given their anatomical and physiological features suggest that they are likely to be glutamatergic projection cells 1 . Cortical projections from the midline thalamus are not homogeneous across nuclei. Indeed, neurons in dorsal structures preferentially target the medial prefrontal cortex, with little input to the medial temporal lobe 5,24 ; neurons in middle regions, project almost exclusively to the medial prefrontal cortex 24,47 ; whereas neurons in ventral areas, project mostly to the parahippocampal cortex, with less significant projections to the medial prefrontal cortex 24,27 . Thus, the anatomical organization of synaptic output from the midline thalamus is not homogeneous, with every nucleus displaying distinct efferent connectivity patterns. Interestingly, our results suggest that a major feature in defining the activity pattern of thalamic neurons is not their nuclear location, but the expression profile of calcium-binding proteins, particularly CR."

(Peng et al., 2021)

Morphological diversity of single neurons in molecularly defined cell types

"Dendritic and axonal morphology reflects the input and output of neurons and is a defining feature of neuronal types1,2, yet our knowledge of its diversity remains limited. Here, to systematically examine complete single-neuron morphologies on a brain-wide scale, we established a pipeline encompassing sparse labelling, whole-brain imaging, reconstruction, registration and analysis. We fully reconstructed 1,741 neurons from cortex, claustrum, thalamus, striatum and other brain regions in mice. We identified 11 major projection neuron types with distinct morphological features and corresponding transcriptomic identities. Extensive projectional diversity was found within each of these major types, on the basis of which some types were clustered into more refined subtypes. This diversity follows a set of generalizable principles that govern long-range axonal projections at different levels, including molecular correspondence, divergent or convergent projection, axon termination pattern, regional specificity, topography, and individual cell variability. Although clear concordance with transcriptomic profiles is evident at the level of major projection type, fine-grained morphological diversity often does not readily correlate with transcriptomic subtypes derived from unsupervised clustering, highlighting the need for single-cell cross-modality studies. Overall, our study demonstrates the crucial need for quantitative description of complete single-cell anatomy in cell-type classification, as single-cell morphological diversity reveals a plethora of ways in which different cell types and their individual members may contribute to the configuration and function of their respective circuits. Sparse labelling and whole-brain imaging are used to reconstruct and classify brain-wide complete morphologies of 1,741 individual neurons in the mouse brain, revealing a dependence on both brain region and transcriptomic profile."

"Dendritic and axonal morphology reflects the input and output of neurons and is a defining feature of neuronal types 1,2 , yet our knowledge of its diversity remains limited. Here, to systematically examine complete single-neuron morphologies on a brain-wide scale, we established a pipeline encompassing sparse labelling, wholebrain imaging, reconstruction, registration and analysis. We fully reconstructed 1,741 neurons from cortex, claustrum, thalamus, striatum and other brain regions in mice. We identified 11 major projection neuron types with distinct morphological features and corresponding transcriptomic identities. Extensive projectional diversity was found within each of these major types, on the basis of which some types were clustered into more refined subtypes. This diversity follows a set of generalizable principles that govern long-range axonal projections at different levels, including molecular correspondence, divergent or convergent projection, axon termination pattern, regional specificity, topography, and individual cell variability. Although clear concordance with transcriptomic profiles is evident at the level of major projection type, fine-grained morphological diversity often does not readily correlate with transcriptomic subtypes derived from unsupervised clustering, highlighting the need for single-cell cross-modality studies. Overall, our study demonstrates the crucial need for quantitative description of complete single-cell anatomy in cell-type classification, as single-cell morphological diversity reveals a plethora of ways in which different cell types and their individual members may contribute to the configuration and function of their respective circuits.

Neurons exhibit extraordinary diversity across molecular, morphological, physiological and connectional features, thus accurate classification and mapping of cell types needs to consider and integrate these distinct yet related cellular properties 1,2 . Single-cell RNA-sequencing (scRNA-seq) has enabled systematic classification at the transcriptomic level [3][4][5] , capturing major cell types with known anatomical and functional properties and revealing many potentially new cell types. Classification of cortical neurons using a combination of transcriptomic, electrophysiological and local morphological properties has also been achieved 6,7 . Brain-wide inter-areal connectivity has been mapped extensively using anterograde and retrograde tracing of projection neuron

populations [8][9][10][11] ."

"At the lowest, single-cell level, the degree of similarity or variability between individual neurons within a given type also varies across types. Within-type individual cell variability is high in cortical and claustral neurons, moderate in thalamic matrix neurons, and low in thalamic core neurons and striatal MSNs.

A major question is how morphological and projectional properties compare and correlate with the neurons' molecular identities. We attempted to address this question with two approaches: using validated driver lines to define the subclass-level molecular identities of reconstructed neurons and using Retro-seq to obtain transcriptomic profiles of neurons projecting to specific targets. Both approaches show that subclasses of neurons have highly distinct morphological and projectional properties; however, within these major types, especially for cortical and claustral neurons, many aspects of morphological diversity cannot be accounted for by currently identified transcriptomic subtypes or clusters in the adult stage. Previous studies showed that L2/3 SSp pyramidal neurons projecting to MOp or SSs have distinct intrinsic and network physiological properties 42,43 . Even though they may not belong to distinct transcriptomic subtypes, it will be interesting to examine gene-expression differences that might correspond to the differential connectional and physiological properties for these neurons, as found for primary visual cortical neurons projecting differentially to medial or lateral higher visual areas 44 . Several mechanisms may explain the origin of the morphological diversity, such as molecular instructions that act transiently during development 45 , activity-dependent cell interactions, or stochastic processes. It will be informative to develop methods that enable complete reconstruction of morphology and in-depth gene-expression profiling to be conducted on the same cell, and apply them to single cells in both adult stage and during brain development, so that the developmental correlations of molecular and morphological and connectional features can be identified."

(Clascá et al., 2012)

Unveiling the diversity of thalamocortical neuron subtypes

"Our current understanding of thalamocortical (TC) circuits is largely based on studies investigating so-called 'specific' thalamic nuclei, which receive and transmit sensory-triggered input to specific cortical target areas. TC neurons in these nuclei have a striking point-to-point topography and a stereotyped laminar pattern of termination in the cortex, which has made them ideal models to study the organization, plasticity, and development of TC circuits. However, despite their experimental importance, neurons within these nuclei only represent a fraction of all thalamic neurons and do not reflect the diversity of the TC neuron population. Here we review the distinct subtypes of projection neurons that populate the thalamus, both within and across anatomically-defined nuclei, with regard to differences in their morphology, input/output connectivity and target specificity, as well as more recent findings on their neuron type-specific gene expression and development. We argue that a detailed understanding of the biology of TC neurons is critical to understand the role of the thalamus in normal and pathological perception, voluntary movement, cognition and attention."

(Phillips et al., 2019)

A repeated molecular architecture across thalamic pathways

"The thalamus is the central communication hub of the forebrain and provides the cerebral cortex with inputs from sensory organs, subcortical systems and the cortex itself. Multiple thalamic regions send convergent information to each cortical region, but the organizational logic of thalamic projections has remained elusive. Through comprehensive transcriptional analyses of retrogradely labeled thalamic neurons in adult mice, we identify three major profiles of thalamic pathways. These profiles exist along a continuum that is repeated across all major projection systems, such as those for vision, motor control and cognition. The largest component of gene expression variation in the mouse thalamus is topographically organized, with features conserved in humans. Transcriptional differences between these thalamic neuronal identities are tied to cellular features that are critical for function, such as axonal morphology and membrane properties. Molecular profiling therefore reveals covariation in the properties of thalamic pathways serving all major input modalities and output targets, thus establishing a molecular framework for understanding the thalamus. This work describes comprehensive transcriptomic sequencing from murine thalamic pathways. By integrating this molecular information with anatomical and functional features, this study reveals a repeated architecture across thalamocortical systems."

(Phillips et al., 2018)

A single spectrum of neuronal identities across thalamus

"A. Schematic of experimental pipeline to obtain transcriptomic atlas of the thalamus. In this example, motor thalamic neurons were retrogradely labelled from their primary projection target (motor cortex), manually dissected and sorted. Viruses expressing green and blue fluorescent proteins (GFP, BFP), respectively, were injected to the deep cerebellar nuclei (DCN) and substantia nigra pars reticulata (SNr) to label motor nuclear subdivisions (ventral lateral (VL) and ventral anterior (VA), respectively) previously identified 8,9 . B. Example labelling from scheme shown in A. Coronal sections. Scale bars = 200 μm. C. Hierarchical clustering of thalamic nuclei using Spearman's correlation of top 500 most differentially expressed genes (Extended Data Fig. 1B) across all 22 nuclei. Major clusters defined as the top 5 branches of cluster dendrogram. D. Topographic localization of gene expression profiles in thalamus. Coronal thalamic section schematics with nuclei colored as in Fig. 1C. Unsampled nuclei are left uncolored. (see Fig. 1B). To understand the pattern of gene expression differences between the thalamic projection profiles, we performed principal component analysis (PCA) on our data. Principal component 1 (PC1, 38% explained variance) separated nuclei into the same major profiles identified via hierarchical clustering (Fig. 2A, Extended Data Fig. 3A). Again, position on this first dimension strongly

correlated with mediolateral position, demonstrating topogenetic architecture in thalamus (Fig. 2B, Extended Data Fig. 3B). Based on their relative order on this first component, we named the three major profiles primary, secondary, and tertiary. The progressive difference from primary to tertiary nuclei was also evident in the number of genes differentially expressed between the groups, with the primary and tertiary nuclei being most distinct, and the other two comparisons being less so (Fig. 2C). This primary axis was prominently enriched in genes encoding neurotransmitter receptors, ion channels, and signaling molecules (Fig. 2D and E)."

"A. Overview of single-cell clusters within each projection system visualized via tSNE. Cells colored by cluster identity. B. Violin plots for cluster marker genes (inclusion criteria were Likelihood Ratio Test Pvalue < 10 -5 and log2 fold change > 0.5 for each cluster). C. Projection of single-cell data onto pooled-cell PC1 from Fig. 2A. Each dot is a single cell colored as in A. D. Topographic distribution of marker genes within 6 major thalamic modalities. Multi-FISH with probes for Calb2 (pink), Tnnt1 (red) and Necab1 (green). See Extended Data Fig. 7 and 8 for expanded views and quantification. Scale bars = 200 μm. E. Multi-FISH within VL thalamus comparing genes marking clusters in single-cell data.

Left panel shows field of view (coronal section, scale bar = 100 μm). Right three panels (scale bars = 50μm) show expansion of the three boxed areas, moving left to right.

Middle box shows intermediate cells expressing both single cell cluster markers. (red = Tnnt1, green = Pvalb, blue = DAPI). Thus far we analyzed thalamic nuclei by pooling projection neurons from specific anatomical positions. The resulting transcriptional profiles could represent homogenous populations, discrete subtypes, or cells with graded differences. To probe these possibilities, we profiled the transcriptomes of individual neurons from motor, somatosensory, visual, auditory, and prefrontal projection classes (Extended Data Table 3). Analysis of this single-cell RNAseq dataset resulted in multiple clusters for each projection class, and cluster markers included many genes that also distinguished nuclei (Fig. 4B). Single-cell clusters also separated along the first principal component derived from our pooled-cell RNAseq dataset (Fig. 4C, Extended Data Fig. 6). "

"Markers for the single-cell clusters were spatially separated at the single-cell level, but with intermediate cells especially prominent near the anatomical boundaries of thalamic nuclei (Fig. 4D, Extended Data Fig. 7). This is consistent with spatially organized heterogeneity or a gradient-like organization rather than intermingled, discrete populations 15,16 . Given the strong relationship between PC1 and anatomical nuclei position (Fig. 2C and Extended Data Fig. 3), the presence of single-cell clusters occupying similar PC1 positions (e.g. clusters 1 and 2 of the motor projections neurons, or clusters 1 and 2 of the somatosensory nuclei, Extended Data Fig. 6) suggested that distinct neuron types could also coexist within anatomical boundaries of nuclei. We examined this possibility by performing multi-color fluorescent in situ hybridization (multi-FISH) for genes which distinguished amongst the major profiles (e.g. Pvalb and Tnnt1). Taking motor thalamus as an example, Pvalb and Tnnt1 expressing cells were found within the anatomical boundaries of a single thalamic nucleus (VL; Fig. 4E). Some individual VL cells expressed both Pvalb and Tnnt1; Pvalb-selective, intermediate, and Tnnt1-selective cells were distributed along the mediolateral axis (Fig. 4E). Therefore, spatially organized transcriptomic differences can exist even within individual thalamic nuclei. A common organizing principle of thalamus divides nuclei into discrete core or matrix subtypes that span modalities 1,7 . To date the strongest evidence for cross-modal organization is that genetic differences can be larger within a sensory modality than between sensory modalities 17 (Extended Data Fig. 9). However, previous studies have focused either on select sensory systems and/or have had limited scope with respect to the molecules investigated 18 . Here, using the full transcriptomes of nearly all thalamic pathways, we confirm cross-modal organization but replace the core/matrix dichotomy with a spectrum of profiles that span a single axis of genetic variance."

(Turner et al., 2025)

Exploring the correspondence between gene expression and thalamic nuclei using the THALMANAC resource

"The thalamus connects the sensory organs and major subcortical brain regions with the neocortex. The thalamus has long been divided into multiple discrete nuclei, based on cytoarchitecture, histochemical stains, and mesoscale connectivity. However, thalamic nuclei do not completely describe thalamic organization. For example, some boundaries between thalamic nuclei are disputed, whereas other nuclei are known to contain subdomains with distinct connectivity and function. Moreover, the correspondence between cellular gene expression and other properties of thalamic projection neurons remains to be established. Spatial analysis of single cell gene expression provides a basis for reevaluating thalamic organization. We present the THALMANAC, (THALamusMERFISH ANalysis and ACcess) a Findable, Accessible, Interoperable, Reusable and Reproducible (FAIRR) resource for exploring and analyzing single-cell transcriptomic variation in the thalamus. The THALMANAC provides streamlined access to thalamic gene expression data registered to the common coordinate framework and tools for quantitative analysis and visualization of these data, all encapsulated in a reproducible, cloud computing platform. Using this resource, we find that gene expression generally supports the parcellation of thalamus into distinct nuclei. Some nuclei, such as the anteromedial nucleus, are additionally composed of discrete subdomains, while other nuclei share patterns of gene expression or are arrayed on a spatial gradient of gene expression. The THALMANAC establishes spatial transcriptomic data as a foundation for delineating thalamic organization. Graphical Abstract"

"This fine-scale structure could correspond to discrete, molecularly defined subdivisions of VM that have not yet been annotated, or to topographic organization of thalamocortical and corticothalamic connectivity.

A major question is how gene expression relates to existing parcellations and thalamic organization in general. Single cell RNA sequencing (scRNA-seq) provides detailed measurements of transcription in large populations of individual neurons (Tasic et al.

2016), which can be used to define transcriptomic cell types ('t-types'). The combination of scRNA-seq with spatial transcriptomics has been used to localize t-types across the whole mouse brain (Langlieb et al. 2023;Yao et al. 2023;Zhang et al. 2023). With t-types mapped into the CCFv3 coordinate space (Wang et al. 2020), we can evaluate how their spatial distributions correspond to anatomical parcellations (Yao et al. 2023).

Here, building on the Allen Brain Cell (ABC) Atlas scRNA-seq and spatial transcriptomics data, we developed a resource, the THALamus MERFISH ANalysis and ACcess (THALMANAC), which provides access to spatial gene expression data in the mouse thalamus; targeted analysis and data visualization of gene expression patterns; a platform for reproducible computations; and a no-code visualization app for data exploration. We demonstrate the utility of the THALMANAC by addressing how thalamic structural organization relates to gene expression. Aligning spatial transcriptomic data and nuclei boundaries reveals a wide range of relationships between t-types and thalamic nuclei: some t-types correspond to a single thalamic nucleus; others subdivide nuclei into multiple parts; many t-types span multiple thalamic nuclei.

Analysis of expression by itself supports these same conclusions: some nuclei boundaries are associated with sharp changes in gene expression; other nuclei contain gene expression gradients; and, in many cases, gene expression patterns span multiple nuclei.

Understanding how information is routed through the thalamus requires integrating insights gained from gene expression with other modalities, including connectivity and neurophysiology."

"The THALMANAC resource provides access to thalamic spatial gene expression data, visualization of the relationships between gene expression, t-types, and CCFv3 anatomical parcellations, and quantitative analysis of targeted quantitative analyses, all encapsulated in a cloud platform that enables reproducible computations and customized exploration of the data.

The THALMANAC is built on a thalamus-specific subset of spatial transcriptomics data from multiplexed error-robust fluorescence in situ hybridization (MERFISH) from the ABC Atlas. The ABC Atlas dataset measures 500 transcripts, chosen to distinguish t-types across the whole mouse brain, in spatially resolved, single cells. MERFISH cells were registered to the CCFv3 coordinate space and mapped to the scRNA-seq data by assigning each MERFISH cell to the label of the most similar t-type cluster. We identified neurons that belong to the thalamus (including the zona incerta and habenula) according to their t-type label (Figure S1A) and CCFv3 parcellation label (Figure S1B). The thalamic dataset analyzed here includes 80,170 MERFISH cells across 14 coronal sections collected from a single male mouse (Methods).

The THALMANAC includes three avenues for engaging with the t-type and gene expression data and accompanying analyses (Figure 1). First, a Streamlit web app (RRID:SCR_024354) provides nocode exploration, including interactive visualizations and on-the-fly di"erential gene expression analysis for thalamic nuclei (Interactive Streamlit App: thalmanac.allenneuraldynamics.org). Second, Jupyter notebooks (RRID:SCR_018315) provide step-by-step demonstrations of the Python code needed to reproduce all results presented here (github.com/AllenNeuralDynamics/thalmanac). Both tools allow, users to customize the thalamic cells that are selected from the full ABC Atlas."

"In the ABC Atlas, transcriptomic variation is used to assign cells to a hierarchy of discrete t-types (from broad to narrow categories: class, subclass, supertype, cluster). Thalamic neurons belong to 5 classes, 58 subclasses, 163 supertypes, and 563 clusters (Table S1; Figure S2) (taxonomy version CCN20230722) (Yao et al. 2023), but these numbers will change as the taxonomy evolves. Whereas the cluster level is based on unsupervised clustering, higher levels of the taxonomy are additionally informed by developmental origin via shared transcription factor expression, anatomical location, and classical (pre-scRNAseq) neuron types (DeFelipe et al. 2013) defined by key molecular markers (e.g. neuropeptides, calcium-binding proteins).

Transcriptomic classes typically correspond to major brain structures. Glutamatergic neurons in thalamus proper form a single class (class 18 TH Glut), whereas habenula neurons belong to a distinct class (class 17 MH-LH Glut) (Figure S2B). The other three classes are not unique to the thalamus: class 12 HY GABA is found in the reticular thalamus, zona incerta and ventral lateral geniculate nucleus (CCFv3 annotations; RT, ZI, and LGv, respectively, see Table S1) but also in the hypothalamus; class 19 MB Glut is found in thalamic nucleus SPA but also in the midbrain; class 20 MB GABA is found in thalamic nuclei LGv and SPF and the midbrain. Transcriptomic subclasses span multiple thalamic nuclei(Figure S2C), whereas transcriptomic supertypes often define individual nuclei (e.g. AM), and several nuclei contain multiple supertypes (e.g. ZI) (Figure S2D). Most nuclei contain multiple transcriptomic clusters (Figure 2A, Figure S2E).

We explored correspondence between clusters and thalamic nuclei."

"A comprehensive understanding of how the thalamus routes information to the cortex will require multiple types of data and a community-wide e"ort. Our work contributes to this program by curating and analyzing the correspondence between state-of-the-art spatially resolved, single-cell transcriptomic data, and widely used parcellations of thalamus into discrete nuclei. In addition, we make our data and analyses available in an accessible, documented, and reproducible bioinformatics resource in the cloud. Using the AM nucleus and the somatosensory nuclei as examples, we illustrate how this interactive resource can be used to identify novel patterns of gene expression and how these patterns correspond to prior parcellations of the thalamus.

The THALMANAC resource streamlines researchers' access to thalamic spatial transcriptomic data. We anticipate that scientists

will extend our analysis based on the ABC atlas data, or apply similar analyses to additional spatial transcriptomics datasets registered to the CCF coordinate system (Langlieb et al. 2023;Zhang et al. 2023). The tool should also be used as a reference to validate recent studies focused on specific nuclei, such as the spatially segregated projections from PF to distinct dorsal striatum regions (Mandelbaum et al. 2019) and spatially segregated inputs to PO (Casas-Torremocha et al. 2022), the molecular organization of the PVT (Gao et al. 2023), and the core/shell organization of neurons in RT (Li et al. 2020). Our interactive Streamlit app can be used to identify marker genes for post-hoc localization of neurophysiological and neuroanatomical experiments, or to select gene panels for multiplexed mFISH experiments. The THALMANAC is already being used in the e"ort to create and characterize enhancerbased genetic tools (Ben-Simon et al. 2025) to target thalamic cell types and regions.

The THALMANAC shows that at the scale of thalamic nuclei, the molecular organization of the thalamus resembles existing parcellations. "

"However, both discrete cell type composition and gene expression gradients vary at finer spatial scales within and across nuclei. Given the evidence for circuit and functional variation at the same length scales, additional data need to be integrated with molecular characterization to understand thalamic organization. We see two promising paths forward for data integration. First, existing data registered to the CCFv3 3D coordinate space, such as the databases of single-neuron morphological reconstruction (Winnubst et al. 2019) or neurophysiology data (Chen et al. 2024), could be linked to cell types and gene expression described in the THALMANAC. Second, targeted multimodal datasets, such as single-neuron morphological reconstructions combined with posthoc mFISH for transcriptomic cell typing would link cell types and gene expression to projection targets at the single cell level.

Beyond data integration, we also recognize the need for thalamus-focused, community annotation, similar to previous e"orts to annotate the fly connectome (Schlegel et al. 2024). We envision a refined parcellation of the thalamus where scientists with specific domain knowledge would combine molecular information from the THALMANAC with neuroanatomical and functional data to reflect progress in understanding of thalamic organization. Diagram illustrating the three workflows that can be launched from the Code Ocean Collection accompanying this paper. The Code Ocean platform provides built-in support for code versioning (via Git), environment replicability (via Docker), data and analysis provenance (via tracked data assets and linked pipeline runs), cloud compute resources (via AWS EC2 instances). Users can launch an interactive, no-code Streamlit application to explore spatial distributions of transcriptomic cell types and gene expression from their browser, as well as visualize locations of annotated cell types for each thalamic nucleus and highlight a specific nucleus or group of related thalamic nuclei. Tutorial Jupyter notebooks and Python code replicate all analyses and visualizations in a cloud workstation. These visualizations can be customized and extended. Users can download an AnnData-formatted .h5ad"

(Zhou et al., 2023)

Brain-wide Correspondence Between Neuronal Epigenomics and Long-Distance Projections

"In any given brain, each neuron contributes uniquely to brain function. Nevertheless, neurons can be grouped into types based on similarities and differences across multiple dimensions, including epigenetic state, gene expression, anatomy, and physiology. Single-cell genomic technologies have been particularly impactful for cell type classification due to their high throughput (millions of cells assayed) and dimensionality (thousands of genes and even more genetic loci) leading to the identification of large numbers of transcriptomic and epigenomic clusters corresponding to possible cell types across the entire mouse brain.

A prominent and distinguishing anatomical feature of many brain neuron types is their long-distance axonal projections. Long-distance projections can be directly related to single neuron gene expression or epigenomes by use of powerful linking technologies, including BARseq 1,2 , Retro-seq 3,4 , and Epi-Retro-Seq 5 . Previous studies have used Retro-Seq and Epi-Retro-Seq to link mouse neocortical [3][4][5] , hypothalamic 6 and thalamic projection cell types 7 to their genetic and epigenetic clusters, revealing complex but predictable relationships. For example, cortical neurons projecting solely to intra-telencephalic (IT) targets fall into different clusters than those that project to extra-telencephalic (ET) targets. On the other hand, cortical layer 2/3 (L2/3) IT neuron types projecting to different cortical areas typically co-cluster despite having quantifiable and predictable genetic and epigenetic differences across the populations 4,5 . In the face of this complexity, how can single-cell genetic and epigenetic assays be used to inform the structure and function of brain cell types and how can neuronal structure predict genetics, epigenetics, and function? Further, can the principles learned from more limited previous studies be extended to the entire brain? Are there different principles linking projection status to epigenetics for different brain areas?

To address these questions we employed Epi-Retro-Seq to assay 33,034 neurons from 225 source target combinations across the entire mouse brain."

"The thalamus is a primary hub in sensory and cortical information processing and also projects to subcortical structures. Similar to hypothalamus, thalamus consists of a large number of nuclei that are organized into multiple functional groups. The main, central regions of the thalamus are composed of exclusively excitatory regions (except for a few local GABAergic interneurons in the dorsal lateral geniculate nucleus (LGd)) that are reciprocally connected with cortical areas 20 . Other more ventral and lateral regions of the thalamus (such as LGv and RT) contain GABAergic inhibitory neurons that are either reciprocally connected with thalamic excitatory neurons or project to subcortical structures such as the basal ganglia and brainstem 20 . In contrast to the hypothalamus, the thalamus had a lower degree of cell type complexity as shown by the smaller number of cell clusters identified through gene expression analysis 16 . Despite both the thalamus and hypothalamus showing a high level of heterogeneity in their anatomical

nuclei and projections, the differences in their cell type complexity prompted us to investigate whether the relationships between cell types, their projections, and their spatial locations in the thalamus differ from those observed in the hypothalamus, as discussed above.

We analyzed thalamic neurons that project to twelve different targets, including nine cortical areas (PFC, MOp, SSp, ACA, AI, AUDp, RSP, PTLp, and VISp), SC, VTA, and P. To gain a comprehensive understanding of these neurons, we combined Epi-Retro-Seq data with unbiased snmC-seq and single-cell RNA-seq data from the thalamus. "

"Through this integration, we identified a total of 58 thalamic neuronal cell clusters (Fig. 4a), of which 33 clusters were enriched for Epi-Retro-Seq neurons (Fig. 4b). It is worth noting that neurons dissected from different anatomical regions within the thalamus were located in distinct sets of clusters 17 (Fig. 4c), as expected from prior descriptions based on analysis of scRNA-seq data 7 , suggesting that these molecularly defined cell clusters also have a spatial organization.

In Figure 4b, we assessed the degree of enrichment of each projection in each cluster. Similar to what was observed in the hypothalamus, each population of thalamic projection neurons exhibited enrichment in distinct subsets of cell clusters, with each cluster showing enrichment for a specific set of projections, sometimes only one. Notably, the clusters enriched for TH□ SC,TH□ VTA, TH□ P,and TH□ cortexwere mostly mutually exclusive. Regarding cortical projections, TH□ PTLpand TH□ VISpneurons exhibited enrichment in a largely overlapping set of clusters, but with varying degrees of enrichment. TH□ MOpand TH□ SSpneurons also shared most of their enriched clusters, which differed from those enriched for TH□ PTLpand TH□ VISp.These results support the notion of a separation of thalamic cell types between the visual and motor pathways in the thalamus and highlight the heterogeneity of cell types within each pathway. Notably, TH□ RSPneurons showed no overlap in enriched clusters with any other cortical projections, and were uniquely enriched in clusters 13, 26 and 47. These clusters were annotated by their gene expression patterns as belonging to the anteroventral (AV) nucleus (clusters 13 and 26), and anterodorsal (AD) nucleus (cluster 47), which is consistent with TH□ RSPprojections originating from anterior thalamic nuclei 21 . In summary, TH neurons projecting to cortex vs. subcortical targets were enriched in distinct sets of clusters. "

"The enriched cell clusters for cortical projections were further segregated by different thalamic pathways, with multiple enriched cell clusters observed for each pathway or projection. These findings highlight the cell type specificity as well as heterogeneity at the level of TH projections.

Such cell type specificity and heterogeneity of TH projection neurons were also reported in transcriptomic analysis of single TH projection neurons. Phillips et al. conducted a single-cell RNA-seq study on thalamic neurons projecting to the prefrontal, motor, somatosensory, auditory, and visual cortices 7 . We will refer to their dataset as Retro-Seq of these thalamocortical (TC) projections. In their study, clustering analysis of Retro-Seq revealed that neurons of each projection were enriched in a specific set of clusters, indicating specific and shared cell types between different TC projections, as well as cell type heterogeneity within each TC projection.

To compare Retro-Seq and Epi-Retro-Seq data for these TC projections, we mapped the Retro-Seq data onto our 58 integrated TH clusters (Extended Data Fig. 8a, Methods). The t-SNE of Retro-Seq and Epi-Retro-Seq neurons from comparisons of the most closely corresponding projections showed that they occupied similar spaces and were enriched in a common set of clusters (Extended Data Fig. 8a,b). However, in addition to the shared projection-enriched clusters, certain clusters were found to be enriched for each projection only in either Retro-Seq or Epi-Retro-Seq. These differences are most likely due to the use of different injection coordinates for each cortical target (Supplementary Table 4), resulting in overlapping yet different populations of retrogradely labeled TH neurons being analyzed in the two data sets. The degree of alignment between Epi-Retro-Seq and Retro-Seq was quantified by the overlap score and cosine distance (Extended Data Fig. 8c, Methods), which revealed that Retro-Seq neurons of each projection were more similar to Epi-Retro-Seq neurons for the corresponding projections than for any other projections."

(Yao et al., 2023)

A high-resolution transcriptomic and spatial atlas of cell types in the whole mouse brain

"The mammalian brain is composed of millions to billions of cells that are organized into numerous cell types with specific spatial distribution patterns and structural and functional properties. An essential step towards understanding brain function is to obtain a parts list, i.e., a catalog of cell types, of the brain. Here, we report a comprehensive and high-resolution transcriptomic and spatial cell type atlas for the whole adult mouse brain. The cell type atlas was created based on the combination of two single-cell-level, whole-brain-scale datasets: a single- cell RNA-sequencing (scRNA-seq) dataset of □ 7million cells profiled, and a spatially resolved transcriptomic dataset of □ 4.3million cells using MERFISH. The atlas is hierarchically organized into five nested levels of classification: 7 divisions, 32 classes, 306 subclasses, 1,045 supertypes and 5,200 clusters. We systematically analyzed the neuronal, non-neuronal, and immature neuronal cell types across the brain and identified a high degree of correspondence between transcriptomic identity and spatial specificity for each cell type. The results reveal unique features of cell type organization in different brain regions, in particular, a dichotomy between the dorsal and ventral parts of the brain: the dorsal part contains relatively fewer yet highly divergent neuronal types, whereas the ventral part contains more numerous neuronal types that are more closely related to each other. We also systematically characterized cell-type specific expression of neurotransmitters, neuropeptides, and transcription factors. The study uncovered extraordinary diversity and heterogeneity in neurotransmitter and neuropeptide expression and co-expression patterns in different cell types across the brain, suggesting they mediate a myriad of modes of

intercellular communications. Finally, we found that transcription factors are major determinants of cell type classification in the adult mouse brain and identified a combinatorial transcription factor code that defines cell types across all parts of the brain. The whole-mouse-brain transcriptomic and spatial cell type atlas establishes a benchmark reference atlas and a foundational resource for deep and integrative investigations of cell type and circuit function, development, and evolution of the mammalian brain."

"In this study, we created a comprehensive, high-resolution transcriptomic cell type atlas for the whole adult mouse brain based on the combination of two whole-brain-scale datasets: a scRNAseq dataset of ~7 million cells and a MERFISH dataset of a similar scale (~4.3 million cells from the AIBS MERFISH brain). We used ~4.1 million high-quality single-cell transcriptomes after stringent QC to create a transcriptomic cell type taxonomy. We used the MERFISH data, which were generated using marker genes derived from the whole-brain transcriptomic taxonomy, to annotate the spatial location of each subclass and each cluster in the taxonomy. We then built a hierarchically organized transcriptomic and spatial cell type atlas with five nested levels: 7 divisions, 32 classes, 306 subclasses, 1,045 supertypes and 5,200 clusters (Figure 1). The neuronal cell type composition in each major brain structure were systematically analyzed (Figure 2) and distinct features in different brain structures identified (Figure 6). We discovered many sets of neuronal types with varying degrees of similarity with each other, including transitional neuronal types across regions as well as highly distinct neuronal types. We also systematically analyzed all divisions of non-neuronal cell types as well as immature neuronal types present in the adult brain, and identified their unique spatial distribution and spatial interaction patterns (Figure 4). Finally, we systematically characterized cell-type specific neurotransmitters, neuropeptides, and transcription factors, and discovered unique characteristics for each as discussed below. This large-scale study allowed us to delineate several principles regarding cell type organization across the whole mouse brain. It provides a benchmark reference cell type atlas as a resource for the community that will enable many more discoveries in the future.

One of the most striking findings from our study is the high degree of correspondence between transcriptomic identity and spatial specificity. Every subclass (and all supertypes and many clusters within each) has a unique and specific spatial localization pattern within the brain. Furthermore, the relative relatedness between transcriptomic types (as revealed in 2D and 3D UMAPs) is strongly correlated with the spatial relationship between them. Transcriptomically related cell types"

(Schulmann et al., 2024)

A conserved cell-type gradient across the human mediodorsal and paraventricular thalamus

"MD was dissected from coronal slabs based on anatomical landmarks, followed by nuclei isolation and snRNA-seq via the 10X Chromium platform (Fig. 1a). Data underwent extensive quality control and filtering, clustering, and cell-type annotation based on marker genes and data from the Human Brain Cell Atlas 15 . Despite differences in tissue processing between the brain banks, both datasets showed the same major cell classes (Fig. 1b, Supplementary fig. 1a-d, see Methods for details). We identified excitatory neurons, marked by expression of SLC17A6 (VGluT2), inhibitory neurons, which comprised nearly 40% of neurons in our samples and expressed SOX14 -consistent with midbrain origin -and several non-neuronal populations (Fig. 1c; Supplementary data table 1).

To investigate thalamic projection neuron subpopulations in MD, we re-clustered the excitatory neurons, integrating HBCC and MSSM data (Fig. 1d, Supplementary fig. 1e-f). We found that most excitatory neurons in MD were part of a transcriptional gradient captured by the first principal component (PC1, Fig. 1e). Genes aligned with PC1 overlapped with and correlated in their loadings with those of a previously described trans-thalamic axis separating primary, secondary and tertiary thalamic cells in mice 8 (Supplementary fig. 1g-h). Thus, projection neurons in human MD and midline thalamus exhibit a gene expression gradient that closely resembles a cell-type gradient with distinct functional and morphological properties in mice. Human MD has traditionally been divided into a magnocellular (MDmc, medial), parvocellular (MDpc, lateral-ventral) and multiform/densocellular (MDdc, lateral-dorsal) part 16 . We hypothesized that these cytoarchitectonic subdivisions could correspond to transcriptionally defined neuronal populations. "

"To map cell populations identified by snRNA-seq onto anatomical space, we applied spatial transcriptomics to two coronal thalamic sections, then deconvoluted the cell types, using our snRNA-seq data as a reference 17 . This revealed prominent spatial separation of neuronal populations. Some excitatory neuron clusters preferentially mapped to MDmc, MDpc, or adjacent nuclei, while inhibitory interneurons appeared to be particularly enriched in MDpc and MDdc (Fig. 2a; Supplementary fig. 2a).

Similarly, projection of the excitatory neuron PC1 from Fig. 1e onto the spatial transcriptomics data showed that the primary end of the gradient mapped to MD, especially MDmc, while the tertiary end mapped to PVT and PT (Supplementary fig. 2b). Among the surrounding nuclei, intralaminar nuclei were on the tertiary end, while motor/somatosensory and lateral dorsal thalamus were on the primary end, consistent with their profiles in mouse thalamus 8 . RNA-FISH confirmed this spatial distribution of marker genes associated with opposite ends of the gradient and showed that numerous cells co-expressed multiple markers (Fig. 2b, Supplementary fig. 3). These analyses suggest that the gene expression gradient of excitatory neurons corresponds to a spatial gradient of neurons between MD and PT/PVT. Having identified a cell-type gradient in human MD and PVT with major similarities to the thalamic cell-type gradient in mice, we sought to better understand the commonalities and differences in MD cellular composition between humans and mice. Since previous snRNA-seq from retrogradely labeled cells did not include enough neurons in MD 8 , we collected additional snRNA-seq from mouse MD based on anatomical dissections. As expected, this identified the same major cell classes as human MD, except that inhibitory neurons were not observed in mouse MD (Supplementary fig. 4). To directly compare thalamic projection neurons between species, we integrated human and mouse excitatory neurons (Fig. 3a, Supplementary fig."

"We tested for association with cell type specificity scores derived from our data using the top 10% most specific gene sets, as previously described 20 . Neuronal populations in the thalamus were significantly associated with genetic risk for severe mental illness: ExN_H2 (MDpc-enriched) was most strongly associated with genetic risk for schizophrenia and bipolar disorder, while thalamic inhibitory neurons and ExN_H3 (PVTenriched) were most strongly associated with genetic risk for major depressive disorder. GWAS for Alzheimer's disease and height, included for control purposes, implicated microglia and vascular cells, respectively, as expected. a. UMAP plot illustrating cross-species integration of human and mouse excitatory neurons in medial thalamus, colored by dataset: human data from HBCC and MSSM, and mouse data from this study based on anatomical dissections and a previous study using retrograde labeling from prefrontal cortex 8 . b. UMAP plot of integrated mouse and human data, with mouse cells labeled by their nuclei of origin inferred from integration with public datasets and Allen Brain Atlas ISH data (see also Supplementary fig. 4 We produced a cell type atlas of adult human MD and adjacent midline nuclei, spatially mapped the major neuronal populations, and compared mouse and human neuronal identities. Recent studies examined cell populations across the entire human brain 15 , fetal human thalamus 11 , and multiple structures within the mouse thalamus 8,13,14,21-23 and habenula 18,24 , but were not focused on the MD, and cross-species comparisons have been limited to visual thalamus 21 , making our study the first in-depth analysis of cell types in human MD.

We identified a major transcriptional gradient between MD and PVT, which closely aligns with a previously-described gradient from primary to tertiary thalamus and likely reflects distinct connectivity and functional properties of these neurons 8 . This gene expression gradient mapped onto anatomical space, and respective clusters were enriched in MDmc, MDpc, and adjacent nuclei. Whilst MD has historically been considered 'higher-order', we identify major primary cell populations within both mouse and human MD. Cross-species analyses show that primary cells in MD are substantially expanded in humans relative to mice."

"Human Brain Cell Atlas integration: Data from the medial thalamus and epithalamus regions of interest of the Human Brain Cell Atlas 15 were downloaded and integrated with the HBCC and MSSM data, respectively, using anchors defined via canonical correlation analysis (CCA) in Seurat (Supplementary fig. 1c). Nearest centroid classification of HBCC and MSSM cells by supercluster label was performed to map corresponding cell types (Supplementary fig. 1d). Brain bank integration: Excitatory neurons from HBCC and MSSM were integrated using CCAbased anchors in Seurat. SNN clustering with a resolution of 0.5 on the top 5 PCs was used to define the clusters. To alleviate clustering being driven by individual samples, the four most closely related clusters were merged, resulting in 10 clusters (Supplementary fig. 1e-f).

Mouse anatomical and retrograde labeling data integration: Mouse thalamic neurons obtained in this study were integrated with those based on retrograde labeling from prefrontal cortex in a previous study 8 using CCA-based anchors in Seurat. SNN clustering with a resolution of 0.8 on the top 10 PCs was used, resulting in 19 clusters (Supplementary fig. 4b). PVT data integration: Five populations previously annotated and validated in the mouse PVT 13 were integrated with the mouse thalamic neuron clusters using anchors based on reciprocal principal component analysis (RPCA) in Seurat. Nearest centroid classification was used to map the PVT clusters onto corresponding excitatory neuron clusters (Supplementary fig. 4c). Habenula data integration: Previously annotated and validated neuron populations from the mouse habenula 18 were re-identified and integrated with the mouse thalamic neurons using RPCA-based anchors. Nearest centroid classification was used to map the re-identified habenula populations onto the excitatory neuron clusters (Supplementary fig. 4d). All integrations were performed on the top 2,000 most variable genes across datasets (SelectIntegrationFeatures in Seurat). Nearest neighbor classifiers were implemented via the lolR library (lol.classify.nearestCentroid)"

(Phillips et al., 2017)

A topographic axis of transcriptional identity in thalamus

"A fundamental goal in neuroscience is to uncover common principles by which different modalities of information are processed. In the mammalian brain, thalamus acts as the essential hub for forebrain circuits handling inputs from sensory, motor, limbic, and cognitive pathways. Whether thalamus imposes common transformations on each of these modalities is unknown. Molecular characterization offers a principled approach to revealing the organization of thalamus. Using near-comprehensive and projection-specific transcriptomic sequencing, we found that almost all thalamic nuclei fit into one of three profiles. These profiles lie on a single axis of genetic variance which is aligned with the mediolateral spatial axis of thalamus. Genes defining this axis of variance include receptors and ion channels, providing a systematic diversification of input/output transformations across the topography of thalamus. Single cell transcriptional profiling revealed graded heterogeneity within individual thalamic nuclei, demonstrating that a spectrum of cell types and potentially diverse input/output transforms exist within a given thalamic nucleus. Together, our data argue for an archetypal organization of pathways serving diverse input modalities, and provides a comprehensive organizational scheme for thalamus."

"We produced a near-comprehensive transcriptomic atlas of murine thalamus via microdissection of retrogradely labeled cells and subsequent RNAseq analysis (22 nuclei, 120 samples). Anterograde tracing of inputs to thalamus was used when identification of nuclear boundaries was ambiguous (Figure 1A,B for an example dissection approach). Labeled neurons from a given thalamic nucleus were pooled and profiled, an approach chosen because nuclear identity and projection target are major definers of cell type. See supplementary tables 1 and 2 for details on labelling strategies.

To identify the relationship between the transcriptomes of thalamic nuclei in an unsupervised manner, we selected the top 500 most differentially expressed genes between thalamic nuclei (via an ANOVA-like test, see methods). Using this differentially expressed

gene set, we grouped the thalamic nuclei via hierarchical clustering (1 -Spearman's correlation as distance metric), identifying five major subdivisions of nuclei (Figure 1C). We termed these profiles, not clusters, as we do not mean to imply discreteness between the classifications (see below). These profiles are not explained by the modality of information processed, since subnuclei of motor, somatosensory and visual complexes all split across different profiles. For example, the blue profile (Figure 1C) included both the cerebellar-recipient motor thalamus and the retina-recipient visual nucleus (lateral geniculate, LGd), whilst the red profile contained the basal ganglia recipient motor thalamus and the superior colliculus-recipient visual nucleus (lateral posterior, LP). Interestingly, the anterior dorsal nucleus (AD) and nucleus reuniens (RE) had distinct profiles and each formed profiles of their own. When schematic sections of thalamus were colored by these profiles, a mediolateral separation was observed, with the 'blue' nuclei being most lateral and 'yellow' nuclei most medial (Figure 1D). Thus, the largest fraction of linear gene expression variance in thalamus is not explained by modality, but rather a topographic axis spanning the structure's mediolateral extent."

"A. Overview of clusters (identified using the Seurat package, see methods) within each modality visualized using tSNE, cells colored by cluster identity. B. Violin plots for top marker genes for each cluster (Likelihood Ratio Test, p-value<10^-5, logFC>0.5, top 4-5 signal-to-noise ratio). C. Projection of single-cell data onto bulk principal component 1 (as in figure 2 Thus far we analyzed thalamic nuclei by pooling projection neurons from their anatomical position. The resulting transcriptional profiles could represent homogenous populations, discrete subtypes, or cells with graded differences. To probe these possibilities, we profiled the transcriptomes of individual neurons from motor, somatosensory, auditory, and prefrontal projection classes (supplementary table 3). Analysis of the single-cell RNAseq dataset resulted in multiple clusters for each projection class which, when projected onto the first principal component of our bulk RNAseq dataset, spanned the same axis of variance identified from bulk data (Figure 4 A,B,C, supplementary figure 4). Markers for this axis were spatially separated at the single-cell level, with clear intermediate cells present (Figure 4D, supplementary figure 5,6,7). However, some individual single-cell clusters had similar PC1 positions (e.g. clusters 1 and 2 of the motor projections neurons, or clusters 1 and 2 of the somatosensory nuclei, supplementary figure 4). This suggests that distinct clusters within a single modality could exist in similar anatomical positions. We examined this possibility by performing RNAscope for genes which distinguished these clusters (Pvalb and Tnnt1) . Using motor thalamus as an example, Pvalb and Tnnt1 expression were both found within the confines of a single thalamic nucleus, VL. While neurons selectively expressing each gene could be found, intermediate cells were also apparent. Pvalb-selective, intermediate, and Tnnt1-selective cells were distributed along the mediolateral axis (Figure 4E). "

"Therefore spatially organized genetic gradients can exist even within individual thalamic nuclei.

In this study we produced a near-comprehensive transcriptomic atlas of thalamus via RNAseq in order to investigate its organizational structure. Though input modality has historically defined thalamic nuclei, several alternative, overlapping organizational classifications have been proposed 1,6 . Early work perturbing thalamus showed that nuclei of the same modality can have strikingly different contributions to behavior, leading to a division between arousal-linked intralaminar nuclei and the rest of thalamus [7][8][9] . More recently, theories developed in the sensory thalamus have subdivided thalamic nuclei into two groups, based upon anatomical or neurochemical features [10][11][12] . However, precise delineation of groups is not consistent between theories, and no theory is comprehensive. If and how these later theories apply beyond sensory thalamus, for example, is unknown.

Here, we reconcile and extend these theories by placing almost all thalamic nuclei into one of three profiles, lying on a single topographic axis identified via unsupervised analysis. This axis extends from the midline/intralaminar groups through to a profile that includes the classic 'relay' sensory nuclei. Though prior theories did not incorporate electrophysiological properties, we found that genes directly affecting these properties were highly differentially expressed between groups, implying conserved, systematic variation of response properties across thalamus.

Understanding how the pattern of intra-thalamic variability identified here is used to support different circuit functions will be an important challenge 13 .

Prior theories of thalamic organization have placed thalamic cells into discrete nuclei, and thalamic nuclei into discrete groups. However, it remains unclear if subtypes of cells exist within nuclei and how discrete vs. continuous the cells type differences are. Recent studies in other non-thalamic areas have argued that gradients of projection neuron identities can exist 14,15 . Our single-cell transcriptomic data revealed heterogeneity within thalamic nuclei and suggest that thalamic genetic profiles exist on a spectrum, with mixtures of genetic profiles often existing in single neurons. Intermediate cell types exist both within and between nuclei, and are topographically localised."

Paraventricular and other midline glutamatergic neurons

Midline thalamic neurons, especially in PVT, show strong molecular and projection heterogeneity organized largely as gradients rather than sharply separate categories. PVT is one of the best-resolved mouse thalamic nuclei at single-cell level, with four to five transcriptomic subtypes reported across studies.

Type/family: midline thalamic glutamatergic projection neurons, especially **PVT** neurons.

Structure: Midline neurons do not show an obvious set of sharply separate somatodendritic morphologies,

but bushy-like and radiate-like dendritic patterns both occur. Axons arise from the soma and show little local branching in filled cells, consistent with projection-neuron identity. (Lara-Vasquez et al., 2016)

Function: PVT regulates many cognitive and behavioral processes and is a behavior-modulating hub. Different PVT subpopulations differ in appetite-related peptide responses, projection patterns, and behavioral effects on feeding. Midline CR-positive neurons also differ physiologically with respect to hippocampal state coupling. (Gao et al., 2023) (Shima et al., 2022) (Lara-Vasquez et al., 2016)

Location: PVT spans anterior-posterior, medio-lateral, and dorso-ventral axes. In both mouse and human medial thalamus, PVT-like populations are enriched toward the tertiary end of the broader trans-thalamic gradient. (Gao et al., 2023) (Schulmann et al., 2024)

Connectivity: PVT projects widely and contains projection-biased subpopulations. Shima et al. found that anterior and posterior PVT subpopulations have nearly non-overlapping axon projection patterns with an intermediate population between them. Gao et al., by integration with ThalamoSeq, found that PVT1 and PVT2 overlap with prefrontal projectors, PVT4 has a small overlap with auditory and visual projectors, PVT3 with prefrontal/visual, and PVT5 with visual. Midline nuclei more broadly send different outputs depending on dorsal, middle, or ventral position, for example to medial prefrontal versus parahippocampal cortex. (Shima et al., 2022) (Gao et al., 2023) (Lara-Vasquez et al., 2016)

Neurotransmitter: glutamate, consistent with projection-neuron identity. (Lara-Vasquez et al., 2016)

Neuropeptides and markers: PVT-enriched signatures include calcium-binding proteins **Calb1** and **Calb2**, and human work notes dopamine-related genes such as **DRD2**, **MOXD1**, **SNCA**, **PRKN** in PVT-enriched populations. Mouse studies reported top marker genes sufficient for multiplex FISH validation but the quoted text does not list the full marker set for all five mouse subtypes. Calretinin/CR expression marks one physiologically distinct midline population. (Lara-Vasquez et al., 2016) (Gao et al., 2023) (Nishioka et al., 2026)

Known heterogeneity:

- **Shima et al. 2022:** four PVT types, but with a continuous transcriptomic distribution whose largest variance follows the anterior-posterior axis; histology also supported continuity. (Shima et al., 2022)
- **Gao et al. 2023:** five molecularly distinct PVT subtypes arranged by combined AP, ML, and DV gradients; these are called **PVT1-5** in the paper. (Gao et al., 2023)
- **Midline CR+ vs CR− physiology:** CR expression cuts across strict nuclear location as a major predictor of physiology in sampled midline neurons. (Lara-Vasquez et al., 2016)

Transcriptomics and related datasets:

- PVT **snRNA-seq** plus multiplex FISH, defining five subtypes; accession number not given in the quote. (Gao et al., 2023)
- PVT **scRNA-seq** defining four continuous types and linking them to projection maps and feeding-related responses; accession number not given in the quote. (Shima et al., 2022)
- Cross-dataset integration with **ThalamoSeq** helped infer cortical projection relationships. (Gao et al., 2023)

Evidence

(Lara-Vasquez et al., 2016)

Midline thalamic neurons are differentially engaged during hippocampus network oscillations

> *"Calretinin and calbindin expression in midline thalamic neurons. We sampled the action potential firing of individual neurons from thalamic nuclei across the midline (Table 1). After recording, neurons were individually labeled with the juxtacellular filling method 38 and classified by their expression of neurochemical markers (Supplementary Table 1). In order to map the location of recorded neurons on the thalamus, we projected all labelled neurons on the neuroanatomical atlas of the mouse brain 39 . Neurons were recovered from most midline thalamic nuclei (Fig. 1a). The somatodendritic distributions of midline thalamic neurons were relatively*

uniform, with no obvious distinct morphological types. However, we could identify cells conforming to the distinct branching patterns of dendrites described in thalamic relay cells, such as "bushy" and "radiate" neurons 1,40 . Indeed, some cells had more angular and multipolar cell bodies, with dendritic trees branching dichotomously, characteristic of "radiate" cells (Fig. 1c). Conversely, other cells had elongated somata, with dendrites longer and branching less frequently, particularly in regions proximal to the soma, and followed a "bushy" organization (Fig. 1c). We were not able to fill the entire extension of axonal arborizations, even in cases where intense somatic labeling was achieved. In fact, most labelled cells did not show evidence of axonal labelling. When labelled, axons arose in all cases from the soma and rarely branched locally (Supplementary Figure 1). Given the above mentioned dendritic branching patterns, as well as axonal branching patterns, with little or no local collaterals, we considered our sample to be entirely comprised of thalamic projection neurons, which are known to be glutamatergic and typically reach the cortical mantle 1 ."

"By recording and labeling individual neurons in anesthetized mice, we show here evidence for functionally distinct neuronal populations in the midline thalamus. Indeed, our anatomical and physiological characterization leads us to propose that at least two different populations of midline thalamic projection cell deliver differential synaptic output to cortical targets according to ongoing brain states as defined by hippocampal activity. Our data show that rather than anatomical location, a major defining feature in the physiological properties of midline thalamic neurons is the expression of CR, a calcium-binding protein. Hence, distinct thalamic neuronal populations, identified by the somatic expression of CR, are differentially regulated by hippocampal pathways during network rhythms associated to behavioural contexts, likely supporting different stages of memory processing. Midline thalamic nuclei give rise to large glutamatergic axon terminals that densely distribute within frontal and temporal cortical lobes 27,45 . It has been reported that a dense fiber plexus located in the entorhinal cortex contains both CR and vesicular glutamate transporter 2 in the same axons. Complementary retrograde tracing experiments found labeled cell bodies in the nucleus reuniens co-expressing CR 46 . Our results show that CR+ cells are widely expressed across the midline thalamus, and given their anatomical and physiological features suggest that they are likely to be glutamatergic projection cells 1 . Cortical projections from the midline thalamus are not homogeneous across nuclei. Indeed, neurons in dorsal structures preferentially target the medial prefrontal cortex, with little input to the medial temporal lobe 5,24 ; neurons in middle regions, project almost exclusively to the medial prefrontal cortex 24,47 ; whereas neurons in ventral areas, project mostly to the parahippocampal cortex, with less significant projections to the medial prefrontal cortex 24,27 . Thus, the anatomical organization of synaptic output from the midline thalamus is not homogeneous, with every nucleus displaying distinct efferent connectivity patterns. Interestingly, our results suggest that a major feature in defining the activity pattern of thalamic neurons is not their nuclear location, but the expression profile of calcium-binding proteins, particularly CR."

(Gao et al., 2023)

Molecular and spatial profiling of the paraventricular nucleus of the thalamus

"The paraventricular nucleus of the thalamus (PVT) is known to regulate various cognitive and behavioral processes. However, while functional diversity among PVT circuits has often been linked to cellular differences, the molecular identity and spatial distribution of PVT cell types remain unclear. To address this gap, here we used single nucleus RNA sequencing (snRNA-seq) and identified five molecularly distinct PVT neuronal subtypes in the mouse brain. Additionally, multiplex fluorescent in situ hybridization of top marker genes revealed that PVT subtypes are organized by a combination of previously unidentified molecular gradients. Lastly, comparing our dataset with a recently published single-cell sequencing atlas of the thalamus yielded novel insight into the PVT's connectivity with the cortex, including unexpected innervation of auditory and visual areas. This comparison also revealed that our data contains a largely non-overlapping transcriptomic map of multiple midline thalamic nuclei. Collectively, our findings uncover previously unknown features of the molecular diversity and anatomical organization of the PVT and provide a valuable resource for future investigations."

"Although this broad classification into aPVT and pPVT subregions has aided advances in our understanding of the functional organization of the PVT, systematic dissections of the local organization of PVT subnetworks are currently lacking (Barson et al., 2020;McGinty and Otis, 2020;Iglesias and Flagel, 2021). Importantly, consistent with the notion that molecular identity can delineate cell types with different anatomical, functional, and electrophysiological properties (Huang and Zeng, 2013;Huang, 2014;Fuzik et al., 2016;Oh et al., 2014;Luo et al., 2008;Luo et al., 2018), studies have recently identified functional and anatomical differences that tie onto genetically defined subpopulations of PVT neurons (Mátyás et al., 2018;Gao et al., 2020;Kessler et al., 2021;Engelke et al., 2021). From this perspective, transcriptional profiling of the PVT could lead to valuable insights into the functional heterogeneity of this thalamic structure (McGinty and Otis, 2020).

In this study, we employ high throughput single-nucleus RNA sequencing and multiplex fluorescent in situ hybridization (ISH) labeling to identify five PVT subtypes that segregate differentially across the antero-posterior, medio-lateral, and dorso-ventral axes of the PVT. Our data highlights novel features about the molecular organization of the PVT, thereby offering a glimpse into how genetic diversity may tie into the various functions associated with this region of the thalamus. In addition, by performing the reciprocal principal component analysis (RPCA)-based integration and comparative analysis with the ThalamoSeq atlas, we find that PVT subtypes differentially innervate cortical areas and that our dataset contains complementary and previously unexplored mouse thalamic single-nuclei transcriptomes, including mediodorsal thalamus (MD) and intermediodorsal (IMD) thalamus (Phillips et al., 2019)."

"To cross-reference our snRNA-seq dataset of the mouse midline thalamus, we performed integration using RPCA together with the recently published ThalamoSeq RNA-Seq atlas (thalamoseq.janelia. org), which contains single-cell sequenced thalamic cells that

project to either motor, auditory, visual, somatosensory, or prefrontal cortices (PFC) (Figure 5, Figure 5-figure supplement 1; Phillips et al., 2019;Hao et al., 2021). Upon clustering analysis and plotting the UMAP of the integrated dataset with all midline-thalamic and BSTpr neurons, we found that our midline-thalamic dataset and the Thal-amoSeq dataset contained mostly non-overlapping transcriptomes (Figure 5-figure supplement 1).

Next, we performed integration of the ThalamoSeq dataset and PVT neurons specifically (Figure 5a-b).

By mapping the integrated dataset based on cortical projection location from the ThalamoSeq cells or their original cluster identity (either PVT1-5 or ThalamoSeq), we could observe the cluster relationship between ThalamoSeq cortical projection location and our five PVT molecular subtypes (Figure 5c-e). DGE analysis on the integrated dataset revealed that our original PVT subtypes (PVT1-5) could be mapped onto clusters 0, 1, 3, 5, and 4, respectively, of the integrated dataset (Supplementary file 1, Figure 5a and c). Following this, comparing the proportion of cortical-projectors in the integrated dataset showed that the clusters representing PVT1 P and PVT2 edge contained overlap with PFCprojectors, whereas, unexpectedly, the cluster representing PVT4 AM had a small amount of overlap with auditory and visual projectors (Figure 5b). The clusters representing PVT3 AL and PVT5 pan exhibited a small degree of overlap with prefrontal and visual cortical projectors, respectively (Figure 5b)."

(Shima et al., 2022)

Distinctiveness and continuity in transcriptome and connectivity in the anterior-posterior axis of the paraventricular nucleus of thalamus

"The paraventricular nucleus of the thalamus (PVT) projects axons to multiple areas and mediates a wide range of behaviors. Heterogeneity of functions and axonal projections in PVT have been reported, but what cell types exist in PVT and how different they are have not been addressed. We applied single-cell RNA sequencing to depict transcriptomic characteristics of mouse PVT neurons. The transcriptome of PVT neurons had a continuous distribution with the largest variance corresponding to the anterior-posterior axis. Although the single-cell transcriptome classified PVT neurons into four types, transcriptomic and histological analyses showed their continuity. Similarly, anterior and posterior subpopulations had nearly non-overlapping axon projection patterns, while another population showed intermediate patterns. In addition, they responded differently to appetite-related neuropeptides, and their chemogenetic activation showed opposing effects in food consumption. Our studies showed contrasts and continuity of PVT neurons underlying their function as a behavior-modulating hub."

(Schulmann et al., 2024)

A conserved cell-type gradient across the human mediodorsal and paraventricular thalamus

"MD was dissected from coronal slabs based on anatomical landmarks, followed by nuclei isolation and snRNA-seq via the 10X Chromium platform (Fig. 1a). Data underwent extensive quality control and filtering, clustering, and cell-type annotation based on marker genes and data from the Human Brain Cell Atlas 15 . Despite differences in tissue processing between the brain banks, both datasets showed the same major cell classes (Fig. 1b, Supplementary fig. 1a-d, see Methods for details). We identified excitatory neurons, marked by expression of SLC17A6 (VGluT2), inhibitory neurons, which comprised nearly 40% of neurons in our samples and expressed SOX14 -consistent with midbrain origin -and several non-neuronal populations (Fig. 1c; Supplementary data table 1).

To investigate thalamic projection neuron subpopulations in MD, we re-clustered the excitatory neurons, integrating HBCC and MSSM data (Fig. 1d, Supplementary fig. 1e-f). We found that most excitatory neurons in MD were part of a transcriptional gradient captured by the first principal component (PC1, Fig. 1e). Genes aligned with PC1 overlapped with and correlated in their loadings with those of a previously described trans-thalamic axis separating primary, secondary and tertiary thalamic cells in mice 8 (Supplementary fig. 1g-h). Thus, projection neurons in human MD and midline thalamus exhibit a gene expression gradient that closely resembles a cell-type gradient with distinct functional and morphological properties in mice. Human MD has traditionally been divided into a magnocellular (MDmc, medial), parvocellular (MDpc, lateral-ventral) and multiform/densocellular (MDdc, lateral-dorsal) part 16 . We hypothesized that these cytoarchitectonic subdivisions could correspond to transcriptionally defined neuronal populations. "

"To map cell populations identified by snRNA-seq onto anatomical space, we applied spatial transcriptomics to two coronal thalamic sections, then deconvoluted the cell types, using our snRNA-seq data as a reference 17 . This revealed prominent spatial separation of neuronal populations. Some excitatory neuron clusters preferentially mapped to MDmc, MDpc, or adjacent nuclei, while inhibitory interneurons appeared to be particularly enriched in MDpc and MDdc (Fig. 2a; Supplementary fig. 2a).

Similarly, projection of the excitatory neuron PC1 from Fig. 1e onto the spatial transcriptomics data showed that the primary end of the gradient mapped to MD, especially MDmc, while the tertiary end mapped to PVT and PT (Supplementary fig. 2b). Among the surrounding nuclei, intralaminar nuclei were on the tertiary end, while motor/somatosensory and lateral dorsal thalamus were on the primary end, consistent with their profiles in mouse thalamus 8 . RNA-FISH confirmed this spatial distribution of marker genes associated with opposite ends of the gradient and showed that numerous cells co-expressed multiple markers (Fig. 2b, Supplementary fig. 3). These analyses suggest that the gene expression gradient of excitatory neurons corresponds to a spatial gradient of neurons between MD and PT/PVT. Having identified a cell-type gradient in human MD and PVT with major similarities to the thalamic cell-type gradient in mice, we sought to better understand the commonalities and differences in MD cellular composition between humans and mice. Since previous snRNA-seq from retrogradely labeled cells did not include enough neurons

in MD 8 , we collected additional snRNA-seq from mouse MD based on anatomical dissections. As expected, this identified the same major cell classes as human MD, except that inhibitory neurons were not observed in mouse MD (Supplementary fig. 4). To directly compare thalamic projection neurons between species, we integrated human and mouse excitatory neurons (Fig. 3a, Supplementary fig."

"We tested for association with cell type specificity scores derived from our data using the top 10% most specific gene sets, as previously described 20 . Neuronal populations in the thalamus were significantly associated with genetic risk for severe mental illness: ExN_H2 (MDpc-enriched) was most strongly associated with genetic risk for schizophrenia and bipolar disorder, while thalamic inhibitory neurons and ExN_H3 (PVTenriched) were most strongly associated with genetic risk for major depressive disorder. GWAS for Alzheimer's disease and height, included for control purposes, implicated microglia and vascular cells, respectively, as expected. a. UMAP plot illustrating cross-species integration of human and mouse excitatory neurons in medial thalamus, colored by dataset: human data from HBCC and MSSM, and mouse data from this study based on anatomical dissections and a previous study using retrograde labeling from prefrontal cortex 8 . b. UMAP plot of integrated mouse and human data, with mouse cells labeled by their nuclei of origin inferred from integration with public datasets and Allen Brain Atlas ISH data (see also Supplementary fig. 4 We produced a cell type atlas of adult human MD and adjacent midline nuclei, spatially mapped the major neuronal populations, and compared mouse and human neuronal identities. Recent studies examined cell populations across the entire human brain 15 , fetal human thalamus 11 , and multiple structures within the mouse thalamus 8,13,14,21-23 and habenula 18,24 , but were not focused on the MD, and cross-species comparisons have been limited to visual thalamus 21 , making our study the first in-depth analysis of cell types in human MD.

We identified a major transcriptional gradient between MD and PVT, which closely aligns with a previously-described gradient from primary to tertiary thalamus and likely reflects distinct connectivity and functional properties of these neurons 8 . This gene expression gradient mapped onto anatomical space, and respective clusters were enriched in MDmc, MDpc, and adjacent nuclei. Whilst MD has historically been considered 'higher-order', we identify major primary cell populations within both mouse and human MD. Cross-species analyses show that primary cells in MD are substantially expanded in humans relative to mice."

"Human Brain Cell Atlas integration: Data from the medial thalamus and epithalamus regions of interest of the Human Brain Cell Atlas 15 were downloaded and integrated with the HBCC and MSSM data, respectively, using anchors defined via canonical correlation analysis (CCA) in Seurat (Supplementary fig. 1c). Nearest centroid classification of HBCC and MSSM cells by supercluster label was performed to map corresponding cell types (Supplementary fig. 1d). Brain bank integration: Excitatory neurons from HBCC and MSSM were integrated using CCAbased anchors in Seurat. SNN clustering with a resolution of 0.5 on the top 5 PCs was used to define the clusters. To alleviate clustering being driven by individual samples, the four most closely related clusters were merged, resulting in 10 clusters (Supplementary fig. 1e-f).

Mouse anatomical and retrograde labeling data integration: Mouse thalamic neurons obtained in this study were integrated with those based on retrograde labeling from prefrontal cortex in a previous study 8 using CCA-based anchors in Seurat. SNN clustering with a resolution of 0.8 on the top 10 PCs was used, resulting in 19 clusters (Supplementary fig. 4b). PVT data integration: Five populations previously annotated and validated in the mouse PVT 13 were integrated with the mouse thalamic neuron clusters using anchors based on reciprocal principal component analysis (RPCA) in Seurat. Nearest centroid classification was used to map the PVT clusters onto corresponding excitatory neuron clusters (Supplementary fig. 4c). Habenula data integration: Previously annotated and validated neuron populations from the mouse habenula 18 were re-identified and integrated with the mouse thalamic neurons using RPCA-based anchors. Nearest centroid classification was used to map the re-identified habenula populations onto the excitatory neuron clusters (Supplementary fig. 4d). All integrations were performed on the top 2,000 most variable genes across datasets (SelectIntegrationFeatures in Seurat). Nearest neighbor classifiers were implemented via the lolR library (lol.classify.nearestCentroid)"

(Nishioka et al., 2026)

Disturbances of paraventricular thalamic nucleus neurons in bipolar disorder revealed by single-nucleus analysis

"Initially, we focused on the thalamic and cortical snRNA-seq data from 20 controls to delineate the physiological cytoarchitecture of the human medial thalamus. Our QC yielded 32 distinct cell clusters (Fig. 1b). One cluster, designated as NeuCyto, was removed from the subsequent analysis as non-nuclei neuronal cytoplasmic components (see Methods). Accordingly, 147,477 nuclei in 31 cell nuclei clusters underwent subsequent analysis. Excitatory and inhibitory neurons were distinctly segregated between the thalamus and cortex, while astrocytes and oligodendrocyte progenitor cells (OPCs) were continuously separated. This observation aligned with Siletti et al. 6 .

The thalamic excitatory neurons (ExN_Tha) predominantly expressed SLC17A6 (coding VGLUT2), whereas the cortical excitatory neurons (ExN_FrC) primarily expressed SLC17A7 (coding VGLUT1) (Supplementary Fig. 1a). RBFOX3, encoding a well-established neuronal marker known as NeuN, exhibited robust expression in ExN_FrC, yet was minimally expressed in ExN_Tha. Hierarchical clustering of the 31 cell clusters, using all expressed genes, indicated that thalamic excitatory neurons were more closely aligned with cortical inhibitory neurons than with cortical excitatory neurons (Fig. 1c), corroborating the findings by Mathys et al. 28 . Furthermore, mapping through cortex reference 29 confirmed that thalamic excitatory neurons corresponded more closely with cortical inhibitory neurons than with their excitatory counterparts (Supplementary Fig. 1b). These observed similarities were represented by specific subtype markers of cortical inhibitory neurons, particularly calcium-binding protein genes, including PVALB,

CALB1, and CALB2, which encode parvalbumin, calbindin, and calretinin, respectively (Fig. 1d and Supplementary Fig. 1c)."

"Among the thalamic excitatory neurons, ExN_Tha_CALB cells in Fig. 1b were putatively supposed to be PVT neurons, as characterized by the expression of CALB1 and CALB2. We used these genes as guidance for locating PVT neuron clusters because they have been employed in previous studies as indicators of PVT neurons in both humans 36 and rodents 37,38 . However, note that CALB1 and CALB2 expressions are not exclusive to PVT neurons, and additional heterogeneity likely exists within the human PVT beyond what can be defined by these genes. These putative PVT neurons were also robustly identified in the Human Brain Cell Atlas (HBCA) dataset 6 , as demonstrated by the integration of our and HBCA datasets (Fig. 2a). ExN_CALB in Fig. 2a corresponded to ExN_Tha_CALB in Fig. 1b. The putative PVT neurons (ExN_CALB cells) were detected from all 20 our samples and 20 HBCA samples (see Methods). This integrated dataset was used for subsequent analyses to enhance generalizability.

As CALB1 and CALB2 expression alone were insufficient to confirm that the ExN_CALB clusters represented PVT neurons, we validated this classification using two approaches: (1) spatial transcriptomic deconvolution and (2) anchor-based label transfer using mouse PVT neuron reference data. We obtained spatial transcriptomic data of the medial thalamus from one control sample as a representative specimen (Fig. 2b). The PVT was characterized through unbiased clustering of spatial transcriptomic data and partially corresponded to the areas immunostained by calretinin, a protein coded by CALB2 36 (Fig. 2c). "

"Deconvolution of thalamic spatial transcriptomic data using the integrated cell clusters by CytoSPACE 39 indicated that ExN_CALB cells were predominantly localized in the paraventricular regions of the thalamus with a clear demarcation from other excitatory neuron populations (Fig. 2d, e), aligning with the anatomical definition of the PVT by Uroz et al. 36 . Oligodendrocytes were concentrated in the lateral side of the thalamus, while astrocytes were primarily found in the medial side. Oligodendrocyte precursor cells (OPCs) and microglia displayed a diffuse distribution throughout the thalamus. Expectedly, ExN_CALB corresponded most closely with mouse PVT neurons when employing anchor transfer with published single-cell datasets from mice 23,40 (Fig. 2f). These data confirmed the validity of ExN_CALB as PVT neurons. Upon closer examination of PVT neuron subtypes, the majority of human PVT neurons, particularly those from our dataset, corresponded to mouse posterior PVT neurons (Fig. 2g). Compared to other thalamic excitatory neurons, PVT neurons exhibited a distinctive expression profile for genes (Supplementary Data 5) associated with dopamine signaling (e.g., MOXD1, SNCA, PRKN) and the regulation of calcium and ion channels (e.g., MCUB, CACNG3, PDE4B, SCN9A), as determined by GSEA (Fig. 2h, Supplementary Data 2). These PVT neuron markers were particularly expressed in the PVT regions identified through spatial transcriptomics (Fig. 2i). Notably, DRD2, highly expressed in the medial thalamus relative to the frontal cortex, was prominent in the PVT compared to other thalamic areas. This observation corroborates the enrichment of dopamine- h GO network visualization for PVT-enriched genes, illustrating major functional pathways related to dopamine metabolism and calcium/ion channels. GO terms with enrichment scores >0.5 and network connectivity >2 are shown. Core genes involved in these pathways are highlighted."

Anterior thalamic nucleus excitatory neurons

The anterior thalamic nuclei contain discrete excitatory clusters that match the classic nuclei, but each nucleus also contains within-nucleus gradients. This makes ATN a clear example where transcriptomic types and anatomical nuclei both matter.

Type/family: excitatory projection neurons of the **anterior thalamic nuclei (ATN)**, including AD, AV, and AM-related populations.

Structure and location: Kapustina et al. identified **three discrete excitatory cell-type clusters** that correspond to the known nuclei of the ATN. They also found graded spatial variation within each nucleus along a dorsomedial-ventrolateral axis. (Kapustina et al., 2024)

Function and connectivity: The ATN are involved in a wide range of functions. Long-range input structure also varies within the ATN: Kapustina et al. showed that the anteroventral nucleus has spatially covarying protein products and long-range inputs. Epi-Retro-Seq further supports projection specificity, showing that thalamus retrosplenial cortex neurons are uniquely enriched in clusters annotated as **AV** and **AD**, consistent with known anterior thalamic connectivity. (Kapustina et al., 2024) (Zhou et al., 2023)

Neurotransmitter: glutamate, as these are excitatory thalamic projection neurons. (Kapustina et al., 2024)

Markers: The quoted summary states that marker genes and molecular pathways were identified, but it does not list the specific markers. At the broader thalamic level, anterior nuclei such as AM can align well with transcriptomic supertypes in ABC/THALMANAC. (Kapustina et al., 2024) (Turner et al., 2025)

Known heterogeneity: ATN show both **discrete** organization across nuclei and **continuous** organization within nuclei. This helps explain why purely nuclear or purely gradient-based schemes each capture only part of

thalamic organization. (Kapustina et al., 2024)

Transcriptomics and related datasets: single-cell RNA-seq, spatial transcriptomics, and multiplexed FISH were combined in the ATN study; accession number is not given in the provided quote. (Kapustina et al., 2024)

Evidence

(Kapustina et al., 2024)

The cell-type-specific spatial organization of the anterior thalamic nuclei of the mouse brain.

"Understanding the cell-type composition and spatial organization of brain regions is crucial for interpreting brain computation and function. In the thalamus, the anterior thalamic nuclei (ATN) are involved in a wide variety of functions, yet the cell-type composition of the ATN remains unmapped at a single-cell and spatial resolution. Combining single-cell RNA sequencing, spatial transcriptomics, and multiplexed fluorescent in situ hybridization, we identify three discrete excitatory cell-type clusters that correspond to the known nuclei of the ATN and uncover marker genes, molecular pathways, and putative functions of these cell types. We further illustrate graded spatial variation along the dorsomedial-ventrolateral axis for all individual nuclei of the ATN and additionally demonstrate that the anteroventral nucleus exhibits spatially covarying protein products and long-range inputs. Collectively, our study reveals discrete and continuous cell-type organizational principles of the ATN, which will help to guide and interpret experiments on ATN computation and function."

(Zhou et al., 2023)

Brain-wide Correspondence Between Neuronal Epigenomics and Long-Distance Projections

"In any given brain, each neuron contributes uniquely to brain function. Nevertheless, neurons can be grouped into types based on similarities and differences across multiple dimensions, including epigenetic state, gene expression, anatomy, and physiology. Single-cell genomic technologies have been particularly impactful for cell type classification due to their high throughput (millions of cells assayed) and dimensionality (thousands of genes and even more genetic loci) leading to the identification of large numbers of transcriptomic and epigenomic clusters corresponding to possible cell types across the entire mouse brain.

A prominent and distinguishing anatomical feature of many brain neuron types is their long-distance axonal projections. Long-distance projections can be directly related to single neuron gene expression or epigenomes by use of powerful linking technologies, including BARseq 1,2 , Retro-seq 3,4 , and Epi-Retro-Seq 5 . Previous studies have used Retro-Seq and Epi-Retro-Seq to link mouse neocortical [3][4][5] , hypothalamic 6 and thalamic projection cell types 7 to their genetic and epigenetic clusters, revealing complex but predictable relationships. For example, cortical neurons projecting solely to intra-telencephalic (IT) targets fall into different clusters than those that project to extra-telencephalic (ET) targets. On the other hand, cortical layer 2/3 (L2/3) IT neuron types projecting to different cortical areas typically co-cluster despite having quantifiable and predictable genetic and epigenetic differences across the populations 4,5 . In the face of this complexity, how can single-cell genetic and epigenetic assays be used to inform the structure and function of brain cell types and how can neuronal structure predict genetics, epigenetics, and function? Further, can the principles learned from more limited previous studies be extended to the entire brain? Are there different principles linking projection status to epigenetics for different brain areas?

To address these questions we employed Epi-Retro-Seq to assay 33,034 neurons from 225 source target combinations across the entire mouse brain."

"The thalamus is a primary hub in sensory and cortical information processing and also projects to subcortical structures. Similar to hypothalamus, thalamus consists of a large number of nuclei that are organized into multiple functional groups. The main, central regions of the thalamus are composed of exclusively excitatory regions (except for a few local GABAergic interneurons in the dorsal lateral geniculate nucleus (LGd)) that are reciprocally connected with cortical areas 20 . Other more ventral and lateral regions of the thalamus (such as LGv and RT) contain GABAergic inhibitory neurons that are either reciprocally connected with thalamic excitatory neurons or project to subcortical structures such as the basal ganglia and brainstem 20 . In contrast to the hypothalamus, the thalamus had a lower degree of cell type complexity as shown by the smaller number of cell clusters identified through gene expression analysis 16 . Despite both the thalamus and hypothalamus showing a high level of heterogeneity in their anatomical nuclei and projections, the differences in their cell type complexity prompted us to investigate whether the relationships between cell types, their projections, and their spatial locations in the thalamus differ from those observed in the hypothalamus, as discussed above.

We analyzed thalamic neurons that project to twelve different targets, including nine cortical areas (PFC, MOp, SSp, ACA, AI, AUDp, RSP, PTLp, and VISp), SC, VTA, and P. To gain a comprehensive understanding of these neurons, we combined Epi-Retro-Seq data with unbiased snmC-seq and single-cell RNA-seq data from the thalamus. "

"Through this integration, we identified a total of 58 thalamic neuronal cell clusters (Fig. 4a), of which 33 clusters were enriched for

Epi-Retro-Seq neurons (Fig. 4b). It is worth noting that neurons dissected from different anatomical regions within the thalamus were located in distinct sets of clusters 17 (Fig. 4c), as expected from prior descriptions based on analysis of scRNA-seq data 7 , suggesting that these molecularly defined cell clusters also have a spatial organization.

In Figure 4b, we assessed the degree of enrichment of each projection in each cluster. Similar to what was observed in the hypothalamus, each population of thalamic projection neurons exhibited enrichment in distinct subsets of cell clusters, with each cluster showing enrichment for a specific set of projections, sometimes only one. Notably, the clusters enriched for TH SC, TH VTA, TH P and TH cortex were mostly mutually exclusive. Regarding cortical projections, TH PTLp and TH VISp neurons exhibited enrichment in a largely overlapping set of clusters, but with varying degrees of enrichment. TH MOp and TH SSp neurons also shared most of their enriched clusters, which differed from those enriched for TH PTLp and TH VISp. These results support the notion of a separation of thalamic cell types between the visual and motor pathways in the thalamus and highlight the heterogeneity of cell types within each pathway. Notably, TH RSP neurons showed no overlap in enriched clusters with any other cortical projections, and were uniquely enriched in clusters 13, 26 and 47. These clusters were annotated by their gene expression patterns as belonging to the anteroventral (AV) nucleus (clusters 13 and 26), and anterodorsal (AD) nucleus (cluster 47), which is consistent with TH RSP projections originating from anterior thalamic nuclei 21 . In summary, TH neurons projecting to cortex vs. subcortical targets were enriched in distinct sets of clusters. "

"The enriched cell clusters for cortical projections were further segregated by different thalamic pathways, with multiple enriched cell clusters observed for each pathway or projection. These findings highlight the cell type specificity as well as heterogeneity at the level of TH projections.

Such cell type specificity and heterogeneity of TH projection neurons were also reported in transcriptomic analysis of single TH projection neurons. Phillips et al. conducted a single-cell RNA-seq study on thalamic neurons projecting to the prefrontal, motor, somatosensory, auditory, and visual cortices 7 . We will refer to their dataset as Retro-Seq of these thalamocortical (TC) projections. In their study, clustering analysis of Retro-Seq revealed that neurons of each projection were enriched in a specific set of clusters, indicating specific and shared cell types between different TC projections, as well as cell type heterogeneity within each TC projection.

To compare Retro-Seq and Epi-Retro-Seq data for these TC projections, we mapped the Retro-Seq data onto our 58 integrated TH clusters (Extended Data Fig. 8a, Methods). The t-SNE of Retro-Seq and Epi-Retro-Seq neurons from comparisons of the most closely corresponding projections showed that they occupied similar spaces and were enriched in a common set of clusters (Extended Data Fig. 8a,b). However, in addition to the shared projection-enriched clusters, certain clusters were found to be enriched for each projection only in either Retro-Seq or Epi-Retro-Seq. These differences are most likely due to the use of different injection coordinates for each cortical target (Supplementary Table 4), resulting in overlapping yet different populations of retrogradely labeled TH neurons being analyzed in the two data sets. The degree of alignment between Epi-Retro-Seq and Retro-Seq was quantified by the overlap score and cosine distance (Extended Data Fig. 8c, Methods), which revealed that Retro-Seq neurons of each projection were more similar to Epi-Retro-Seq neurons for the corresponding projections than for any other projections."

(Turner et al., 2025)

Exploring the correspondence between gene expression and thalamic nuclei using the THALMANAC resource

"The thalamus connects the sensory organs and major subcortical brain regions with the neocortex. The thalamus has long been divided into multiple discrete nuclei, based on cytoarchitecture, histochemical stains, and mesoscale connectivity. However, thalamic nuclei do not completely describe thalamic organization. For example, some boundaries between thalamic nuclei are disputed, whereas other nuclei are known to contain subdomains with distinct connectivity and function. Moreover, the correspondence between cellular gene expression and other properties of thalamic projection neurons remains to be established. Spatial analysis of single cell gene expression provides a basis for reevaluating thalamic organization. We present the THALMANAC, (THALamusMERFISH ANalysis and ACcess) a Findable, Accessible, Interoperable, Reusable and Reproducible (FAIRR) resource for exploring and analyzing single-cell transcriptomic variation in the thalamus. The THALMANAC provides streamlined access to thalamic gene expression data registered to the common coordinate framework and tools for quantitative analysis and visualization of these data, all encapsulated in a reproducible, cloud computing platform. Using this resource, we find that gene expression generally supports the parcellation of thalamus into distinct nuclei. Some nuclei, such as the anteromedial nucleus, are additionally composed of discrete subdomains, while other nuclei share patterns of gene expression or are arrayed on a spatial gradient of gene expression. The THALMANAC establishes spatial transcriptomic data as a foundation for delineating thalamic organization. Graphical Abstract"

"This fine-scale structure could correspond to discrete, molecularly defined subdivisions of VM that have not yet been annotated, or to topographic organization of thalamocortical and corticothalamic connectivity.

A major question is how gene expression relates to existing parcellations and thalamic organization in general. Single cell RNA sequencing (scRNA-seq) provides detailed measurements of transcription in large populations of individual neurons (Tasic et al. 2016), which can be used to define transcriptomic cell types ('t-types'). The combination of scRNA-seq with spatial transcriptomics has been used to localize t-types across the whole mouse brain (Langlieb et al. 2023;Yao et al. 2023;Zhang et al. 2023). With t-types mapped into the CCFv3 coordinate space (Wang et al. 2020), we can evaluate how their spatial distributions correspond to

anatomical parcellations (Yao et al. 2023).

Here, building on the Allen Brain Cell (ABC) Atlas scRNA-seq and spatial transcriptomics data, we developed a resource, the THALamus MERFISH ANalysis and ACcess (THALMANAC), which provides access to spatial gene expression data in the mouse thalamus; targeted analysis and data visualization of gene expression patterns; a platform for reproducible computations; and a no-code visualization app for data exploration. We demonstrate the utility of the THALMANAC by addressing how thalamic structural organization relates to gene expression. Aligning spatial transcriptomic data and nuclei boundaries reveals a wide range of relationships between t-types and thalamic nuclei: some t-types correspond to a single thalamic nucleus; others subdivide nuclei into multiple parts; many t-types span multiple thalamic nuclei.

Analysis of expression by itself supports these same conclusions: some nuclei boundaries are associated with sharp changes in gene expression; other nuclei contain gene expression gradients; and, in many cases, gene expression patterns span multiple nuclei.

Understanding how information is routed through the thalamus requires integrating insights gained from gene expression with other modalities, including connectivity and neurophysiology."

"The THALMANAC resource provides access to thalamic spatial gene expression data, visualization of the relationships between gene expression, t-types, and CCFv3 anatomical parcellations, and quantitative analysis of targeted quantitative analyses, all encapsulated in a cloud platform that enables reproducible computations and customized exploration of the data.

The THALMANAC is built on a thalamus-specific subset of spatial transcriptomics data from multiplexed error-robust fluorescence in situ hybridization (MERFISH) from the ABC Atlas. The ABC Atlas dataset measures 500 transcripts, chosen to distinguish t-types across the whole mouse brain, in spatially resolved, single cells. MERFISH cells were registered to the CCFv3 coordinate space and mapped to the scRNA-seq data by assigning each MERFISH cell to the label of the most similar t-type cluster. We identified neurons that belong to the thalamus (including the zona incerta and habenula) according to their t-type label (Figure S1A) and CCFv3 parcellation label (Figure S1B). The thalamic dataset analyzed here includes 80,170 MERFISH cells across 14 coronal sections collected from a single male mouse (Methods).

The THALMANAC includes three avenues for engaging with the t-type and gene expression data and accompanying analyses (Figure 1). First, a Streamlit web app (RRID:SCR_024354) provides nocode exploration, including interactive visualizations and on-the-fly di"erential gene expression analysis for thalamic nuclei (Interactive Streamlit App: thalmanac.allenneuraldynamics.org). Second, Jupyter notebooks (RRID:SCR_018315) provide step-by-step demonstrations of the Python code needed to reproduce all results presented here (github.com/AllenNeuralDynamics/thalmanac). Both tools allow, users to customize the thalamic cells that are selected from the full ABC Atlas."

"In the ABC Atlas, transcriptomic variation is used to assign cells to a hierarchy of discrete t-types (from broad to narrow categories: class, subclass, supertype, cluster). Thalamic neurons belong to 5 classes, 58 subclasses, 163 supertypes, and 563 clusters (Table S1; Figure S2) (taxonomy version CCN20230722) (Yao et al. 2023), but these numbers will change as the taxonomy evolves. Whereas the cluster level is based on unsupervised clustering, higher levels of the taxonomy are additionally informed by developmental origin via shared transcription factor expression, anatomical location, and classical (pre-scRNAseq) neuron types (DeFelipe et al. 2013) defined by key molecular markers (e.g. neuropeptides, calcium-binding proteins).

Transcriptomic classes typically correspond to major brain structures. Glutamatergic neurons in thalamus proper form a single class (class 18 TH Glut), whereas habenula neurons belong to a distinct class (class 17 MH-LH Glut) (Figure S2B). The other three classes are not unique to the thalamus: class 12 HY GABA is found in the reticular thalamus, zona incerta and ventral lateral geniculate nucleus (CCFv3 annotations; RT, ZI, and LGv, respectively, see Table S1) but also in the hypothalamus; class 19 MB Glut is found in thalamic nucleus SPA but also in the midbrain; class 20 MB GABA is found in thalamic nuclei LGv and SPF and the midbrain. Transcriptomic subclasses span multiple thalamic nuclei(Figure S2C), whereas transcriptomic supertypes often define individual nuclei (e.g. AM), and several nuclei contain multiple supertypes (e.g. ZI) (Figure S2D). Most nuclei contain multiple transcriptomic clusters (Figure 2A, Figure S2E).

We explored correspondence between clusters and thalamic nuclei."

"A comprehensive understanding of how the thalamus routes information to the cortex will require multiple types of data and a community-wide e"ort. Our work contributes to this program by curating and analyzing the correspondence between state-of-the-art spatially resolved, single-cell transcriptomic data, and widely used parcellations of thalamus into discrete nuclei. In addition, we make our data and analyses available in an accessible, documented, and reproducible bioinformatics resource in the cloud. Using the AM nucleus and the somatosensory nuclei as examples, we illustrate how this interactive resource can be used to identify novel patterns of gene expression and how these patterns correspond to prior parcellations of the thalamus.

The THALMANAC resource streamlines researchers' access to thalamic spatial transcriptomic data. We anticipate that scientists will extend our analysis based on the ABC atlas data, or apply similar analyses to additional spatial transcriptomics datasets registered to the CCF coordinate system (Langlieb et al. 2023;Zhang et al. 2023). The tool should also be used as a reference to validate recent studies focused on specific nuclei, such as the spatially segregated projections from PF to distinct dorsal striatum

regions (Mandelbaum et al. 2019) and spatially segregated inputs to PO (Casas-Torremocha et al. 2022), the molecular organization of the PVT (Gao et al. 2023), and the core/shell organization of neurons in RT (Li et al. 2020). Our interactive Streamlit app can be used to identify marker genes for post-hoc localization of neurophysiological and neuroanatomical experiments, or to select gene panels for multiplexed mFISH experiments. The THALMANAC is already being used in the e"ort to create and characterize enhancerbased genetic tools (Ben-Simon et al. 2025) to target thalamic cell types and regions.

The THALMANAC shows that at the scale of thalamic nuclei, the molecular organization of the thalamus resembles existing parcellations. "

"However, both discrete cell type composition and gene expression gradients vary at finer spatial scales within and across nuclei. Given the evidence for circuit and functional variation at the same length scales, additional data need to be integrated with molecular characterization to understand thalamic organization. We see two promising paths forward for data integration. First, existing data registered to the CCFv3 3D coordinate space, such as the databases of single-neuron morphological reconstruction (Winnubst et al. 2019) or neurophysiology data (Chen et al. 2024), could be linked to cell types and gene expression described in the THALMANAC. Second, targeted multimodal datasets, such as single-neuron morphological reconstructions combined with posthoc mFISH for transcriptomic cell typing would link cell types and gene expression to projection targets at the single cell level.

Beyond data integration, we also recognize the need for thalamus-focused, community annotation, similar to previous e"orts to annotate the fly connectome (Schlegel et al. 2024). We envision a refined parcellation of the thalamus where scientists with specific domain knowledge would combine molecular information from the THALMANAC with neuroanatomical and functional data to reflect progress in understanding of thalamic organization. Diagram illustrating the three workflows that can be launched from the Code Ocean Collection accompanying this paper. The Code Ocean platform provides built-in support for code versioning (via Git), environment replicability (via Docker), data and analysis provenance (via tracked data assets and linked pipeline runs), cloud compute resources (via AWS EC2 instances). Users can launch an interactive, no-code Streamlit application to explore spatial distributions of transcriptomic cell types and gene expression from their browser, as well as visualize locations of annotated cell types for each thalamic nucleus and highlight a specific nucleus or group of related thalamic nuclei. Tutorial Jupyter notebooks and Python code replicate all analyses and visualizations in a cloud workstation. These visualizations can be customized and extended. Users can download an AnnData-formatted .h5ad"

dLGN principal neurons and local inhibitory cells

The mouse dLGN is unusual because clear physiological and projection differences exist, but adult transcriptomic differences among principal relay neurons are subtle and largely gradient-like. The main molecular axis in mouse dLGN aligns with core versus shell organization, while local inhibitory interneurons are present but sparse compared with principal glutamatergic cells.

Type/family 1: glutamatergic principal relay neurons of **dLGN/LGd**.

- **Structure/function:** dLGN conveys retinal information to visual cortex. Across mammals, thalamocortical neurons here differ in morphology, connectivity, and visual signal content. (Bakken et al., 2021) (Duménieu et al., 2021)
- **Location:** mouse dLGN core and shell regions. (Bakken et al., 2021)
- **Connectivity:** mouse shell neurons align more with primate koniocellular-like signatures, while core neurons align more with primate M/P-like signatures. (Bakken et al., 2021)
- **Neurotransmitter:** glutamate, typical of principal thalamocortical neurons. (Bakken et al., 2021)
- **Markers: Pvalb** is enriched in the core; **Calb1** and **Necab1** are enriched in the shell. (Bakken et al., 2021)
- **Known heterogeneity:** in adult mouse dLGN, transcriptomic diversity is modest; the dominant signal is a graded core-shell axis rather than sharply separated discrete transcriptomic types. This fits the broader thalamic pattern of graded heterogeneity. (Bakken et al., 2021) (Duménieu et al., 2021)
- **Transcriptomics:** single-nucleus and single-cell RNA-seq were used across mouse, macaque, and human; accession number is not given in the quote. (Bakken et al., 2021)

Type/family 2: local GABAergic interneurons of dLGN.

- **Location/connectivity:** sparse interneurons exist in dLGN, unlike most central thalamus proper. (Zhou et al., 2023)
- **Neurotransmitter:** GABA. (Zhou et al., 2023)
- **Markers/heterogeneity:** the supplied quotes do not detail mouse dLGN interneuron markers or subclasses. Their presence is clear, but the provided sources focus much more on principal relay cells. (Zhou et al.,

2023)

Paper	Neuron Classification Scheme	Anatomical Location	Cell Morphology	Functional Role	Connectivity Pattern	Neurotransmitter Identity	Neuropeptide Expression	Molecular Markers	Transcriptomic Evidence	Know Hete gene
Single-cell and single-nucleus RNA-seq uncovers shared and distinct axes of variation in dorsal LGN neurons in mice, non-human primates, and humans	Uses core/shell and principal relay-type transcriptomic labels; primate K, M, and P types are distinguished, with mouse core/shell alignment and homology-based comparisons across species.	Primates have K neurons in distinct laminar populations; mouse TC neurons show a core-versus-shell gradient rather than discrete types.	Principal dLGN cell-type differences in mature mammals are subtle transcriptomically despite distinct morphologies and projection targets.	Visual relay: dLGN thalamocortical neurons convey visual information from the retina to primary visual cortex.	Primate K neurons have distinct laminar distributions and V1 projection patterns, and are more similar to inferior pulvinar neurons; mouse dLGN shows core-shell wiring differences rather than discrete TC types.	Excitatory TC neurons are glutamatergic; primate dLGN also contains expanded diversity of GABAergic neurons.	No specific neuropeptide expression patterns are reported in the provided text.	Marker genes were conserved across species; Pvalb marked core/M-P neurons, while Calb1 and Necab1 marked shell/K neurons in opposite gradients.	Single-cell and single-nucleus RNA-seq identified four primate TC neuron types and mainly graded, non-discrete mouse dLGN variation.	With type diver y is subtl in matu dLGN neur s: prim K neur s for distin subc ses, while mou TC neur s sho grad core- shell hete eneit and stron cross spec diffe ces.

Paper	Neuron Classification Scheme	Anatomical Location	Cell Morphology	Functional Role	Connectivity Pattern	Neurotransmitter Identity	Neuropeptide Expression	Molecular Markers	Transcriptomic Evidence	Know Hete gene
Mechanisms of Plasticity in Subcortical Visual Areas	It notes that transcriptomic differences between dLGN principal cell types are subtle despite distinct morphology and projection targets, indicating low gene diversity.	Distinct molecular profiles follow a topographical shift across lateral, intermediate, and medial thalamic nuclei; dLGN neurons are in the most lateral thalamus.	Thalamo-cortical neurons show distinct morphologies, but transcriptomic differences in the mature dLGN are subtle relative to these structural differences.	N/A	Connectivity patterns vary across thalamo-cortical neuron types; dLGN neurons show subtle transcriptomic differences despite distinct morphology and cortical projection targets.	Not stated in the provided snippets.	N/A	dLGN neuron genes show low diversity, with only subtle transcriptomic differences between principal cell types.	Transcriptomic profiling revealed only subtle differences among mature dLGN principal cell types, indicating low gene diversity rather than clearly discrete types.	The pape repo low diver y in dLGN neur gene with subt trans ptom diffe ces betw n princ al ce types and prog sive topo phica mole lar sl in thala c nucl

Paper	Neuron Classification Scheme	Anatomical Location	Cell Morphology	Functional Role	Connectivity Pattern	Neurotransmitter Identity	Neuropeptide Expression	Molecular Markers	Transcriptomic Evidence	Know Hete gene
Brain-wide Correspondence Between Neuronal Epigenomics and Long-Distance Projections	N/A	Not specified for dLGN; the paper only notes that thalamic neurons from different anatomical regions occupy distinct clusters.	Long-distance axonal projection pattern is a distinguishing anatomical feature used to classify neurons and link them to epigenomic cell types.	Thalamic neurons are involved in sensory and cortical information processing and thalamocortical signaling.	Distinct projection targets define thalamic cell clusters; cortical vs subcortical targets show mutually exclusive enrichment, with pathway-specific overlap among cortical projections.	N/A	N/A	N/A	Single-cell RNA-seq and single-nucleus RNA-seq were used to define and compare thalamic neuron types, supporting discrete clusters with projection-specific enrichment.	Know heter eneit inclu s grad cont a, discr e subc ses, core- shell axes and cross spec mole lar/ funct nal varia n.

Evidence

(Bakken et al., 2021)

Single-cell and single-nucleus RNA-seq uncovers shared and distinct axes of variation in dorsal LGN neurons in mice, non-human primates, and humans

> *"Abundant evidence supports the presence of at least three distinct types of thalamocortical (TC) neurons in the primate dorsal lateral geniculate nucleus (dLGN) of the thalamus, the brain region that conveys visual information from the retina to the primary visual cortex (V1). Different types of TC neurons in mice, humans, and macaques have distinct morphologies, distinct connectivity patterns, and convey different aspects of visual information to the cortex. To investigate the molecular underpinnings of these cell types, and how these relate to differences in dLGN between human, macaque, and mice, we profiled gene expression in single nuclei and cells using RNA-sequencing. These efforts identified four distinct types of TC neurons in the primate dLGN: magnocellular (M) neurons, parvocellular (P) neurons, and two types of koniocellular (K) neurons. Despite extensively documented morphological and physiological differences between M and P neurons, we identified few genes with significant differential expression between*

transcriptomic cell types corresponding to these two neuronal populations. Likewise, the dominant feature of TC neurons of the adult mouse dLGN is high transcriptomic similarity, with an axis of heterogeneity that aligns with core vs. shell portions of mouse dLGN. Together, these data show that transcriptomic differences between principal cell types in the mature mammalian dLGN are subtle relative to the observed differences in morphology and cortical projection targets. Finally, alignment of transcriptome profiles across species highlights expanded diversity of GABAergic neurons in primate versus mouse dLGN and homologous types of TC neurons in primates that are distinct from TC neurons in mouse."

"We used unsupervised clustering to define transcriptomic cell types in a well-studied part of the mammalian thalamus, dLGN, for human, macaque, and mouse. We examined the correspondence of excitatory TC neurons to previously described morphological, connectional, and physiological differences (Hendry and Reid, 2000;Krahe et al., 2011;Cruz-Martı´n et al., 2014) and identified celltype homologies across species.

We find that primate K neurons are transcriptomically clearly distinguishable from their M and P counterparts, consistent with previous results (reviewed in Hendry and Reid, 2000). Moreover, we identify two transcriptomic types of K neurons that may correspond to two populations reported in macaque that have distinct laminar distributions in dLGN and projection patterns in V1 (Casagrande et al., 2007). We also find that K neurons are more transcriptomically similar to inferior pulvinar neurons than to M or P neurons. This is consistent with their shared inputs from retina and superior colliculus and cortical projection targets (Huo et al., 2019) and supports a close functional relationship between K neurons and inferior pulvinar. M and P neurons are transcriptomically similar to each other in both macaque and human.

In contrast to primate, TC neurons in mouse dLGN cannot be grouped into discrete types based on gene expression, although many genes show graded expression differences between core and shell regions. Similar graded expression heterogeneity has been reported within and across many first-and higher-order thalamic nuclei (Phillips et al., 2019). Alignment of primate and mouse TC neurons revealed cell-type homologies between pulvinar and LP neurons, as expected. Mouse shell neurons aligned more with primate K neurons and core neurons aligned with M and P neurons, and some marker genes were conserved across species. Genes for calcium binding proteins were expressed in opposing gradients in TC neurons across mouse dLGN: Pvalb-expressing neurons were enriched in the core (and primate M and P neurons), Calb1-and Necab1-expressing neurons were enriched in the shell (and K neurons)."

(Duménieu et al., 2021)

Mechanisms of Plasticity in Subcortical Visual Areas

"Different types of thalamo-cortical neurons with distinct morphologies, connectivity patterns and conveying different aspects of visual information to the cortex have been identified in the dLGN of mice, cats and primates. However, this functional categorization is difficult to translate into molecular terms.

In thalamic nuclei, three major molecular profiles of thalamo-cortical neurons have recently been identified using RNA sequencing [127]. This categorization cannot be explained by cortical projections but reflects a progressive topographical shift across laterally, intermediate and medially localized thalamic nuclei. Notably, ion channels and receptor profiles were organized along the same lines. As a result, differentially projecting thalamic neurons such as thalamo-cortical neurons of the dLGN and ventrobasal sensory neurons, all located in the most lateral part of the thalamus, express a common specific set of genes [127]. Moreover, gene profiling in the mature dLGN showed that transcriptomic differences between principal cell types are subtle relative to the observed differences in morphology and cortical projection targets [128]. In conclusion, these two studies intriguingly reveal a low diversity in dLGN neuron genes."

(Zhou et al., 2023)

Brain-wide Correspondence Between Neuronal Epigenomics and Long-Distance Projections

"In any given brain, each neuron contributes uniquely to brain function. Nevertheless, neurons can be grouped into types based on similarities and differences across multiple dimensions, including epigenetic state, gene expression, anatomy, and physiology. Single-cell genomic technologies have been particularly impactful for cell type classification due to their high throughput (millions of cells assayed) and dimensionality (thousands of genes and even more genetic loci) leading to the identification of large numbers of transcriptomic and epigenomic clusters corresponding to possible cell types across the entire mouse brain.

A prominent and distinguishing anatomical feature of many brain neuron types is their long-distance axonal projections. Long-distance projections can be directly related to single neuron gene expression or epigenomes by use of powerful linking technologies, including BARseq 1,2 , Retro-seq 3,4 , and Epi-Retro-Seq 5 . Previous studies have used Retro-Seq and Epi-Retro-Seq to link mouse neocortical [3][4][5] , hypothalamic 6 and thalamic projection cell types 7 to their genetic and epigenetic clusters, revealing complex but predictable relationships. For example, cortical neurons projecting solely to intra-telencephalic (IT) targets fall into different clusters than those that project to extra-telencephalic (ET) targets. On the other hand, cortical layer 2/3 (L2/3) IT neuron types projecting to different cortical areas typically co-cluster despite having quantifiable and predictable genetic and epigenetic differences across the populations 4,5 . In the face of this complexity, how can single-cell genetic and epigenetic assays be used to inform the structure and function of brain cell types and how can neuronal structure predict genetics, epigenetics, and function? Further, can the principles learned from more limited previous studies be extended to the entire brain? Are there different principles

linking projection status to epigenetics for different brain areas?

To address these questions we employed Epi-Retro-Seq to assay 33,034 neurons from 225 source–target combinations across the entire mouse brain."

"The thalamus is a primary hub in sensory and cortical information processing and also projects to subcortical structures. Similar to hypothalamus, thalamus consists of a large number of nuclei that are organized into multiple functional groups. The main, central regions of the thalamus are composed of exclusively excitatory regions (except for a few local GABAergic interneurons in the dorsal lateral geniculate nucleus (LGd)) that are reciprocally connected with cortical areas 20 . Other more ventral and lateral regions of the thalamus (such as LGv and RT) contain GABAergic inhibitory neurons that are either reciprocally connected with thalamic excitatory neurons or project to subcortical structures such as the basal ganglia and brainstem 20 . In contrast to the hypothalamus, the thalamus had a lower degree of cell type complexity as shown by the smaller number of cell clusters identified through gene expression analysis 16 . Despite both the thalamus and hypothalamus showing a high level of heterogeneity in their anatomical nuclei and projections, the differences in their cell type complexity prompted us to investigate whether the relationships between cell types, their projections, and their spatial locations in the thalamus differ from those observed in the hypothalamus, as discussed above.

We analyzed thalamic neurons that project to twelve different targets, including nine cortical areas (PFC, MOp, SSp, ACA, AI, AUDp, RSP, PTLp, and VISp), SC, VTA, and P. To gain a comprehensive understanding of these neurons, we combined Epi-Retro-Seq data with unbiased snmC-seq and single-cell RNA-seq data from the thalamus. "

"Through this integration, we identified a total of 58 thalamic neuronal cell clusters (Fig. 4a), of which 33 clusters were enriched for Epi-Retro-Seq neurons (Fig. 4b). It is worth noting that neurons dissected from different anatomical regions within the thalamus were located in distinct sets of clusters 17 (Fig. 4c), as expected from prior descriptions based on analysis of scRNA-seq data 7 , suggesting that these molecularly defined cell clusters also have a spatial organization.

In Figure 4b, we assessed the degree of enrichment of each projection in each cluster. Similar to what was observed in the hypothalamus, each population of thalamic projection neurons exhibited enrichment in distinct subsets of cell clusters, with each cluster showing enrichment for a specific set of projections, sometimes only one. Notably, the clusters enriched for TH→SC, TH→VTA, TH→P and TH→cortex were mostly mutually exclusive. Regarding cortical projections, TH→PTLp and TH→VISp neurons exhibited enrichment in a largely overlapping set of clusters, but with varying degrees of enrichment. TH→MOp and TH→SSp neurons also shared most of their enriched clusters, which differed from those enriched for TH→PTLp and TH→VISp. These results support the notion of a separation of thalamic cell types between the visual and motor pathways in the thalamus and highlight the heterogeneity of cell types within each pathway. Notably, TH→RSP neurons showed no overlap in enriched clusters with any other cortical projections, and were uniquely enriched in clusters 13, 26 and 47. These clusters were annotated by their gene expression patterns as belonging to the anteroventral (AV) nucleus (clusters 13 and 26), and anterodorsal (AD) nucleus (cluster 47), which is consistent with TH→RSP projections originating from anterior thalamic nuclei 21 . In summary, TH neurons projecting to cortex vs. subcortical targets were enriched in distinct sets of clusters. "

"The enriched cell clusters for cortical projections were further segregated by different thalamic pathways, with multiple enriched cell clusters observed for each pathway or projection. These findings highlight the cell type specificity as well as heterogeneity at the level of TH projections.

Such cell type specificity and heterogeneity of TH projection neurons were also reported in transcriptomic analysis of single TH projection neurons. Phillips et al. conducted a single-cell RNA-seq study on thalamic neurons projecting to the prefrontal, motor, somatosensory, auditory, and visual cortices 7 . We will refer to their dataset as Retro-Seq of these thalamocortical (TC) projections. In their study, clustering analysis of Retro-Seq revealed that neurons of each projection were enriched in a specific set of clusters, indicating specific and shared cell types between different TC projections, as well as cell type heterogeneity within each TC projection.

To compare Retro-Seq and Epi-Retro-Seq data for these TC projections, we mapped the Retro-Seq data onto our 58 integrated TH clusters (Extended Data Fig. 8a, Methods). The t-SNE of Retro-Seq and Epi-Retro-Seq neurons from comparisons of the most closely corresponding projections showed that they occupied similar spaces and were enriched in a common set of clusters (Extended Data Fig. 8a,b). However, in addition to the shared projection-enriched clusters, certain clusters were found to be enriched for each projection only in either Retro-Seq or Epi-Retro-Seq. These differences are most likely due to the use of different injection coordinates for each cortical target (Supplementary Table 4), resulting in overlapping yet different populations of retrogradely labeled TH neurons being analyzed in the two data sets. The degree of alignment between Epi-Retro-Seq and Retro-Seq was quantified by the overlap score and cosine distance (Extended Data Fig. 8c, Methods), which revealed that Retro-Seq neurons of each projection were more similar to Epi-Retro-Seq neurons for the corresponding projections than for any other projections."

GABAergic thalamic-region neurons: TRN, LGv/IGL, and related inhibitory populations

The best-studied inhibitory population in the mouse thalamic region is the TRN, where neurons form a transcriptomic gradient linked to anatomy, physiology, connectivity, and sleep function. Other inhibitory neurons are found in LGv/IGL and related nuclei, and developmental work indicates they arise from a lineage distinct from most thalamic glutamatergic neurons.

Type/family 1: thalamic reticular nucleus (TRN/RT) neurons

- **Structure:** TRN neurons form core-like and shell-like extremes across a continuous transcriptomic gradient. (Li et al., 2020)
- **Function:** TRN regulates thalamocortical interactions important for sensory processing, attention, cognition, and sleep; selective perturbation of molecularly defined TRN-thalamus subnetworks altered sleep. (Li et al., 2020)
- **Location:** TRN surrounding dorsal thalamus. In ABC-derived taxonomies, RT belongs to a GABAergic class shared with some hypothalamic-like regions. (Li et al., 2020) (Turner et al., 2025)
- **Connectivity:** two TRN subpopulations connect differentially with first-order versus higher-order thalamic nuclei. (Li et al., 2020)
- **Neurotransmitter:** GABA. (Li et al., 2020)
- **Markers:** opposite ends of the gradient are marked by largely mutually exclusive **Spp1** and **Ecel1** expression. (Li et al., 2020)
- **Heterogeneity:** continuous gradient rather than sharply discrete classes, though the ends are biologically distinct. (Li et al., 2020)
- **Transcriptomics:** integrative single-cell RNA-seq plus electrophysiology; accession number not given in the quote. (Li et al., 2020)

Type/family 2: LGv/IGL inhibitory neurons

- **Development/location:** embryonic single-cell work found a Shh-responsive rostral thalamic progenitor lineage that gives rise to GABAergic neurons in **VLG/LGv** and **IGL**. (Govek et al., 2022)
- **Neurotransmitter:** GABA. (Govek et al., 2022)
- **Markers and connectivity:** the provided quotes do not give fine subtype markers or projection maps for mouse LGv/IGL neurons. In ABC analyses, LGv contributes to midbrain-like and hypothalamus-like GABAergic classes, which already suggests heterogeneity. (Turner et al., 2025)

Type/family 3: inhibitory neurons in other thalamic-region nuclei

- **SPF and related regions:** ABC/THALMANAC notes a midbrain GABA class in **LGv** and **SPF**, and a hypothalamus-like GABA class in **RT**, **ZI**, and **LGv**. This shows that inhibitory neurons in the thalamic region do not form one uniform molecular family. (Turner et al., 2025)
- **Species note:** human medial thalamus contains SOX14-positive and SOX14-negative inhibitory populations, but mouse MD appears to lack the human-like inhibitory abundance; this cautions against overgeneralizing human inhibitory diversity to mouse dorsal thalamus. (Schulmann et al., 2024) (Trujillo et al., 2026)

Paper	Neuron Nomenclature / Classification Scheme	Anatomical Location / Nucleus Of Origin	Developmental Origin	Transcriptomic Identity / Molecular Signature	Neurotransmitter Phenotype	Neuropeptide / Co-transmitter Profile	Morphology / Structural Organization	Connectivity / Projection Targets	Functional Role	Hetegene / Subte Struc re

Paper	Neuron Nomenclature / Classification Scheme	Anatomical Location / Nucleus Of Origin	Developmental Origin	Transcriptomic Identity / Molecular Signature	Neurotransmitter Phenotype	Neuropeptide / Co-transmitter Profile	Morphology / Structural Organization	Connectivity / Projection Targets	Functional Role	Hetegene / Subte Struc re
Distinct subnetworks of the thalamic reticular nucleus	An integrative, transcriptomic-based classification: TRN neurons are organized by a transcriptomic gradient with mutually exclusive markers and distinct anatomical/ electrophysiological subpopulations.	Thalamic reticular nucleus (TRN).	N/A	A transcriptomic gradient of two negatively correlated gene-expression profiles defines two TRN subpopulations with mutually exclusive markers, Spp1 and Ecel1.	GABAergic inhibitory neurons; the TRN is the major source of thalamic inhibition.	No neuropeptides or additional co-expressed signaling molecules are reported in the provided text.	TRN neurons at transcriptomic extremes exhibit core- or shell-like anatomical structure.	The two TRN subpopulations make differential connections with first-order and higher-order thalamic nuclei, forming distinct TRN–thalamus subnetworks.	Differentially regulates sleep through distinct TRN-thalamus subnetworks.	Cellu heter eneit is orga ed as trans ptom grad t with two nega ely corre ed gene expr ion profi , yield mutu y excl ve TR subp ulati s.

Paper	Neuron Nomenclature / Classification Scheme	Anatomical Location / Nucleus Of Origin	Developmental Origin	Transcriptomic Identity / Molecular Signature	Neurotransmitter Phenotype	Neuropeptide / Co-transmitter Profile	Morphology / Structural Organization	Connectivity / Projection Targets	Functional Role	Hete gene / Subt e Struc re
Exploring the correspondence between gene expression and thalamic nuclei using the THALMANAC resource	An integrative transcriptomic taxonomy: thalamic neurons are assigned to classes, subclasses, supertypes, and clusters using transcriptomic variation, developmental origin, anatomy, and classical markers.	It is the specific thalamic nucleus or subregion where a neuron type is found, and many transcriptomic types map to or span multiple nuclei.	Developmental origin is one factor informing higher-type taxonomy levels, via shared transcription factor expression and anatomical location.	It varies across the thalamus: some nuclei show sharp gene-expression boundaries, others gradients, and many patterns span multiple nuclei.	N/A	N/A	Thalamic structural organization generally matches gene-expression-based parcellation, but finer-scale variation and subdomains exist within nuclei.	Not established here; connectivity/projection targets are mentioned as needing integration with gene expression, not analyzed directly.	N/A	Thal ic nucl show hete enec subt e struc e: disc e subd main shar gene expr ion patte s, an cont ous grad ts withi and acro nucl

Paper	Neuron Nomenclature / Classification Scheme	Anatomical Location / Nucleus Of Origin	Developmental Origin	Transcriptomic Identity / Molecular Signature	Neurotransmitter Phenotype	Neuropeptide / Co-transmitter Profile	Morphology / Structural Organization	Connectivity / Projection Targets	Functional Role	Hetegene / Subte Struc re
A conserved cell-type gradient across the human mediodorsal and paraventricular thalamus	Inhibitory thalamic-region neurons were classified transcriptomically by snRNA-seq clustering and marker-gene annotation, integrated with Human Brain Cell Atlas labels.	Neuronal populations mapped to MDmc, MDpc, adjacent nuclei, and the gradient extended between MD and PVT/PT.	Human and mouse MD excitatory neurons show a conserved transcriptional gradient; inhibitory neurons in human MD express SOX14, consistent with midbrain origin.	MD projection neurons show a continuous transcriptional gradient, with PC1-linked gene sets marking primary-to-tertiary thalamic identities and distinct excitatory subclasses.	Excitatory neurons expressed SLC17A6 (VGluT2); inhibitory neurons expressed SOX14 and comprised nearly 40% of neurons.	Neuropeptide/co-transmitter information is not reported in the provided text.	A continuous transcriptional and spatial gradient links MDmc, MDpc, and PVT, corresponding to distinct structural/anatomical organization.	A transcriptional gradient maps between MD and PVT/PT, suggesting distinct connectivity; primary cells in MD are expanded in humans relative to mice.	Distinct connectivity and functional properties, with primary-to-tertiary gradients likely reflecting circuit-level roles.	Neur diver y is large cont ous, form a MD to-PV trans ption grad t with regio l clust ng a cons ved prim -to- terti subty es acro spec .

Paper	Neuron Nomenclature / Classification Scheme	Anatomical Location / Nucleus Of Origin	Developmental Origin	Transcriptomic Identity / Molecular Signature	Neurotransmitter Phenotype	Neuropeptide / Co-transmitter Profile	Morphology / Structural Organization	Connectivity / Projection Targets	Functional Role	Hete gene / Subt e Struc re

Evidence

(Li et al., 2020)

Distinct subnetworks of the thalamic reticular nucleus

"The thalamic reticular nucleus (TRN), the major source of thalamic inhibition, regulates thalamocortical interactions that are critical for sensory processing, attention and cognition1–5. TRN dysfunction has been linked to sensory abnormality, attention deficit and sleep disturbance across multiple neurodevelopmental disorders6–9. However, little is known about the organizational principles that underlie its divergent functions. Here we performed an integrative study linking single-cell molecular and electrophysiological features of the mouse TRN to connectivity and systems-level function. We found that cellular heterogeneity in the TRN is characterized by a transcriptomic gradient of two negatively correlated gene-expression profiles, each containing hundreds of genes. Neurons in the extremes of this transcriptomic gradient express mutually exclusive markers, exhibit core or shell-like anatomical structure and have distinct electrophysiological properties. The two TRN subpopulations make differential connections with the functionally distinct first-order and higher-order thalamic nuclei to form molecularly defined TRN–thalamus subnetworks. Selective perturbation of the two subnetworks in vivo revealed their differential role in regulating sleep. In sum, our study provides a comprehensive atlas of TRN neurons at single-cell resolution and links molecularly defined subnetworks to the functional organization of thalamocortical circuits. A study integrating single-cell RNA-sequencing and electrophysiology data shows that in mouse, the cellular repertoire of the thalamic reticular nucleus is characterized by a transcriptomic gradient defined at its extremes by mutually exclusive expression of Spp1 and Ecel1, providing insights into the organizational principles underlying the divergent functions of this brain region."

(Turner et al., 2025)

Exploring the correspondence between gene expression and thalamic nuclei using the THALMANAC resource

"The thalamus connects the sensory organs and major subcortical brain regions with the neocortex. The thalamus has long been divided into multiple discrete nuclei, based on cytoarchitecture, histochemical stains, and mesoscale connectivity. However, thalamic nuclei do not completely describe thalamic organization. For example, some boundaries between thalamic nuclei are disputed, whereas other nuclei are known to contain subdomains with distinct connectivity and function. Moreover, the correspondence between cellular gene expression and other properties of thalamic projection neurons remains to be established. Spatial analysis of single cell gene expression provides a basis for reevaluating thalamic organization. We present the THALMANAC, (THALamusMERFISH ANalysis and ACcess) a Findable, Accessible, Interoperable, Reusable and Reproducible (FAIRR) resource for exploring and analyzing single-cell transcriptomic variation in the thalamus. The THALMANAC provides streamlined access to thalamic gene expression data registered to the common coordinate framework and tools for quantitative analysis and visualization of these data, all encapsulated in a reproducible, cloud computing platform. Using this resource, we find that gene expression generally supports the parcellation of thalamus into distinct nuclei. Some nuclei, such as the anteromedial nucleus, are additionally composed of discrete subdomains, while other nuclei share patterns of gene expression or are arrayed on a spatial gradient of gene expression. The THALMANAC establishes spatial transcriptomic data as a foundation for delineating thalamic organization. Graphical Abstract"

"This fine-scale structure could correspond to discrete, molecularly defined subdivisions of VM that have not yet been annotated, or to topographic organization of thalamocortical and corticothalamic connectivity.

A major question is how gene expression relates to existing parcellations and thalamic organization in general. Single cell RNA sequencing (scRNA-seq) provides detailed measurements of transcription in large populations of individual neurons (Tasic et al. 2016), which can be used to define transcriptomic cell types ('t-types'). The combination of scRNA-seq with spatial transcriptomics has been used to localize t-types across the whole mouse brain (Langlieb et al. 2023;Yao et al. 2023;Zhang et al. 2023). With t-types mapped into the CCFv3 coordinate space (Wang et al. 2020), we can evaluate how their spatial distributions correspond to anatomical parcellations (Yao et al. 2023).

Here, building on the Allen Brain Cell (ABC) Atlas scRNA-seq and spatial transcriptomics data, we developed a resource, the THALamus MERFISH ANalysis and ACcess (THALMANAC), which provides access to spatial gene expression data in the mouse

thalamus; targeted analysis and data visualization of gene expression patterns; a platform for reproducible computations; and a no-code visualization app for data exploration. We demonstrate the utility of the THALMANAC by addressing how thalamic structural organization relates to gene expression. Aligning spatial transcriptomic data and nuclei boundaries reveals a wide range of relationships between t-types and thalamic nuclei: some t-types correspond to a single thalamic nucleus; others subdivide nuclei into multiple parts; many t-types span multiple thalamic nuclei.

Analysis of expression by itself supports these same conclusions: some nuclei boundaries are associated with sharp changes in gene expression; other nuclei contain gene expression gradients; and, in many cases, gene expression patterns span multiple nuclei.

Understanding how information is routed through the thalamus requires integrating insights gained from gene expression with other modalities, including connectivity and neurophysiology."

"The THALMANAC resource provides access to thalamic spatial gene expression data, visualization of the relationships between gene expression, t-types, and CCFv3 anatomical parcellations, and quantitative analysis of targeted quantitative analyses, all encapsulated in a cloud platform that enables reproducible computations and customized exploration of the data.

The THALMANAC is built on a thalamus-specific subset of spatial transcriptomics data from multiplexed error-robust fluorescence in situ hybridization (MERFISH) from the ABC Atlas. The ABC Atlas dataset measures 500 transcripts, chosen to distinguish t-types across the whole mouse brain, in spatially resolved, single cells. MERFISH cells were registered to the CCFv3 coordinate space and mapped to the scRNA-seq data by assigning each MERFISH cell to the label of the most similar t-type cluster. We identified neurons that belong to the thalamus (including the zona incerta and habenula) according to their t-type label (Figure S1A) and CCFv3 parcellation label (Figure S1B). The thalamic dataset analyzed here includes 80,170 MERFISH cells across 14 coronal sections collected from a single male mouse (Methods).

The THALMANAC includes three avenues for engaging with the t-type and gene expression data and accompanying analyses (Figure 1). First, a Streamlit web app (RRID:SCR_024354) provides nocode exploration, including interactive visualizations and on-the-fly di"erential gene expression analysis for thalamic nuclei (Interactive Streamlit App: thalmanac.allenneuraldynamics.org). Second, Jupyter notebooks (RRID:SCR_018315) provide step-by-step demonstrations of the Python code needed to reproduce all results presented here (github.com/AllenNeuralDynamics/thalmanac). Both tools allow, users to customize the thalamic cells that are selected from the full ABC Atlas."

"In the ABC Atlas, transcriptomic variation is used to assign cells to a hierarchy of discrete t-types (from broad to narrow categories: class, subclass, supertype, cluster). Thalamic neurons belong to 5 classes, 58 subclasses, 163 supertypes, and 563 clusters (Table S1; Figure S2) (taxonomy version CCN20230722) (Yao et al. 2023), but these numbers will change as the taxonomy evolves. Whereas the cluster level is based on unsupervised clustering, higher levels of the taxonomy are additionally informed by developmental origin via shared transcription factor expression, anatomical location, and classical (pre-scRNAseq) neuron types (DeFelipe et al. 2013) defined by key molecular markers (e.g. neuropeptides, calcium-binding proteins).

Transcriptomic classes typically correspond to major brain structures. Glutamatergic neurons in thalamus proper form a single class (class 18 TH Glut), whereas habenula neurons belong to a distinct class (class 17 MH-LH Glut) (Figure S2B). The other three classes are not unique to the thalamus: class 12 HY GABA is found in the reticular thalamus, zona incerta and ventral lateral geniculate nucleus (CCFv3 annotations; RT, ZI, and LGv, respectively, see Table S1) but also in the hypothalamus; class 19 MB Glut is found in thalamic nucleus SPA but also in the midbrain; class 20 MB GABA is found in thalamic nuclei LGv and SPF and the midbrain. Transcriptomic subclasses span multiple thalamic nuclei(Figure S2C), whereas transcriptomic supertypes often define individual nuclei (e.g. AM), and several nuclei contain multiple supertypes (e.g. ZI) (Figure S2D). Most nuclei contain multiple transcriptomic clusters (Figure 2A, Figure S2E).

We explored correspondence between clusters and thalamic nuclei."

"A comprehensive understanding of how the thalamus routes information to the cortex will require multiple types of data and a community-wide e"ort. Our work contributes to this program by curating and analyzing the correspondence between state-of-the-art spatially resolved, single-cell transcriptomic data, and widely used parcellations of thalamus into discrete nuclei. In addition, we make our data and analyses available in an accessible, documented, and reproducible bioinformatics resource in the cloud. Using the AM nucleus and the somatosensory nuclei as examples, we illustrate how this interactive resource can be used to identify novel patterns of gene expression and how these patterns correspond to prior parcellations of the thalamus.

The THALMANAC resource streamlines researchers' access to thalamic spatial transcriptomic data. We anticipate that scientists will extend our analysis based on the ABC atlas data, or apply similar analyses to additional spatial transcriptomics datasets registered to the CCF coordinate system (Langlieb et al. 2023;Zhang et al. 2023). The tool should also be used as a reference to validate recent studies focused on specific nuclei, such as the spatially segregated projections from PF to distinct dorsal striatum regions (Mandelbaum et al. 2019) and spatially segregated inputs to PO (Casas-Torremocha et al. 2022), the molecular organization of the PVT (Gao et al. 2023), and the core/shell organization of neurons in RT (Li et al. 2020). Our interactive Streamlit app can be used to identify marker genes for post-hoc localization of neurophysiological and neuroanatomical experiments, or to select gene panels for multiplexed mFISH experiments. The THALMANAC is already being used in the e"ort to create and characterize

enhancerbased genetic tools (Ben-Simon et al. 2025) to target thalamic cell types and regions.

The THALMANAC shows that at the scale of thalamic nuclei, the molecular organization of the thalamus resembles existing parcellations. "

"However, both discrete cell type composition and gene expression gradients vary at finer spatial scales within and across nuclei. Given the evidence for circuit and functional variation at the same length scales, additional data need to be integrated with molecular characterization to understand thalamic organization. We see two promising paths forward for data integration. First, existing data registered to the CCFv3 3D coordinate space, such as the databases of single-neuron morphological reconstruction (Winnubst et al. 2019) or neurophysiology data (Chen et al. 2024), could be linked to cell types and gene expression described in the THALMANAC. Second, targeted multimodal datasets, such as single-neuron morphological reconstructions combined with posthoc mFISH for transcriptomic cell typing would link cell types and gene expression to projection targets at the single cell level.

Beyond data integration, we also recognize the need for thalamus-focused, community annotation, similar to previous e"orts to annotate the fly connectome (Schlegel et al. 2024). We envision a refined parcellation of the thalamus where scientists with specific domain knowledge would combine molecular information from the THALMANAC with neuroanatomical and functional data to reflect progress in understanding of thalamic organization. Diagram illustrating the three workflows that can be launched from the Code Ocean Collection accompanying this paper. The Code Ocean platform provides built-in support for code versioning (via Git), environment replicability (via Docker), data and analysis provenance (via tracked data assets and linked pipeline runs), cloud compute resources (via AWS EC2 instances). Users can launch an interactive, no-code Streamlit application to explore spatial distributions of transcriptomic cell types and gene expression from their browser, as well as visualize locations of annotated cell types for each thalamic nucleus and highlight a specific nucleus or group of related thalamic nuclei. Tutorial Jupyter notebooks and Python code replicate all analyses and visualizations in a cloud workstation. These visualizations can be customized and extended. Users can download an AnnData-formatted .h5ad"

(Govek et al., 2022)

Developmental trajectories of thalamic progenitors revealed by single-cell transcriptome profiling and Shh perturbation

"Our study provides a comprehensive transcriptome-wide analysis of thalamic progenitors and their trajectories into thalamic nuclei during embryonic stages of brain development. We demonstrate that molecular signatures of thalamic neuronal subtypes can be readily distinguished as early as E12.5, prior to their aggregation into histologically distinct thalamic nuclei. 26 These data lend support to the outside-in model of thalamic neurogenesis, whereby early-born neurons contribute to lateral thalamic nuclei and later-born neurons contribute to medial thalamic nuclei. 36,58 Our work further extends this model with molecular insights into the mechanisms by which diverse thalamic nuclei acquire their identities. By following single-cell trajectories over developmental time, our data are consistent with a model in which thalamic nuclei arise from three distinct cellular lineages. In this model, Shh-responsive cTh.Pro1 progenitors give rise to glutamatergic neurons (cTh.N1) within sensory (DLG, VPM/VPL, vMG), motor (VA/VL, VM), and anterior (AD, AV, AM) nuclei that occupy predominantly ventrolateral regions of the thalamus. Shh-responsive rTh.Pro progenitors give rise to GABAergic neurons (rTh.N) in VLG and IGL thalamic nuclei, in agreement with previous findings. 12,13 In contrast, Shh-independent cTh.Pro2 progenitors give rise to neurons (cTh.N2) within nuclei (e.g., PV, MD, CM) that form at more dorsomedial positions of the thalamus.

The bifurcation of transcriptomic trajectories according to thalamic progenitor identity is in general agreement with results from in vivo clonal analyses. 36,59 These studies revealed that individual thalamic progenitors give rise to many neurons that populate multiple thalamic nuclei. They also demonstrated that the clonal relationship between thalamic nuclei is determined primarily by the rostro-caudal and dorsoventral positions of thalamic progenitors."

(Schulmann et al., 2024)

A conserved cell-type gradient across the human mediodorsal and paraventricular thalamus

"MD was dissected from coronal slabs based on anatomical landmarks, followed by nuclei isolation and snRNA-seq via the 10X Chromium platform (Fig. 1a). Data underwent extensive quality control and filtering, clustering, and cell-type annotation based on marker genes and data from the Human Brain Cell Atlas 15 . Despite differences in tissue processing between the brain banks, both datasets showed the same major cell classes (Fig. 1b, Supplementary fig. 1a-d, see Methods for details). We identified excitatory neurons, marked by expression of SLC17A6 (VGluT2), inhibitory neurons, which comprised nearly 40% of neurons in our samples and expressed SOX14 -consistent with midbrain origin -and several non-neuronal populations (Fig. 1c; Supplementary data table 1).

To investigate thalamic projection neuron subpopulations in MD, we re-clustered the excitatory neurons, integrating HBCC and MSSM data (Fig. 1d, Supplementary fig. 1e-f). We found that most excitatory neurons in MD were part of a transcriptional gradient captured by the first principal component (PC1, Fig. 1e). Genes aligned with PC1 overlapped with and correlated in their loadings with those of a previously described trans-thalamic axis separating primary, secondary and tertiary thalamic cells in mice 8 (Supplementary fig. 1g-h). Thus, projection neurons in human MD and midline thalamus exhibit a gene expression gradient that closely resembles a cell-type gradient with distinct functional and morphological properties in mice. Human MD has traditionally been divided into a magnocellular (MDmc, medial), parvocellular (MDpc, lateral-ventral) and multiform/densocellular (MDdc, lateral-dorsal) part 16 . We hypothesized that these cytoarchitectonic subdivisions could correspond to transcriptionally defined neuronal

populations. "

"To map cell populations identified by snRNA-seq onto anatomical space, we applied spatial transcriptomics to two coronal thalamic sections, then deconvoluted the cell types, using our snRNA-seq data as a reference 17 . This revealed prominent spatial separation of neuronal populations. Some excitatory neuron clusters preferentially mapped to MDmc, MDpc, or adjacent nuclei, while inhibitory interneurons appeared to be particularly enriched in MDpc and MDdc (Fig. 2a; Supplementary fig. 2a).

Similarly, projection of the excitatory neuron PC1 from Fig. 1e onto the spatial transcriptomics data showed that the primary end of the gradient mapped to MD, especially MDmc, while the tertiary end mapped to PVT and PT (Supplementary fig. 2b). Among the surrounding nuclei, intralaminar nuclei were on the tertiary end, while motor/somatosensory and lateral dorsal thalamus were on the primary end, consistent with their profiles in mouse thalamus 8 . RNA-FISH confirmed this spatial distribution of marker genes associated with opposite ends of the gradient and showed that numerous cells co-expressed multiple markers (Fig. 2b, Supplementary fig. 3). These analyses suggest that the gene expression gradient of excitatory neurons corresponds to a spatial gradient of neurons between MD and PT/PVT. Having identified a cell-type gradient in human MD and PVT with major similarities to the thalamic cell-type gradient in mice, we sought to better understand the commonalities and differences in MD cellular composition between humans and mice. Since previous snRNA-seq from retrogradely labeled cells did not include enough neurons in MD 8 , we collected additional snRNA-seq from mouse MD based on anatomical dissections. As expected, this identified the same major cell classes as human MD, except that inhibitory neurons were not observed in mouse MD (Supplementary fig. 4). To directly compare thalamic projection neurons between species, we integrated human and mouse excitatory neurons (Fig. 3a, Supplementary fig."

"We tested for association with cell type specificity scores derived from our data using the top 10% most specific gene sets, as previously described 20 . Neuronal populations in the thalamus were significantly associated with genetic risk for severe mental illness: ExN_H2 (MDpc-enriched) was most strongly associated with genetic risk for schizophrenia and bipolar disorder, while thalamic inhibitory neurons and ExN_H3 (PVTenriched) were most strongly associated with genetic risk for major depressive disorder. GWAS for Alzheimer's disease and height, included for control purposes, implicated microglia and vascular cells, respectively, as expected. a. UMAP plot illustrating cross-species integration of human and mouse excitatory neurons in medial thalamus, colored by dataset: human data from HBCC and MSSM, and mouse data from this study based on anatomical dissections and a previous study using retrograde labeling from prefrontal cortex 8 . b. UMAP plot of integrated mouse and human data, with mouse cells labeled by their nuclei of origin inferred from integration with public datasets and Allen Brain Atlas ISH data (see also Supplementary fig. 4 We produced a cell type atlas of adult human MD and adjacent midline nuclei, spatially mapped the major neuronal populations, and compared mouse and human neuronal identities. Recent studies examined cell populations across the entire human brain 15 , fetal human thalamus 11 , and multiple structures within the mouse thalamus 8,13,14,21-23 and habenula 18,24 , but were not focused on the MD, and cross-species comparisons have been limited to visual thalamus 21 , making our study the first in-depth analysis of cell types in human MD.

We identified a major transcriptional gradient between MD and PVT, which closely aligns with a previously-described gradient from primary to tertiary thalamus and likely reflects distinct connectivity and functional properties of these neurons 8 . This gene expression gradient mapped onto anatomical space, and respective clusters were enriched in MDmc, MDpc, and adjacent nuclei. Whilst MD has historically been considered 'higher-order', we identify major primary cell populations within both mouse and human MD. Cross-species analyses show that primary cells in MD are substantially expanded in humans relative to mice."

"Human Brain Cell Atlas integration: Data from the medial thalamus and epithalamus regions of interest of the Human Brain Cell Atlas 15 were downloaded and integrated with the HBCC and MSSM data, respectively, using anchors defined via canonical correlation analysis (CCA) in Seurat (Supplementary fig. 1c). Nearest centroid classification of HBCC and MSSM cells by supercluster label was performed to map corresponding cell types (Supplementary fig. 1d). Brain bank integration: Excitatory neurons from HBCC and MSSM were integrated using CCAbased anchors in Seurat. SNN clustering with a resolution of 0.5 on the top 5 PCs was used to define the clusters. To alleviate clustering being driven by individual samples, the four most closely related clusters were merged, resulting in 10 clusters (Supplementary fig. 1e-f).

Mouse anatomical and retrograde labeling data integration: Mouse thalamic neurons obtained in this study were integrated with those based on retrograde labeling from prefrontal cortex in a previous study 8 using CCA-based anchors in Seurat. SNN clustering with a resolution of 0.8 on the top 10 PCs was used, resulting in 19 clusters (Supplementary fig. 4b). PVT data integration: Five populations previously annotated and validated in the mouse PVT 13 were integrated with the mouse thalamic neuron clusters using anchors based on reciprocal principal component analysis (RPCA) in Seurat. Nearest centroid classification was used to map the PVT clusters onto corresponding excitatory neuron clusters (Supplementary fig. 4c). Habenula data integration: Previously annotated and validated neuron populations from the mouse habenula 18 were re-identified and integrated with the mouse thalamic neurons using RPCA-based anchors. Nearest centroid classification was used to map the re-identified habenula populations onto the excitatory neuron clusters (Supplementary fig. 4d). All integrations were performed on the top 2,000 most variable genes across datasets (SelectIntegrationFeatures in Seurat). Nearest neighbor classifiers were implemented via the lolR library (lol.classify.nearestCentroid)"

(Trujillo et al., 2026)

Selective vulnerability and resilience of thalamic neuronal populations in Frontotemporal Dementia

"To define the neuronal populations underlying selective thalamic vulnerability, we subclustered thalamic neurons and identified four major neuronal populations composed of two excitatory and two inhibitory neuronal subtypes (Figure 2A-C). Cell-type annotation was guided by previously reported human thalamic transcriptional signatures (Bakken et al., 2021;Marsan et al., 2023;Mathys et al., 2024;Siletti et al., 2023), and refined using the large scale and cross-cohort structure of our integrated dataset. These neuronal populations were distinguished by robust transcriptional signatures, including pan-inhibitory expression of GAD2 and MEIS2 together with differential expression of SOX14, NR2F2, ERBB4, LAMP5, and BCL11B across inhibitory neuronal populations. While InNeu SOX14+ neurons correspond to midbrain-derived interneurons, the InNeu SOX14population is composed of ganglionic eminence-derived subtypes, encompassing medial ganglionic eminence (MGE; marked by LHX6, SST, and NPY) and caudal ganglionic eminence (CGE; marked by CXCL14, VIP, and LAMP5) lineages (Supplementary Figure 2A). In contrast, excitatory neuronal populations were defined by SLC17A6 expression, consistent with thalamic projection neuron identity, and differential RNF220 expression (Figure 2A,B). We therefore annotated these populations as InNeu SOX14+, InNeu SOX14-, ExNeu RNF220+, and ExNeu RNF220-(Figure 2C). Integration with human spatial transcriptomic datasets generated by Schulmann et al. (Schulmann et al., 2024) (Schulmann et al. under revision) showed that InNeu SOX14+ and ExNeu RNF220+ populations were present across several nuclei in the human medial thalamus (Figure 2D)."

Molecular markers linked to classic core, matrix, and projection styles

Several marker systems recur across thalamus, especially calcium-binding proteins such as PVALB, CALB1, and CALB2, but they do not map perfectly onto one universal classification. In mouse, these markers often reflect gradients, projection styles, or nucleus-biased subpopulations rather than fixed pan-thalamic classes.

A recurring molecular theme is the use of calcium-binding proteins and related markers to label projection styles.

- In mouse dLGN, **Pvalb** marks core-enriched principal neurons, while **Calb1** and **Necab1** mark shell-enriched neurons. (Bakken et al., 2021)
- Across the thalamus more broadly, **Pvalb**, **Tnnt1**, **Calb2**, and **Necab1** separate points along the major transcriptional axis, but cells can co-express markers and intermediate cells are common. (Phillips et al., 2018) (Phillips et al., 2017)
- Midline neurons with **calretinin/CR (Calb2)** form a physiologically distinct population independent of strict nuclear borders. (Lara-Vasquez et al., 2016)
- In primates, core/matrix is often described with **PV** and **CB**, and new mouse MD morphology work on **Calb1**-defined neurons argues that many are matrix-type because they project broadly to superficial cortical layers, but this is currently specific to one MD population rather than a full mouse-wide rule. (Zhao et al., 2026)

These data support a careful view: marker genes are useful, but they do not resolve all thalamic neuron types by themselves. A given marker can define a projection-biased subgroup in one nucleus, a gradient end in another, or a broader class only at coarse scale. (Roy et al., 2022) (Turner et al., 2025)

Evidence

(Bakken et al., 2021)

Single-cell and single-nucleus RNA-seq uncovers shared and distinct axes of variation in dorsal LGN neurons in mice, non-human primates, and humans

"Abundant evidence supports the presence of at least three distinct types of thalamocortical (TC) neurons in the primate dorsal lateral geniculate nucleus (dLGN) of the thalamus, the brain region that conveys visual information from the retina to the primary visual cortex (V1). Different types of TC neurons in mice, humans, and macaques have distinct morphologies, distinct connectivity patterns, and convey different aspects of visual information to the cortex. To investigate the molecular underpinnings of these cell types, and how these relate to differences in dLGN between human, macaque, and mice, we profiled gene expression in single nuclei and cells using RNA-sequencing. These efforts identified four distinct types of TC neurons in the primate dLGN: magnocellular (M) neurons, parvocellular (P) neurons, and two types of koniocellular (K) neurons. Despite extensively documented morphological and physiological differences between M and P neurons, we identified few genes with significant differential expression between transcriptomic cell types corresponding to these two neuronal populations. Likewise, the dominant feature of TC neurons of the adult mouse dLGN is high transcriptomic similarity, with an axis of heterogeneity that aligns with core vs. shell portions of mouse dLGN. Together, these data show that transcriptomic differences between principal cell types in the mature mammalian dLGN are subtle relative to the observed differences in morphology and cortical projection targets. Finally, alignment of transcriptome profiles across species highlights expanded diversity of GABAergic neurons in primate versus mouse dLGN and homologous types of TC

neurons in primates that are distinct from TC neurons in mouse."

"We used unsupervised clustering to define transcriptomic cell types in a well-studied part of the mammalian thalamus, dLGN, for human, macaque, and mouse. We examined the correspondence of excitatory TC neurons to previously described morphological, connectional, and physiological differences (Hendry and Reid, 2000;Krahe et al., 2011;Cruz-Martı´n et al., 2014) and identified celltype homologies across species.

We find that primate K neurons are transcriptomically clearly distinguishable from their M and P counterparts, consistent with previous results (reviewed in Hendry and Reid, 2000). Moreover, we identify two transcriptomic types of K neurons that may correspond to two populations reported in macaque that have distinct laminar distributions in dLGN and projection patterns in V1 (Casagrande et al., 2007). We also find that K neurons are more transcriptomically similar to inferior pulvinar neurons than to M or P neurons. This is consistent with their shared inputs from retina and superior colliculus and cortical projection targets (Huo et al., 2019) and supports a close functional relationship between K neurons and inferior pulvinar. M and P neurons are transcriptomically similar to each other in both macaque and human.

In contrast to primate, TC neurons in mouse dLGN cannot be grouped into discrete types based on gene expression, although many genes show graded expression differences between core and shell regions. Similar graded expression heterogeneity has been reported within and across many first-and higher-order thalamic nuclei (Phillips et al., 2019). Alignment of primate and mouse TC neurons revealed cell-type homologies between pulvinar and LP neurons, as expected. Mouse shell neurons aligned more with primate K neurons and core neurons aligned with M and P neurons, and some marker genes were conserved across species. Genes for calcium binding proteins were expressed in opposing gradients in TC neurons across mouse dLGN: Pvalb-expressing neurons were enriched in the core (and primate M and P neurons), Calb1-and Necab1-expressing neurons were enriched in the shell (and K neurons)."

(Phillips et al., 2018)

A single spectrum of neuronal identities across thalamus

"A. Schematic of experimental pipeline to obtain transcriptomic atlas of the thalamus. In this example, motor thalamic neurons were retrogradely labelled from their primary projection target (motor cortex), manually dissected and sorted. Viruses expressing green and blue fluorescent proteins (GFP, BFP), respectively, were injected to the deep cerebellar nuclei (DCN) and substantia nigra pars reticulata (SNr) to label motor nuclear subdivisions (ventral lateral (VL) and ventral anterior (VA), respectively) previously identified 8,9 . B. Example labelling from scheme shown in A. Coronal sections. Scale bars = 200 μm. C. Hierarchical clustering of thalamic nuclei using Spearman's correlation of top 500 most differentially expressed genes (Extended Data Fig. 1B) across all 22 nuclei. Major clusters defined as the top 5 branches of cluster dendrogram. D. Topographic localization of gene expression profiles in thalamus. Coronal thalamic section schematics with nuclei colored as in Fig. 1C. Unsampled nuclei are left uncolored. (see Fig. 1B). To understand the pattern of gene expression differences between the thalamic projection profiles, we performed principal component analysis (PCA) on our data. Principal component 1 (PC1, 38% explained variance) separated nuclei into the same major profiles identified via hierarchical clustering (Fig. 2A, Extended Data Fig. 3A). Again, position on this first dimension strongly correlated with mediolateral position, demonstrating topogenetic architecture in thalamus (Fig. 2B, Extended Data Fig. 3B). Based on their relative order on this first component, we named the three major profiles primary, secondary, and tertiary. The progressive difference from primary to tertiary nuclei was also evident in the number of genes differentially expressed between the groups, with the primary and tertiary nuclei being most distinct, and the other two comparisons being less so (Fig. 2C). This primary axis was prominently enriched in genes encoding neurotransmitter receptors, ion channels, and signaling molecules (Fig. 2D and E)."

"A. Overview of single-cell clusters within each projection system visualized via tSNE. Cells colored by cluster identity. B. Violin plots for cluster marker genes (inclusion criteria were Likelihood Ratio Test Pvalue < 10 -5 and log2 fold change > 0.5 for each cluster). C. Projection of single-cell data onto pooled-cell PC1 from Fig. 2A. Each dot is a single cell colored as in A. D. Topographic distribution of marker genes within 6 major thalamic modalities. Multi-FISH with probes for Calb2 (pink), Tnnt1 (red) and Necab1 (green). See Extended Data Fig. 7 and 8 for expanded views and quantification. Scale bars = 200 μm. E. Multi-FISH within VL thalamus comparing genes marking clusters in single-cell data.

Left panel shows field of view (coronal section, scale bar = 100 μm). Right three panels (scale bars = 50μm) show expansion of the three boxed areas, moving left to right.

Middle box shows intermediate cells expressing both single cell cluster markers. (red = Tnnt1, green = Pvalb, blue = DAPI). Thus far we analyzed thalamic nuclei by pooling projection neurons from specific anatomical positions. The resulting transcriptional profiles could represent homogenous populations, discrete subtypes, or cells with graded differences. To probe these possibilities, we profiled the transcriptomes of individual neurons from motor, somatosensory, visual, auditory, and prefrontal projection classes (Extended Data Table 3). Analysis of this single-cell RNAseq dataset resulted in multiple clusters for each projection class, and cluster markers included many genes that also distinguished nuclei (Fig. 4B). Single-cell clusters also separated along the first principal component derived from our pooled-cell RNAseq dataset (Fig. 4C, Extended Data Fig. 6). "

"Markers for the single-cell clusters were spatially separated at the single-cell level, but with intermediate cells especially prominent near the anatomical boundaries of thalamic nuclei (Fig. 4D, Extended Data Fig. 7). This is consistent with spatially organized

heterogeneity or a gradient-like organization rather than intermingled, discrete populations 15,16 . Given the strong relationship between PC1 and anatomical nuclei position (Fig. 2C and Extended Data Fig. 3), the presence of single-cell clusters occupying similar PC1 positions (e.g. clusters 1 and 2 of the motor projections neurons, or clusters 1 and 2 of the somatosensory nuclei, Extended Data Fig. 6) suggested that distinct neuron types could also coexist within anatomical boundaries of nuclei. We examined this possibility by performing multi-color fluorescent in situ hybridization (multi-FISH) for genes which distinguished amongst the major profiles (e.g. Pvalb and Tnnt1). Taking motor thalamus as an example, Pvalb and Tnnt1 expressing cells were found within the anatomical boundaries of a single thalamic nucleus (VL; Fig. 4E). Some individual VL cells expressed both Pvalb and Tnnt1; Pvalb-selective, intermediate, and Tnnt1-selective cells were distributed along the mediolateral axis (Fig. 4E). Therefore, spatially organized transcriptomic differences can exist even within individual thalamic nuclei. A common organizing principle of thalamus divides nuclei into discrete core or matrix subtypes that span modalities 1,7 . To date the strongest evidence for cross-modal organization is that genetic differences can be larger within a sensory modality than between sensory modalities 17 (Extended Data Fig. 9). However, previous studies have focused either on select sensory systems and/or have had limited scope with respect to the molecules investigated 18 . Here, using the full transcriptomes of nearly all thalamic pathways, we confirm cross-modal organization but replace the core/matrix dichotomy with a spectrum of profiles that span a single axis of genetic variance."

(Phillips et al., 2017)

A topographic axis of transcriptional identity in thalamus

"A fundamental goal in neuroscience is to uncover common principles by which different modalities of information are processed. In the mammalian brain, thalamus acts as the essential hub for forebrain circuits handling inputs from sensory, motor, limbic, and cognitive pathways. Whether thalamus imposes common transformations on each of these modalities is unknown. Molecular characterization offers a principled approach to revealing the organization of thalamus. Using near-comprehensive and projection-specific transcriptomic sequencing, we found that almost all thalamic nuclei fit into one of three profiles. These profiles lie on a single axis of genetic variance which is aligned with the mediolateral spatial axis of thalamus. Genes defining this axis of variance include receptors and ion channels, providing a systematic diversification of input/output transformations across the topography of thalamus. Single cell transcriptional profiling revealed graded heterogeneity within individual thalamic nuclei, demonstrating that a spectrum of cell types and potentially diverse input/output transforms exist within a given thalamic nucleus. Together, our data argue for an archetypal organization of pathways serving diverse input modalities, and provides a comprehensive organizational scheme for thalamus."

"We produced a near-comprehensive transcriptomic atlas of murine thalamus via microdissection of retrogradely labeled cells and subsequent RNAseq analysis (22 nuclei, 120 samples). Anterograde tracing of inputs to thalamus was used when identification of nuclear boundaries was ambiguous (Figure 1A,B for an example dissection approach). Labeled neurons from a given thalamic nucleus were pooled and profiled, an approach chosen because nuclear identity and projection target are major definers of cell type. See supplementary tables 1 and 2 for details on labelling strategies.

To identify the relationship between the transcriptomes of thalamic nuclei in an unsupervised manner, we selected the top 500 most differentially expressed genes between thalamic nuclei (via an ANOVA-like test, see methods). Using this differentially expressed gene set, we grouped the thalamic nuclei via hierarchical clustering (1 -Spearman's correlation as distance metric), identifying five major subdivisions of nuclei (Figure 1C). We termed these profiles, not clusters, as we do not mean to imply discreteness between the classifications (see below). These profiles are not explained by the modality of information processed, since subnuclei of motor, somatosensory and visual complexes all split across different profiles. For example, the blue profile (Figure 1C) included both the cerebellar-recipient motor thalamus and the retina-recipient visual nucleus (lateral geniculate, LGd), whilst the red profile contained the basal ganglia recipient motor thalamus and the superior colliculus-recipient visual nucleus (lateral posterior, LP). Interestingly, the anterior dorsal nucleus (AD) and nucleus reuniens (RE) had distinct profiles and each formed profiles of their own. When schematic sections of thalamus were colored by these profiles, a mediolateral separation was observed, with the 'blue' nuclei being most lateral and 'yellow' nuclei most medial (Figure 1D). Thus, the largest fraction of linear gene expression variance in thalamus is not explained by modality, but rather a topographic axis spanning the structure's mediolateral extent."

"A. Overview of clusters (identified using the Seurat package, see methods) within each modality visualized using tSNE, cells colored by cluster identity. B. Violin plots for top marker genes for each cluster (Likelihood Ratio Test, p-value<10^-5, logFC>0.5, top 4-5 signal-to-noise ratio). C. Projection of single-cell data onto bulk principal component 1 (as in figure 2 Thus far we analyzed thalamic nuclei by pooling projection neurons from their anatomical position. The resulting transcriptional profiles could represent homogenous populations, discrete subtypes, or cells with graded differences. To probe these possibilities, we profiled the transcriptomes of individual neurons from motor, somatosensory, auditory, and prefrontal projection classes (supplementary table 3). Analysis of the single-cell RNAseq dataset resulted in multiple clusters for each projection class which, when projected onto the first principal component of our bulk RNAseq dataset, spanned the same axis of variance identified from bulk data (Figure 4 A,B,C, supplementary figure 4). Markers for this axis were spatially separated at the single-cell level, with clear intermediate cells present (Figure 4D, supplementary figure 5,6,7). However, some individual single-cell clusters had similar PC1 positions (e.g. clusters 1 and 2 of the motor projections neurons, or clusters 1 and 2 of the somatosensory nuclei, supplementary figure 4). This suggests that distinct clusters within a single modality could exist in similar anatomical positions. We examined this possibility by performing RNAscope for genes which distinguished these clusters (Pvalb and Tnnt1) . Using motor thalamus as an example, Pvalb and Tnnt1 expression were both found within the confines of a single thalamic nucleus, VL. While neurons selectively expressing each gene could be found, intermediate cells were also apparent. Pvalb-selective, intermediate, and Tnnt1-selective cells were distributed along

the mediolateral axis (Figure 4E). "

"Therefore spatially organized genetic gradients can exist even within individual thalamic nuclei.

In this study we produced a near-comprehensive transcriptomic atlas of thalamus via RNAseq in order to investigate its organizational structure. Though input modality has historically defined thalamic nuclei, several alternative, overlapping organizational classifications have been proposed 1,6 . Early work perturbing thalamus showed that nuclei of the same modality can have strikingly different contributions to behavior, leading to a division between arousal-linked intralaminar nuclei and the rest of thalamus [7][8][9] . More recently, theories developed in the sensory thalamus have subdivided thalamic nuclei into two groups, based upon anatomical or neurochemical features [10][11][12] . However, precise delineation of groups is not consistent between theories, and no theory is comprehensive. If and how these later theories apply beyond sensory thalamus, for example, is unknown.

Here, we reconcile and extend these theories by placing almost all thalamic nuclei into one of three profiles, lying on a single topographic axis identified via unsupervised analysis. This axis extends from the midline/intralaminar groups through to a profile that includes the classic 'relay' sensory nuclei. Though prior theories did not incorporate electrophysiological properties, we found that genes directly affecting these properties were highly differentially expressed between groups, implying conserved, systematic variation of response properties across thalamus.

Understanding how the pattern of intra-thalamic variability identified here is used to support different circuit functions will be an important challenge 13 .

Prior theories of thalamic organization have placed thalamic cells into discrete nuclei, and thalamic nuclei into discrete groups. However, it remains unclear if subtypes of cells exist within nuclei and how discrete vs. continuous the cells type differences are. Recent studies in other non-thalamic areas have argued that gradients of projection neuron identities can exist 14,15 . Our single-cell transcriptomic data revealed heterogeneity within thalamic nuclei and suggest that thalamic genetic profiles exist on a spectrum, with mixtures of genetic profiles often existing in single neurons. Intermediate cell types exist both within and between nuclei, and are topographically localised."

(Lara-Vasquez et al., 2016)

Midline thalamic neurons are differentially engaged during hippocampus network oscillations

"Calretinin and calbindin expression in midline thalamic neurons. We sampled the action potential firing of individual neurons from thalamic nuclei across the midline (Table 1). After recording, neurons were individually labeled with the juxtacellular filling method 38 and classified by their expression of neurochemical markers (Supplementary Table 1). In order to map the location of recorded neurons on the thalamus, we projected all labelled neurons on the neuroanatomical atlas of the mouse brain 39 . Neurons were recovered from most midline thalamic nuclei (Fig. 1a). The somatodendritic distributions of midline thalamic neurons were relatively uniform, with no obvious distinct morphological types. However, we could identify cells conforming to the distinct branching patterns of dendrites described in thalamic relay cells, such as "bushy" and "radiate" neurons 1,40 . Indeed, some cells had more angular and multipolar cell bodies, with dendritic trees branching dichotomously, characteristic of "radiate" cells (Fig. 1c). Conversely, other cells had elongated somata, with dendrites longer and branching less frequently, particularly in regions proximal to the soma, and followed a "bushy" organization (Fig. 1c). We were not able to fill the entire extension of axonal arborizations, even in cases where intense somatic labeling was achieved. In fact, most labelled cells did not show evidence of axonal labelling. When labelled, axons arose in all cases from the soma and rarely branched locally (Supplementary Figure 1). Given the above mentioned dendritic branching patterns, as well as axonal branching patterns, with little or no local collaterals, we considered our sample to be entirely comprised of thalamic projection neurons, which are known to be glutamatergic and typically reach the cortical mantle 1 ."

"By recording and labeling individual neurons in anesthetized mice, we show here evidence for functionally distinct neuronal populations in the midline thalamus. Indeed, our anatomical and physiological characterization leads us to propose that at least two different populations of midline thalamic projection cell deliver differential synaptic output to cortical targets according to ongoing brain states as defined by hippocampal activity. Our data show that rather than anatomical location, a major defining feature in the physiological properties of midline thalamic neurons is the expression of CR, a calcium-binding protein. Hence, distinct thalamic neuronal populations, identified by the somatic expression of CR, are differentially regulated by hippocampal pathways during network rhythms associated to behavioural contexts, likely supporting different stages of memory processing. Midline thalamic nuclei give rise to large glutamatergic axon terminals that densely distribute within frontal and temporal cortical lobes 27,45 . It has been reported that a dense fiber plexus located in the entorhinal cortex contains both CR and vesicular glutamate transporter 2 in the same axons. Complementary retrograde tracing experiments found labeled cell bodies in the nucleus reuniens co-expressing CR 46 . Our results show that CR+ cells are widely expressed across the midline thalamus, and given their anatomical and physiological features suggest that they are likely to be glutamatergic projection cells 1 . Cortical projections from the midline thalamus are not homogeneous across nuclei. Indeed, neurons in dorsal structures preferentially target the medial prefrontal cortex, with little input to the medial temporal lobe 5,24 ; neurons in middle regions, project almost exclusively to the medial prefrontal cortex 24,47 ; whereas neurons in ventral areas, project mostly to the parahippocampal cortex, with less significant projections to the medial prefrontal cortex 24,27 . Thus, the anatomical organization of synaptic output from the midline thalamus is not homogeneous, with every nucleus displaying distinct efferent connectivity patterns. Interestingly, our results suggest that a major feature in defining the activity pattern of thalamic neurons is not their nuclear location, but the expression profile of calcium-binding proteins, particularly

CR."

(Zhao et al., 2026)

Projectomic profiles of calbindin D28K-expressing neurons in mediodorsal thalamic nucleus of mouse.

"Expression of calbindin D28k (CB) and parvalbumin (PV) distinguishes matrix and core type thalamic neurons, respectively, in primates. Whether the nice correspondence of intracellular calcium binding proteins and matrix/core neuron types translates to mouse remains unknown. Sixty reconstructed 3D mediodorsal thalamic nuclear neurons (MDCalb1 neurons), defined by calb1, the gene encoding CB in mouse MD, were mapped onto Allen Mouse Brain Common Coordinate Framework (CCFv3). Most MDCalb1 neurons (54/60) were isocortex-projecting ones, possessing collateral connections with caudate putamen (CP). Collectively, MDCalb1 neuron ensemble linked 93 anatomically-defined brain structures/regions with their axons; and of these 93 axon-connected structures/regions, the single MDCalb1 neurons had projection areas (receiving ≥ 2 axon terminals) ranging from 3 to 34 in number. Distinct projectomic profiles defined roughly 8 neuron subgroups, which differed from each other in composition of axon targets involving cortical and subcortical regions. Isocortical areas in prefrontal and sensorimotor cortices such as agranular insular cortex (AI), secondary motor cortex (MOs) and orbital cortex (ORB) etc., as well as structures of striatopallidal and olfactory systems such as CP, nucleus accumbens (ACB) and main olfactory bulb (MOB), among others, constituted the main axonal projection targets. Intriguingly, some proximal axon branches from over a dozen MDCalb1 neurons (14/60) were seen to be spatially confined to MD, most frequently within dendritic territories of their source neurons, suggesting intra-MD innervation. For thalamocortical connectivity, most neurons (49/54, i.e., about 91%) had widespread, tangentially distributed axons terminating in layers 1-3 (L1-3) of multiple cortical areas. Thus, these neurons were of thalamic matrix type, per core-matrix-intralaminar classifying framework. Our morphometric data support the role of MDCalb1 neurons for coordinated signal processing in networks integrating broad co"

(Roy et al., 2022)

Thalamic subnetworks as units of function

"To appreciate the levels of organization discussed in this Review, we need to explicitly define the three key terms that we use throughout this paper. We refer to the term 'cell type' as a feature that is identified via a singular approach. Within individual thalamic nuclei, cell types have traditionally been defined on the basis of morphology, neurochemistry or physiology. For example, morphology can identify neurons based on differences in soma size and dendritic branching patterns 1,3 , whereas neurochemistry indicates that cell types can be either excitatory or inhibitory (for example, by the expression of enzymes for the synthesis and packaging of glutamate or GABA) 3 . Physiology, on the other hand, can identify neurons using distinct sensory response patterns (for example, in the LGN) 2 . The advent of modern genetic profiling has enabled the identification of 'subpopulations': groups of cells that share a particular pattern of gene expression, morphology, physiology and fine-grained connectivity. In this Review, we define one or more subpopulations serving a particular function as a 'subnetwork' (Fig. 2). Because each neuron or subpopulation often has multiple inputs and outputs, using the same subpopulation in different functional subnetworks is likely to be a common and efficient approach for functional circuit assembly. For instance, subpopulation A may form a subnetwork with subpopulation B for a given function, but forms a subnetwork with subpopulation X to serve a different function. Admittedly, this definition is forward-looking and will require future experiments to continue to validate and refine it.

One example of a putative thalamic subnetwork came from examining neural activity patterns within the TRN 15 , which, up to that point, had been considered a functional monolith 16 . A combination of electrophysiological recordings and connectivity-based optical tagging revealed the existence of subpopulations that share activity patterns depending on the principal thalamic structures to which they project 15 . More recently, state-of-the-art single-cell RNA sequencing (scRNA-seq) has indicated that these 'TRN subnetworks' could be further specified using gene expression gradients 17 subpopulations in several brain regions, such as"

"brainstem, histaminergic afferents from the hypothalamus and noradrenergic afferents from the parabrachial region have heterogeneous terminals in the thalamus 1,2 .

Physiology. Building on these findings, researchers have consistently observed electrophysiological differences in thalamic nuclei.

Whereas thalamic principal cells are thought to have comparable intrinsic properties across sensory nuclei, there is substantial heterogeneity in their action potential parameters [39][40][41][42][43] . Recently, whole-cell recordings from three motor-related thalamic nuclei (central medial (CM), ventral anterior (VA) and ventral lateral (VL) nuclei) showed that electrophysiological membrane and synaptic properties varied along a gradient from CM to VA to VL nuclei 44 . Whereas previous studies revealed electrophysiological heterogeneity of thalamic neurons, this study 44 found that single-cell physiological properties across thalamic nuclei exist along a continuum rather than forming unique, discrete profiles (Fig. 3b). However, the functional implications of this observation have not been revealed. This study also linked these physiological observations to graded transcriptional profiles (Fig. 3c) and morphological differences, suggesting that cellular morphology, terminal arborization, input type and electrophysiological properties are effective approaches for characterizing thalamic cellular diversity. Regarding the fundamental intrinsic feature of thalamic principal neurons 45-47 their two distinct firing modes (tonic versus bursting)-heterogeneity in these modes has been associated with somatodendritic morphological differences 27,48,49 .

Neurotransmitters and receptors. Finally, differences in the localization of neurotransmitters and their receptors [50][51][52][53][54]

[55][56][57] led to the idea that variation in gene expression profiles may serve as an organizing principle across thalamic nuclei. One group identified four transcription factor genes that showed differential expression within excitatory thalamic nuclei 58 that these candidate genes might reflect distinct subpopulations that are shared across excitatory or inhibitory nuclei, many groups attempted to identify thalamic nucleus-and subpopulation-specific molecular markers ["

"Although studies that have identified subnetworks within the thalamus up to now have used rodents, there is substantial evidence to suggest that heterogeneity among thalamic neurons is conserved from rodents to primates. Here, we briefly discuss several primate studies that support the idea of diverse thalamic subpopulations or show that some cellular properties of primate thalamic neurons are comparable to those in rodents. Neurotransmitter expression patterns are usually conserved across species; for instance, nicotinic binding sites are enriched in the LGN, the MD nucleus and the rhomboid nucleus in mice and monkeys 52 . Similarly, a study examined the expression of genes in the monkey thalamus that were enriched in different mouse thalamic nuclei and found highly consistent patterns 63 . Several genes were expressed in excitatory nuclei in monkeys but not in rodents. A study using high-density oligonucleotide arrays to identify thalamic nuclei-specific gene expression in adult monkeys 120 showed that over 550 genes were selectively expressed in the anterior thalamus or the CM, MD or ventral posterior nucleus. This study identified a marker for excitatory nuclei (coding for transcription factor 7-like 2), a marker for excitatory nuclei but not the CM nucleus or the PF (coding for Purkinje cell protein 4), a CM-PF-specific marker (coding for cerebellin 1 precursor) and an

LGN marker (Spp1). These markers have comparable expression patterns in mice 69 . A study that showed distinct subpopulations labeled by PV or somatostatin (SST) in the mouse TRN showed that these two subpopulations are also found in human TRN samples 87 . Additionally, in primates, SST-expressing neurons exhibit differential distribution in the visual sector of the TRN 121 . As in rodents, the LGN in monkeys has distinct subpopulations based on single-cell electrophysiological properties, spatial localization and inputs from distinct RGC subtypes 122,123 .

Tracing studies have also revealed subpopulations within thalamic nuclei in monkeys 109,124,125 . For example, the PF in monkeys contains projection-specific subpopulations 90 , including neurons that project to striatal regions and neurons that project"

(Turner et al., 2025)

Exploring the correspondence between gene expression and thalamic nuclei using the THALMANAC resource

"The thalamus connects the sensory organs and major subcortical brain regions with the neocortex. The thalamus has long been divided into multiple discrete nuclei, based on cytoarchitecture, histochemical stains, and mesoscale connectivity. However, thalamic nuclei do not completely describe thalamic organization. For example, some boundaries between thalamic nuclei are disputed, whereas other nuclei are known to contain subdomains with distinct connectivity and function. Moreover, the correspondence between cellular gene expression and other properties of thalamic projection neurons remains to be established. Spatial analysis of single cell gene expression provides a basis for reevaluating thalamic organization. We present the THALMANAC, (THALamusMERFISH ANalysis and ACcess) a Findable, Accessible, Interoperable, Reusable and Reproducible (FAIRR) resource for exploring and analyzing single-cell transcriptomic variation in the thalamus. The THALMANAC provides streamlined access to thalamic gene expression data registered to the common coordinate framework and tools for quantitative analysis and visualization of these data, all encapsulated in a reproducible, cloud computing platform. Using this resource, we find that gene expression generally supports the parcellation of thalamus into distinct nuclei. Some nuclei, such as the anteromedial nucleus, are additionally composed of discrete subdomains, while other nuclei share patterns of gene expression or are arrayed on a spatial gradient of gene expression. The THALMANAC establishes spatial transcriptomic data as a foundation for delineating thalamic organization. Graphical Abstract"

"This fine-scale structure could correspond to discrete, molecularly defined subdivisions of VM that have not yet been annotated, or to topographic organization of thalamocortical and corticothalamic connectivity.

A major question is how gene expression relates to existing parcellations and thalamic organization in general. Single cell RNA sequencing (scRNA-seq) provides detailed measurements of transcription in large populations of individual neurons (Tasic et al. 2016), which can be used to define transcriptomic cell types ('t-types'). The combination of scRNA-seq with spatial transcriptomics has been used to localize t-types across the whole mouse brain (Langlieb et al. 2023;Yao et al. 2023;Zhang et al. 2023). With t-types mapped into the CCFv3 coordinate space (Wang et al. 2020), we can evaluate how their spatial distributions correspond to anatomical parcellations (Yao et al. 2023).

Here, building on the Allen Brain Cell (ABC) Atlas scRNA-seq and spatial transcriptomics data, we developed a resource, the THALamus MERFISH ANalysis and ACcess (THALMANAC), which provides access to spatial gene expression data in the mouse thalamus; targeted analysis and data visualization of gene expression patterns; a platform for reproducible computations; and a no-code visualization app for data exploration. We demonstrate the utility of the THALMANAC by addressing how thalamic structural organization relates to gene expression. Aligning spatial transcriptomic data and nuclei boundaries reveals a wide range of relationships between t-types and thalamic nuclei: some t-types correspond to a single thalamic nucleus; others subdivide nuclei into multiple parts; many t-types span multiple thalamic nuclei.

Analysis of expression by itself supports these same conclusions: some nuclei boundaries are associated with sharp changes in gene expression; other nuclei contain gene expression gradients; and, in many cases, gene expression patterns span multiple nuclei.

Understanding how information is routed through the thalamus requires integrating insights gained from gene expression with other modalities, including connectivity and neurophysiology."

"The THALMANAC resource provides access to thalamic spatial gene expression data, visualization of the relationships between gene expression, t-types, and CCFv3 anatomical parcellations, and quantitative analysis of targeted quantitative analyses, all encapsulated in a cloud platform that enables reproducible computations and customized exploration of the data.

The THALMANAC is built on a thalamus-specific subset of spatial transcriptomics data from multiplexed error-robust fluorescence in situ hybridization (MERFISH) from the ABC Atlas. The ABC Atlas dataset measures 500 transcripts, chosen to distinguish t-types across the whole mouse brain, in spatially resolved, single cells. MERFISH cells were registered to the CCFv3 coordinate space and mapped to the scRNA-seq data by assigning each MERFISH cell to the label of the most similar t-type cluster. We identified neurons that belong to the thalamus (including the zona incerta and habenula) according to their t-type label (Figure S1A) and CCFv3 parcellation label (Figure S1B). The thalamic dataset analyzed here includes 80,170 MERFISH cells across 14 coronal sections collected from a single male mouse (Methods).

The THALMANAC includes three avenues for engaging with the t-type and gene expression data and accompanying analyses (Figure 1). First, a Streamlit web app (RRID:SCR_024354) provides nocode exploration, including interactive visualizations and on-the-fly di"erential gene expression analysis for thalamic nuclei (Interactive Streamlit App: thalmanac.allenneuraldynamics.org). Second, Jupyter notebooks (RRID:SCR_018315) provide step-by-step demonstrations of the Python code needed to reproduce all results presented here (github.com/AllenNeuralDynamics/thalmanac). Both tools allow, users to customize the thalamic cells that are selected from the full ABC Atlas."

"In the ABC Atlas, transcriptomic variation is used to assign cells to a hierarchy of discrete t-types (from broad to narrow categories: class, subclass, supertype, cluster). Thalamic neurons belong to 5 classes, 58 subclasses, 163 supertypes, and 563 clusters (Table S1; Figure S2) (taxonomy version CCN20230722) (Yao et al. 2023), but these numbers will change as the taxonomy evolves. Whereas the cluster level is based on unsupervised clustering, higher levels of the taxonomy are additionally informed by developmental origin via shared transcription factor expression, anatomical location, and classical (pre-scRNAseq) neuron types (DeFelipe et al. 2013) defined by key molecular markers (e.g. neuropeptides, calcium-binding proteins).

Transcriptomic classes typically correspond to major brain structures. Glutamatergic neurons in thalamus proper form a single class (class 18 TH Glut), whereas habenula neurons belong to a distinct class (class 17 MH-LH Glut) (Figure S2B). The other three classes are not unique to the thalamus: class 12 HY GABA is found in the reticular thalamus, zona incerta and ventral lateral geniculate nucleus (CCFv3 annotations; RT, ZI, and LGv, respectively, see Table S1) but also in the hypothalamus; class 19 MB Glut is found in thalamic nucleus SPA but also in the midbrain; class 20 MB GABA is found in thalamic nuclei LGv and SPF and the midbrain. Transcriptomic subclasses span multiple thalamic nuclei(Figure S2C), whereas transcriptomic supertypes often define individual nuclei (e.g. AM), and several nuclei contain multiple supertypes (e.g. ZI) (Figure S2D). Most nuclei contain multiple transcriptomic clusters (Figure 2A, Figure S2E).

We explored correspondence between clusters and thalamic nuclei."

"A comprehensive understanding of how the thalamus routes information to the cortex will require multiple types of data and a community-wide e"ort. Our work contributes to this program by curating and analyzing the correspondence between state-of-the-art spatially resolved, single-cell transcriptomic data, and widely used parcellations of thalamus into discrete nuclei. In addition, we make our data and analyses available in an accessible, documented, and reproducible bioinformatics resource in the cloud. Using the AM nucleus and the somatosensory nuclei as examples, we illustrate how this interactive resource can be used to identify novel patterns of gene expression and how these patterns correspond to prior parcellations of the thalamus.

The THALMANAC resource streamlines researchers' access to thalamic spatial transcriptomic data. We anticipate that scientists will extend our analysis based on the ABC atlas data, or apply similar analyses to additional spatial transcriptomics datasets registered to the CCF coordinate system (Langlieb et al. 2023;Zhang et al. 2023). The tool should also be used as a reference to validate recent studies focused on specific nuclei, such as the spatially segregated projections from PF to distinct dorsal striatum regions (Mandelbaum et al. 2019) and spatially segregated inputs to PO (Casas-Torremocha et al. 2022), the molecular organization of the PVT (Gao et al. 2023), and the core/shell organization of neurons in RT (Li et al. 2020). Our interactive Streamlit app can be used to identify marker genes for post-hoc localization of neurophysiological and neuroanatomical experiments, or to select gene panels for multiplexed mFISH experiments. The THALMANAC is already being used in the e"ort to create and characterize enhancerbased genetic tools (Ben-Simon et al. 2025) to target thalamic cell types and regions.

The THALMANAC shows that at the scale of thalamic nuclei, the molecular organization of the thalamus resembles existing parcellations. "

"However, both discrete cell type composition and gene expression gradients vary at finer spatial scales within and across nuclei. Given the evidence for circuit and functional variation at the same length scales, additional data need to be integrated with molecular characterization to understand thalamic organization. We see two promising paths forward for data integration. First,

existing data registered to the CCFv3 3D coordinate space, such as the databases of single-neuron morphological reconstruction (Winnubst et al. 2019) or neurophysiology data (Chen et al. 2024), could be linked to cell types and gene expression described in the THALMANAC. Second, targeted multimodal datasets, such as single-neuron morphological reconstructions combined with posthoc mFISH for transcriptomic cell typing would link cell types and gene expression to projection targets at the single cell level.

Beyond data integration, we also recognize the need for thalamus-focused, community annotation, similar to previous e"orts to annotate the fly connectome (Schlegel et al. 2024). We envision a refined parcellation of the thalamus where scientists with specific domain knowledge would combine molecular information from the THALMANAC with neuroanatomical and functional data to reflect progress in understanding of thalamic organization. Diagram illustrating the three workflows that can be launched from the Code Ocean Collection accompanying this paper. The Code Ocean platform provides built-in support for code versioning (via Git), environment replicability (via Docker), data and analysis provenance (via tracked data assets and linked pipeline runs), cloud compute resources (via AWS EC2 instances). Users can launch an interactive, no-code Streamlit application to explore spatial distributions of transcriptomic cell types and gene expression from their browser, as well as visualize locations of annotated cell types for each thalamic nucleus and highlight a specific nucleus or group of related thalamic nuclei. Tutorial Jupyter notebooks and Python code replicate all analyses and visualizations in a cloud workstation. These visualizations can be customized and extended. Users can download an AnnData-formatted .h5ad"

Developmental origin of adult thalamic neuron types

Adult thalamic neuron diversity begins early, before mature nuclei are fully visible. Developmental single-cell studies support separate lineage routes for many lateral glutamatergic nuclei, more medial glutamatergic nuclei, and rostral-thalamic GABAergic nuclei.

Developmental data help explain adult neuron types and why some adult molecular classes cut across nuclei.

Govek et al. found that molecular signatures of thalamic neuronal subtypes are already distinguishable by **E12.5**, before neurons aggregate into histologically clear nuclei. Their data support an outside-in neurogenesis model and three broad lineage streams: **cTh.Pro1** progenitors give rise to glutamatergic neurons in many lateral sensory, motor, and anterior nuclei; **rTh.Pro** progenitors give rise to GABAergic neurons in **VLG/LGv** and **IGL**; and **cTh.Pro2** progenitors give rise to neurons in more dorsomedial nuclei such as **PV, MD, and CM**. (Govek et al., 2022)

Older developmental work had already divided the embryonic thalamus into major progenitor domains such as **pTH-C** and **pTH-R**, and argued that combinations of transcription factors likely foreshadow mature nuclei better than single genes do. (Price et al., 2012)

Human developmental single-cell and MERFISH data show two major glutamatergic neuronal subtypes and several GABAergic subtypes in developing thalamus, with a medial-lateral maturity gradient and substantial inhibitory-cell expansion relative to mouse. Although this is human work, it supports the broader point that transmitter class, lineage, and spatial gradients are basic organizing features of thalamic neuron identity. (Kim et al., 2023)

Evidence

(Govek et al., 2022)

Developmental trajectories of thalamic progenitors revealed by single-cell transcriptome profiling and Shh perturbation

"Our study provides a comprehensive transcriptome-wide analysis of thalamic progenitors and their trajectories into thalamic nuclei during embryonic stages of brain development. We demonstrate that molecular signatures of thalamic neuronal subtypes can be readily distinguished as early as E12.5, prior to their aggregation into histologically distinct thalamic nuclei. 26 These data lend support to the outside-in model of thalamic neurogenesis, whereby early-born neurons contribute to lateral thalamic nuclei and later-born neurons contribute to medial thalamic nuclei. 36,58 Our work further extends this model with molecular insights into the mechanisms by which diverse thalamic nuclei acquire their identities. By following single-cell trajectories over developmental time, our data are consistent with a model in which thalamic nuclei arise from three distinct cellular lineages. In this model, Shh-responsive cTh.Pro1 progenitors give rise to glutamatergic neurons (cTh.N1) within sensory (DLG, VPM/VPL, vMG), motor (VA/VL, VM), and anterior (AD, AV, AM) nuclei that occupy predominantly ventrolateral regions of the thalamus. Shh-responsive rTh.Pro progenitors give rise to GABAergic neurons (rTh.N) in VLG and IGL thalamic nuclei, in agreement with previous findings. 12,13 In contrast, Shh-independent cTh.Pro2 progenitors give rise to neurons (cTh.N2) within nuclei (e.g., PV, MD, CM) that form at more

dorsomedial positions of the thalamus.

The bifurcation of transcriptomic trajectories according to thalamic progenitor identity is in general agreement with results from in vivo clonal analyses. 36,59 These studies revealed that individual thalamic progenitors give rise to many neurons that populate multiple thalamic nuclei. They also demonstrated that the clonal relationship between thalamic nuclei is determined primarily by the rostro-caudal and dorsoventral positions of thalamic progenitors."

(Price et al., 2012)

The Importance of Combinatorial Gene Expression in Early Mammalian Thalamic Patterning and Thalamocortical Axonal Guidance

"A study by Vue et al. (2007) defined two major progenitor domains in the thalamus, one large posterior domain named pTH-C that expresses transcription factors such as Ngn2 and Olig2 and a much smaller anterior domain named pTH-R that expresses transcription factors such as Nkx2.2 and Mash1 (Figure 3). At a finer level, several detailed and elegant expression analyses have shown that from an age before thalamocortical axons start to navigate to the cortex thalamic progenitors and their newly generated postmitotic neurons are subdivided molecularly by the expression of transcription factors and other molecules into regions that are likely to be the forerunners of the thalamic nuclei of the mature structure (Nakagawa and O'Leary, 2001;Vue et al., 2007Vue et al., , 2009;;Kataoka and Shimogori, 2008;Suzuki-Hirano et al., 2011;Yuge et al., 2011). More work is required to determine the relationship between these early patterns of thalamic gene expression and specific mature thalamic nuclei. Most previous studies looked for possible correspondence between the early expression of individual genes and the later development of specific thalamic nuclei and further insights might be gained by examining the early patterning conferred by combinations of transcription factors, since cells that co-express one particular combination are likely to develop differently to cells that express a different combination even if some of the individual genes expressed are the same. In the following section we describe an analysis using existing data on the expression of individual genes to deduce patterns of co-expression of these genes. The results of a theoretical analysis such as this might guide us toward combinations of genes that might be most worth studying experimentally. The Allen Developing Mouse Brain Atlas (ADMBA) contains images of both Feulgen Yellow DNA counterstained 20 μm-thick tissue coronal sections through mouse embryonic brains and in situ hybridizations providing data on the expression of a wide range of genes that pattern the early thalamus. Its detail and highresolution images give it the potential to be a powerful resource for the analysis of gene expression and co-expression patterns. Such data has many possible uses."

(Kim et al., 2023)

Spatiotemporal molecular dynamics of the developing human thalamus

"The thalamus plays a central coordinating role in the brain. Thalamic neurons are organized into spatially distinct nuclei, but the molecular architecture of thalamic development is poorly understood, especially in humans. To begin to delineate the molecular trajectories of cell fate specification and organization in the developing human thalamus, we used single-cell and multiplexed spatial transcriptomics. We show that molecularly defined thalamic neurons differentiate in the second trimester of human development and that these neurons organize into spatially and molecularly distinct nuclei. We identified major subtypes of glutamatergic neuron subtypes that are differentially enriched in anatomically distinct nuclei and six subtypes of γ-aminobutyric acid–mediated (GABAergic) neurons that are shared and distinct across thalamic nuclei. Description INTRODUCTION The thalamus represents a critical node of communication between the brain and the outside world. The thalamus is organized into regions called nuclei and emerges from progenitors in the caudal forebrain during embryonic development. Although the number of thalamic nuclei and projection patterns is largely conserved in mammals, the molecular trajectories of cell fate specification and their organization into nuclei are largely unknown, especially in humans. We sought to better understand the molecular diversity and spatial organization of cell types in the developing human thalamus using single-cell and spatial transcriptomics. RATIONALE Much of our understanding of the molecular diversity and spatial organization of cell types in the mammalian thalamus comes from rodent models, and yet the degree to which this is conserved in humans remains unclear, especially in the context of development. By integrating previously published datasets, we generated a reference atlas of the first-trimester thalamus, spanning Carnegie stages 6 to 10 and represented by five individuals and 27,362 cells, and a reference atlas of the second-trimester thalamus, spanning gestational weeks 16 to 25 and represented by 10 individuals and 137,007 "

"Spatially resolved cell types in the developing human thalamus. Integrative analysis of single-cell and spatial transcriptomics was used to profile the molecular diversity and spatial organization of cell types in the developing human thalamus. The schematic on the right depicts the biased distribution of neuronal subtypes in thalamic nuclei. EN1 and EN2, excitatory neurons 1 and 2; IN, inhibitory neuron.

T he thalamus can be anatomically subdivided into multiple nuclei with distinct patterns of neuronal projections (1). Many thalamic subregions are hypothesized to have undergone extensive remodeling and reorganization in recent evolution, including the pulvinar nucleus that is extensively expanded in humans and primates compared with mice (2,3). Thalamic neurons emerge from progenitor cells located in the diencephalon of the prenatally developing nervous system (4). Recent studies have begun to apply single-cell genomics approaches to uncover the molecular dynamics of their differentiation in mice (5), but our understanding of thalamic neuron differentiation in humans is less well characterized. To better understand their molecular and spatial organization, we investigated the developing human thalamus during the first and second trimesters of development [gestational weeks (GW) 6 to 25] using single-cell and spatial transcriptomics."

"cells. Spatial validation of cell types was performed using the MERFISH spatial transcriptomics platform. RESULTS Our study provides the first comprehensive characterization of cell types and their spatial organization in the developing human thalamus. Neurogenesis in the thalamus and the neighboring prethalamus occurs during the first trimester, giving rise to glutamatergic and γ-aminobutyric acid–mediated (GABAergic) neurons. Glutamatergic neurons differentiate into two major subtypes that organize into spatially and molecularly distinct nuclei during the second trimester. The molecular identities and spatial distribution of glutamatergic neurons in the human thalamus appear to be similar to those that have been reported in mice. By contrast, we observe a significant expansion in the number and spatial distribution of GABAergic neurons in the human thalamus, the majority of which originate from brain regions outside the thalamus and migrate in during the second trimester. Transcriptomic profiling reveals an unexpected degree of GABAergic neuron diversity and confirms the presence of what may be human-specific GABAergic neuron subtypes that originate from a neighboring region of the developing forebrain known as the ganglionic eminences. The onset of gliogenesis occurs during the second trimester, giving rise to glial progenitor cells, astrocytes, and ependymal cells. Our study revealed spatially resolved patterns of glial cell types. We observe a maturity gradient along the medial-lateral axis during the second trimester, with glial progenitor cells showing enrichment in more-medial regions of the thalamus and astrocytes showing enrichment in more-lateral regions of the thalamus. We identified two major subtypes of astrocytes, one showing enrichment within the thalamus and another showing enrichment in neighboring brain regions adjacent to the thalamus. Ependymal cells line the outer edges of the thalamus. Other nonneuronal cell types"

", including cells of oligodendrocyte lineage, microglia, and cells of the neurovasculature, were found throughout the thalamus. CONCLUSION Our study defines the molecular identities and spatial organization of cell types in the developing human thalamus and highlights an understudied GABAergic neuronal population that may contribute to human evolution. Spatially resolved cell types in the developing human thalamus. Integrative analysis of single-cell and spatial transcriptomics was used to profile the molecular diversity and spatial organization of cell types in the developing human thalamus. The schematic on the right depicts the biased distribution of neuronal subtypes in thalamic nuclei. EN1 and EN2, excitatory neurons 1 and 2; IN, inhibitory neuron."

"We performed MERFISH spatial transcriptomics on a coronal section from the GW10 rostral human forebrain to spatially identify cells from the glutamatergic thalamic lineage identified from scRNA-seq (Fig. 1D). We identified three distinct regions based on a hematoxylin and eosin (H&E) staining of an adjacent slice: the medial and lateral ganglionic eminences (GEs), the thalamic germinal zone (GZ), and the thalamic mantle zone (MZ) (Fig. 1D). Our MERFISH analysis confirmed regional identity on the basis of mutually exclusive expression of TCF7L2 and FOXG1 in the thalamus and GE, respectively (Fig. 1H). The GZ is enriched for OLIG3, and the MZ is enriched for SLC17A6 (Fig. 1H and fig. S1F). MKI67 expression segregated the ventricular zone (VZ) from the subventricular zone (SVZ) (Fig. 1H). We performed clustering analysis of our MERFISH dataset to relate the scRNA-seq populations to the regional annotations (Fig. 1, E and F). We identified four clusters that corresponded to IPC1, nEN, EN, and IN3 from our single-cell sequencing dataset (Fig. 1F). IPC1, which coexpress OLIG3 and EYA2, reside in the GZ (Fig. 1, G and H, and fig. S1F). Newborn neurons (EN2) expressing GBX2 and EYA2 are localized to the GZ and MZ, with bias for the GZ (Fig. 1, F to H, and fig. S1F). Mature LHX9and SLC17A6-expressing glutamatergic neurons (EN) are predominantly found in the MZ (Fig. 1, F to H, and fig. S1F). "

"We also observed the presence of SOX14and NKX2.2-expressing GABAergic neurons (IN3) associated with rostral thalamus origin in the MZ (Fig. 1, F to H, and fig. S1F). PAX6and OLIG3-expressing radial glia are observed at the VZ (Fig. 1H and fig. S1F). However, because of damage at the VZ, there were likely too few of these cells in our MERFISH data to form a discrete cluster.

Collectively, our analysis uncovers the cellular diversity of the human thalamus and their spatial distribution during the first trimester. We identify OLIG3-positive thalamic radial glia and IPCs and the glutamatergic neurons that they likely give rise to that classify broadly into two molecular subtypes. In addition, we define ID3and ASCL1-expressing neural progenitors and their presumed GABAergic neurons progeny, which can be classified into three subtypes of diencephalon-derived GABAergic neurons. These cell types express similar patterns of marker genes compared to what has been previously observed in other species in addition to genes not previously associated with thalamic cell type-enriched expression. These findings suggest that the overall molecular blueprint of first-trimester diencephalic cell types is conserved in humans. The majority of GABAergic neurons in the thalamus has been reported to originate from extrathalamic regions, including the midbrain in mice (50,53) and possibly the GEs in humans (54,55). However, we do not observe evidence for their presence in the first trimester, which suggests that these migratory streams develop later. Spatially, the thalamus has not yet partitioned into nuclei based on cytoarchitecture revealed by Nissl staining or by molecular profiles revealed by MERFISH (Fig. 1, D to F) (56). As expected for this stage of development, thalamic cells segregate into layers that relate to the germinal zones, containing radial glia and IPCs, and the mantle zone, containing migrating and maturing neurons."

"Next, to gain insight into the molecular mechanisms and cellular composition of early thalamic nuclei, we analyzed scRNA-seq and single-nucleus RNA-seq (snRNA-seq) datasets from the second trimester of development, which were derived from 10 specimens (four female and six male) between GW16 and GW25. This analysis retained 137,761 high-quality cells or nuclei. For a subset of the samples, metadata information for thalamic subdivisions had been retained from dissections (6) based on microdissections along the dorsalventral and rostral-caudal axes and included the pulvinar nucleus (fig. S4A). We identified 23 cell clusters, including all major cell classes: glutamatergic neurons (EN1 and EN2), GABAergic neurons (IN1 to IN6), IPCs, astrocytes (AC1 and AC2), glial progenitor cells, dividing cells, microglia, ependymal cells, oligodendrocyte progenitor cells, and oligodendrocytes (Fig. 2A). We also identified several vascular cell classes including endothelial cells, pericytes, and fibroblasts that were annotated based on our recent transcriptomic atlas of the human cerebrovasculature (Fig. 2A) (23).

Neural and glial progenitors were identified on the basis of their expression of HES1 (fig. S3D). However, unlike in the first trimester, we did not detect OLIG3+ clusters associated with thalamic progenitor cells (fig. S3D), in line with the early window of neurogenesis observed in the mouse (57)(58)(59). However, we identified MKI67+ dividing cells and NEUROD1/LHX1+ IPCs that were enriched for MEIS1 and PAX3, markers of pretectal progenitors (Fig. 2B and fig. S3D) (5). This observation suggests ongoing neurogenesis in the pretectum, consistent with prior findings in the diencephalon (58)."

"EN1 was enriched for neurotensin (encoded by NTS), calcyclin (encoded by S100A6), and SOX2, whereas EN2 was enriched for RNF220, CRTAC1, and FOXP2 (Fig. 2B and Fig. 3B). EN1 and EN2 likely relate to the SOX2and FOXP2-expressing subtypes recently observed in the embryonic mouse thalamus (5). Three classification schemes exist for thalamic glutamatergic neurons in the adult brain for which molecular markers have been identified. The core-matrix model classifies thalamic neurons on the basis of whether they project to layer IV or layer I in the cortex and in primates can be molecularly identified based on their expression of CALB1 or PVALB (71).

The first-order or higher-order model relates to thalamic nuclei and their involvement in sensory relay pathways or transthalamic corticocortical communication (72), and their transcriptional signatures have been ascertained in mice Area:

Fraction of cell in group (%)"

"The pulvinar, which represents an evolutionarily expanded thalamic nucleus (79), showed an overall enrichment for EN1 in the lateral-most section and an enrichment for EN2 in more medial sections (Fig. 3E). This difference within the pulvinar may relate to its functional organization, whereby the lateral pulvinar is involved with the visual pathway and the medial pulvinar is involved with multisensory, associative, and higher-order cognitive func-tion (3,80). The overall biased distribution of EN1 and EN2 neurons is consistent with the embryonic mouse thalamus (5). The epithalamus, which is a neighboring diencephalic region derived from prosomere 2 like the thalamus, contained glutamatergic neurons associated with EN1 signatures (Fig. 3E). This is in line with previously reported similarities in gene expression profiles between the epithalamus and thalamus (29,81). However, other neighboring regions, including the subthalamic nucleus and midbrain, contained glutamatergic neurons that clustered separately from EN1 and EN2 (EN-STN and EN-MB) (Fig. 3E).

We next analyzed the distribution of GABAergic neuron subtypes revealed by our MERFISH data (Fig. 4A S7C). Abundant IN2 population was also present in the MGE and basal ganglia structures that were visible in our sections, such as the caudate and globus pallidus, consistent with their telencephalic identity (Fig. 4B). IN5 was enriched in the epithalamus (fig. S7). Finally, a FOXG1-positive IN6 cluster was observed in the caudate but not the thalamus, which suggests that this sparse population may have been accidentally included in the scRNA-seq and snRNA-seq data generation as a result of inaccurate dissection (Fig. 4B). We also could not detect IN3, which we hypothesized were pretectal GABAergic neurons, likely because of a lack of representation of pretectum in our spatial transcriptomics datasets."

Transcriptomic and multimodal resources for mouse thalamic cell types

The strongest transcriptomic resources for mouse thalamus are now whole-brain atlas data, projection-linked RNA-seq, PVT- and ATN-focused single-cell studies, and Epi-Retro-Seq integration. The supplied sources rarely give accession numbers, so resource names and identifiers are often the most reliable way to point to the datasets.

Key resources for mouse thalamic neuron typing include:

- **ABC Atlas / THALMANAC**: whole-brain scRNA-seq plus MERFISH mapped to CCFv3, with a thalamus-focused interface. THALMANAC uses 80,170 thalamic MERFISH cells from 14 coronal sections, taxonomy version **CCN20230722**, and provides a web app (**RRID:SCR_024354**) and notebooks (**RRID:SCR_018315**). The quote does not provide GEO/SRA-style accession numbers. (Turner et al., 2025)
- **Allen Brain Cell Atlas reference atlas**: ~7 million scRNA-seq cells and ~4.3 million MERFISH cells for the adult mouse brain, including hierarchical thalamic taxonomy. The quote does not list accessions. (Yao et al., 2023)
- **ThalamoSeq / Retro-Seq**: projection-defined thalamic bulk and single-cell RNA-seq across multiple nuclei and cortical targets. Later papers explicitly refer to this resource as **ThalamoSeq** and use it for integration. The quote gives the website **thalamoseq.janelia.org**, but not a formal accession number. (Gao et al., 2023) (Phillips et al., 2019)
- **PVT-focused datasets**: mouse PVT **scRNA-seq** with four continuous types and mouse PVT **snRNA-seq** with five subtypes plus multiplex FISH validation; accession numbers are not given in the provided text. (Shima et al., 2022) (Gao et al., 2023)
- **ATN-focused dataset**: scRNA-seq, spatial transcriptomics, and multiplex FISH defining three excitatory clusters and within-nucleus gradients; accession number not given in the quote. (Kapustina et al., 2024)

- **Epi-Retro-Seq integration**: 33,034 neurons from 225 source target combinations, integrated with thalamic scRNA-seq and snmC-seq to define 58 thalamic neuronal clusters and link cell type to projection target. No accession number is given in the quote. (Zhou et al., 2023)

A main practical conclusion from these resources is that no single modality is enough. The clearest picture of mouse thalamic neuron types comes from combining spatial transcriptomics, projection mapping, morphology, physiology, and development. (Turner et al., 2025) (Peng et al., 2021) (Roy et al., 2022)

Paper	Modality Scope	Thalamic Coverage	Cell Type Resolution	Classification Framework	Projection Identity	Transcriptomic Markers	Spatial Anatomical Position	Functional Properties	Multimodal Validation	Cell Type Heterogene

Paper	Modality Scope	Thalamic Coverage	Cell Type Resolution	Classification Framework	Projection Identity	Transcriptomic Markers	Spatial Anatomical Position	Functional Properties	Multimodal Validation	Cell Type Hete gene
Exploring the correspondence between gene expression and thalamic nuclei using the THALMANAC resource	Spatial transcriptomics/ MERFISH and scRNA-seq were used; the paper also notes multimodal integration with CCFv3 anatomical parcellations.	Atlas-wide thalamic sampling across multiple nuclei, using a thalamus-specific subset of spatial transcriptomics data.	The paper reports broad classes, subclasses, supertypes, and clusters, with many nuclei containing multiple transcriptomic clusters and some showing continuous gene-expression gradients.	The paper uses the ABC Atlas hierarchical transcriptomic taxonomy: class, subclass, supertype, and cluster, with clusters being unsupervised and higher levels informed by developmental origin, anatomy, and classical neuron types.	Projection identity is not directly defined; the paper emphasizes gene expression and notes connectivity as an additional modality for future integration.	They distinguish neuron types and reveal thalamic nuclei boundaries, subdomains, gradients, and shared patterns of expression.	Neuron types are reported by nucleus, subnuclear domain, topographic gradients, or laminar-like spatial organization within the thalamus.	Functional properties are not directly characterized; the paper notes they may require integration with neurophysiology and connectivity data.	Transcriptomic cell types are partially corroborated by other data types, but finer-scale validation requires additional multimodal data integration.	Evide e for withi type diver y is pres : son nucle cont discr e subd main and gene expr ion grad ts, ar many types span mult e nucle

Paper	Modality Scope	Thalamic Coverage	Cell Type Resolution	Classification Framework	Projection Identity	Transcriptomic Markers	Spatial Anatomical Position	Functional Properties	Multimodal Validation	Cell Type Hete gene
A high-resolution transcriptomic and spatial atlas of cell types in the whole mouse brain	Whole-brain cell types were defined by multimodal integration of scRNA-seq and MERFISH spatial transcriptomics.	Atlas-wide thalamic sampling across the whole mouse brain, not limited to one nucleus or projection-defined population.	A highly granular hierarchy: 7 divisions, 32 classes, 306 subclasses, 1,045 supertypes, and 5,200 clusters.	Hierarchically organized into 7 divisions, 32 classes, 306 subclasses, 1,045 supertypes, and 5,200 clusters, combining scRNA-seq and MERFISH.	N/A	Transcription factors are major determinants of cell type classification, and a combinatorial transcription factor code defines cell types across the brain.	Each neuron type has a unique, specific spatial localization pattern within the brain, including thalamic subnuclear and laminar-like organization.	Functional properties are mentioned only as part of cell types' structural and functional properties, without specific details.	Transcriptomic cell types show high correspondence with spatial specificity and are corroborated by MERFISH-based spatial data.	Show exter ve hete eneit inclu g trans onal types spati grad ts, ar varyi relat ness that com cate discr e class catio

Paper	Modality Scope	Thalamic Coverage	Cell Type Resolution	Classification Framework	Projection Identity	Transcriptomic Markers	Spatial Anatomical Position	Functional Properties	Multimodal Validation	Cell Type Heterogene
Molecular and spatial profiling of the paraventricular nucleus of the thalamus	snRNA-seq with multiplex fluorescent in situ hybridization and RPCA-based multimodal integration were used to define and spatially annotate five PVT neuronal subtypes.	Focused on the paraventricular nucleus of the thalamus, with additional comparison to other midline thalamic nuclei and atlas datasets.	Five molecularly distinct PVT neuronal subtypes, organized by previously unidentified molecular gradients.	Identifies five molecularly distinct PVT neuronal subtypes and maps them to a largely non-overlapping transcriptomic atlas, with projection-based cortical overlaps.	Projection identity is partly defined by target-specific cortical innervation; PVT subtypes differentially innervate cortical areas, including auditory and visual regions.	Five molecularly distinct PVT neuronal subtypes were identified, organized by previously unidentified molecular gradients.	PVT neuron types show subnuclear spatial segregation across antero-posterior, medio-lateral, and dorso-ventral axes, organized by molecular gradients.	Functional diversity is linked to cellular differences, but the PVT's physiological/computational properties are not directly characterized here; the study focuses on molecular subtypes and cortical connectivity.	Transcriptomic PVT subtypes were corroborated by multiplex fluorescent in situ hybridization, revealing distinct spatial gradients.	PVT neuro s sho five molec lar subty es orga ed along previ sly unide ified spati grad ts, indic ng withi type hete eneit

Paper	Modality Scope	Thalamic Coverage	Cell Type Resolution	Classification Framework	Projection Identity	Transcriptomic Markers	Spatial Anatomical Position	Functional Properties	Multimodal Validation	Cell Type Hete gene
A repeated molecular architecture across thalamic pathways	Retrograde tracing was used with comprehensive transcriptional analyses; no other modalities are specified in the snippet.	Atlas-wide sampling of thalamic pathways across major projection systems, centered on retrogradely labeled projection-defined populations.	Three major profiles arranged along a continuous gradient, repeated across major projection systems.	A repeated, continuum-based molecular framework organizes thalamic pathways across major projection systems.	Projection identity is defined by source-to-target thalamic pathway specificity, identified through retrogradely labeled thalamic neurons across major projection systems.	They reveal three major thalamic pathway profiles along a repeated continuum across major projection systems, distinguishing neuronal identities.	Topographically organized across the mouse thalamus, with features conserved in humans.	Transcriptional differences are tied to cellular features critical for function, including membrane properties.	Transcriptomic identities are corroborated by cellular features such as axonal morphology and membrane properties, integrating anatomical and functional features.	Thal ic path ys show cont um o mole lar profi , indic ng withi type hete eneit rathe than discr e class .

Paper	Modality Scope	Thalamic Coverage	Cell Type Resolution	Classification Framework	Projection Identity	Transcriptomic Markers	Spatial Anatomical Position	Functional Properties	Multimodal Validation	Cell Type Hete gene
Distinctiveness and continuity in transcriptome and connectivity in the anterior-posterior axis of the paraventricular nucleus of thalamus	scRNA-seq, transcriptomic and histological analyses, and axon projection analyses; no other modalities are mentioned.	Focused on one nucleus / subregion: the paraventricular nucleus of the thalamus (PVT).	A continuous distribution with four transcriptomic types, but with continuity rather than sharply distinct cell types.	Single-cell transcriptomics classified PVT neurons into four types, but transcriptomic and histological analyses showed they are continuous.	Strong; PVT neurons show nearly non-overlapping axon projection patterns along the anterior-posterior axis.	PVT neurons showed no discrete transcriptomic markers; single-cell transcriptome classified them into four types, but transcriptomic and histological analyses showed continuity, with strongest variation along the anterior-posterior axis.	PVT neuron transcriptomes showed a continuous distribution with the largest variance along the anterior-posterior axis, indicating topographic continuity rather than discrete spatial compartments.	PVT neuron subpopulations responded differently to appetite-related neuropeptides and have opposing effects on food consumption.	Transcriptomic cell types were corroborated by histological, axon projection, neuropeptide response, and chemogenetic data, showing continuity as well as distinct subpopulations.	PVT neur s sho a cont ous trans ptom distr tion with ante - post or grad ts, desp four class ed type indic ng cont ity rathe than discr e cel type

Paper	Modality Scope	Thalamic Coverage	Cell Type Resolution	Classification Framework	Projection Identity	Transcriptomic Markers	Spatial Anatomical Position	Functional Properties	Multimodal Validation	Cell Type Heterogene
The cell-type-specific spatial organization of the anterior thalamic nuclei of the mouse brain.	Single-cell RNA sequencing, spatial transcriptomics, and multiplexed fluorescent in situ hybridization.	Focused on the anterior thalamic nuclei, not atlas-wide thalamic sampling.	Three discrete excitatory cell-type clusters corresponding to the known ATN nuclei, plus continuous graded spatial variation along the dorsomedial-ventrolateral axis.	Identifies three discrete excitatory cell-type clusters corresponding to the known ATN nuclei.	Projection identity is not directly defined; the paper instead maps ATN cell types and notes long-range inputs to the anteroventral nucleus.	Three discrete excitatory cell-type clusters correspond to ATN nuclei, with identified marker genes and molecular pathways.	Discrete nuclei with graded dorsomedial-ventrolateral spatial variation and laminar-like organization in the anteroventral nucleus.	Putative functions of the ATN cell types were identified, but specific functional properties are not detailed in the snippet.	Transcriptomic cell types are corroborated by spatial transcriptomics, multiplexed FISH, and long-range anatomical inputs.	The ATN shows graded spatial variation along the dorsomedial-ventrolateral axis, indicating continuous within-type heterogeneit

Paper	Modality Scope	Thalamic Coverage	Cell Type Resolution	Classification Framework	Projection Identity	Transcriptomic Markers	Spatial Anatomical Position	Functional Properties	Multimodal Validation	Cell Type Heterogene
Brain-wide Correspondence Between Neuronal Epigenomics and Long-Distance Projections	Thalamic cell types were defined by multimodal integration of Epi-Retro-Seq with unbiased snmC-seq and single-cell RNA-seq data.	Atlas-wide thalamic sampling across multiple nuclei and twelve projection targets.	Clusters and subtypes; 58 thalamic neuronal cell clusters were identified, with projection-specific enrichment in distinct subsets.	Thalamic neurons were classified by integrated transcriptomic / epigenomic clusters and projection-enriched labels, with anatomical nuclei organization aligned to these clusters.	Projection identity is strongly defined by source target connectivity; neurons are grouped by distinct, target-specific projection patterns.	N/A	Thalamic cell clusters showed spatial organization by nucleus, subnuclear domain, and pathway-specific topography, with distinct clusters for different anatomical regions.	The snippets do not provide specific functional properties beyond projection-related roles such as sensory and cortical information processing.	Cell types were corroborated by anatomical tracing and cross-validated with transcriptomic, epigenomic, and spatial data.	Thalamic projection neurons sho cell type heterogeneit with each projection enriched in distin some mes overlapping clusts and multiple enriched clusts per pathway.

Paper	Modality Scope	Thalamic Coverage	Cell Type Resolution	Classification Framework	Projection Identity	Transcriptomic Markers	Spatial Anatomical Position	Functional Properties	Multimodal Validation	Cell Type Heterogene
Morphological diversity of single neurons in molecularly defined cell types	Brain-wide morphology plus transcriptomic profiling were used; clear concordance with transcriptomic profiles was found at major-type level, but not fine-grained subtypes.	Atlas-wide thalamic sampling across multiple thalamic regions, not a single nucleus or subregion.	Broad major types align with transcriptomic identities, but finer morphological subtypes often do not map cleanly to transcriptomic clusters.	No specific classification framework is described; the paper emphasizes major projection types and transcriptomic identities, with some refined subtypes.	Projection identity is a major defining feature: neurons are classified by distinct long-range axonal projections, target specificity, and divergent/convergent projection patterns.	Major projection types show clear concordance with transcriptomic profiles, but fine-grained morphological diversity often does not match transcriptomic subtypes.	Thalamic neuron types show reported spatial organization within nuclei, subnuclear domains, topographic gradients, and laminar-like arrangements.	Functional properties are not directly measured; only related intrinsic and network physiological properties are mentioned as prior findings for some neuron groups.	Transcriptomic major types are corroborated by morphology, but fine-grained subtypes often are not.	Stron within type heter eneit inclu g exter ve proje onal diver y and refin subc ters; varia ty diffe by ce type.

Paper	Modality Scope	Thalamic Coverage	Cell Type Resolution	Classification Framework	Projection Identity	Transcriptomic Markers	Spatial Anatomical Position	Functional Properties	Multimodal Validation	Cell Type Heterogene
Thalamic subnetworks as units of function	It spans multiple modalities, including scRNA-seq, electrophysiology, morphology, connectivity-based optical tagging, and gene-expression profiling.	Atlas-wide, spanning multiple thalamic nuclei and subregions, with some focus on projection-defined populations.	Cell type resolution can range from broad classes and nuclei to clusters, subtypes, or continuous gradients of states.	It is a naming and organization system for classifying thalamic neurons using hierarchical atlases, projection labels, transcriptomic clusters, or cross-modal consensus taxonomies.	Projection identity is used to define subpopulations by source-to-target connectivity, including projection-specific and target-specific subpopulations.	They are used to distinguish neuron types by defining cell- or nucleus-specific gene expression signatures, including canonical markers and differentially expressed genes.	It is used to define subpopulations by their reported thalamic location, including nuclei, subnuclear domains, and gradients.	Thalamic neurons show heterogeneous physiological properties, often along gradients rather than discrete types, including different action-potential parameters and firing modes.	Transcriptomic cell types are corroborated by other modalities such as FISH, morphology, electrophysiology, and anatomical tracing.	Suppted: thalac neurs shwithitype heteeneitinclug gradcontum-liphysogicaprofiand distisubpulatis/subcters.

Evidence

(Turner et al., 2025)

Exploring the correspondence between gene expression and thalamic nuclei using the THALMANAC resource

"The thalamus connects the sensory organs and major subcortical brain regions with the neocortex. The thalamus has long been divided into multiple discrete nuclei, based on cytoarchitecture, histochemical stains, and mesoscale connectivity. However, thalamic nuclei do not completely describe thalamic organization. For example, some boundaries between thalamic nuclei are disputed, whereas other nuclei are known to contain subdomains with distinct connectivity and function. Moreover, the correspondence between cellular gene expression and other properties of thalamic projection neurons remains to be established. Spatial analysis of single cell gene expression provides a basis for reevaluating thalamic organization. We present the THALMANAC, (THALamusMERFISH ANalysis and ACcess) a Findable, Accessible, Interoperable, Reusable and Reproducible (FAIRR) resource for exploring and analyzing single-cell transcriptomic variation in the thalamus. The THALMANAC provides streamlined access to thalamic gene expression data registered to the common coordinate framework and tools for quantitative analysis and visualization

of these data, all encapsulated in a reproducible, cloud computing platform. Using this resource, we find that gene expression generally supports the parcellation of thalamus into distinct nuclei. Some nuclei, such as the anteromedial nucleus, are additionally composed of discrete subdomains, while other nuclei share patterns of gene expression or are arrayed on a spatial gradient of gene expression. The THALMANAC establishes spatial transcriptomic data as a foundation for delineating thalamic organization. Graphical Abstract"

"This fine-scale structure could correspond to discrete, molecularly defined subdivisions of VM that have not yet been annotated, or to topographic organization of thalamocortical and corticothalamic connectivity.

A major question is how gene expression relates to existing parcellations and thalamic organization in general. Single cell RNA sequencing (scRNA-seq) provides detailed measurements of transcription in large populations of individual neurons (Tasic et al. 2016), which can be used to define transcriptomic cell types ('t-types'). The combination of scRNA-seq with spatial transcriptomics has been used to localize t-types across the whole mouse brain (Langlieb et al. 2023;Yao et al. 2023;Zhang et al. 2023). With t-types mapped into the CCFv3 coordinate space (Wang et al. 2020), we can evaluate how their spatial distributions correspond to anatomical parcellations (Yao et al. 2023).

Here, building on the Allen Brain Cell (ABC) Atlas scRNA-seq and spatial transcriptomics data, we developed a resource, the THALamus MERFISH ANalysis and ACcess (THALMANAC), which provides access to spatial gene expression data in the mouse thalamus; targeted analysis and data visualization of gene expression patterns; a platform for reproducible computations; and a no-code visualization app for data exploration. We demonstrate the utility of the THALMANAC by addressing how thalamic structural organization relates to gene expression. Aligning spatial transcriptomic data and nuclei boundaries reveals a wide range of relationships between t-types and thalamic nuclei: some t-types correspond to a single thalamic nucleus; others subdivide nuclei into multiple parts; many t-types span multiple thalamic nuclei.

Analysis of expression by itself supports these same conclusions: some nuclei boundaries are associated with sharp changes in gene expression; other nuclei contain gene expression gradients; and, in many cases, gene expression patterns span multiple nuclei.

Understanding how information is routed through the thalamus requires integrating insights gained from gene expression with other modalities, including connectivity and neurophysiology."

"The THALMANAC resource provides access to thalamic spatial gene expression data, visualization of the relationships between gene expression, t-types, and CCFv3 anatomical parcellations, and quantitative analysis of targeted quantitative analyses, all encapsulated in a cloud platform that enables reproducible computations and customized exploration of the data.

The THALMANAC is built on a thalamus-specific subset of spatial transcriptomics data from multiplexed error-robust fluorescence in situ hybridization (MERFISH) from the ABC Atlas. The ABC Atlas dataset measures 500 transcripts, chosen to distinguish t-types across the whole mouse brain, in spatially resolved, single cells. MERFISH cells were registered to the CCFv3 coordinate space and mapped to the scRNA-seq data by assigning each MERFISH cell to the label of the most similar t-type cluster. We identified neurons that belong to the thalamus (including the zona incerta and habenula) according to their t-type label (Figure S1A) and CCFv3 parcellation label (Figure S1B). The thalamic dataset analyzed here includes 80,170 MERFISH cells across 14 coronal sections collected from a single male mouse (Methods).

The THALMANAC includes three avenues for engaging with the t-type and gene expression data and accompanying analyses (Figure 1). First, a Streamlit web app (RRID:SCR_024354) provides nocode exploration, including interactive visualizations and on-the-fly di"erential gene expression analysis for thalamic nuclei (Interactive Streamlit App: thalmanac.allenneuraldynamics.org). Second, Jupyter notebooks (RRID:SCR_018315) provide step-by-step demonstrations of the Python code needed to reproduce all results presented here (github.com/AllenNeuralDynamics/thalmanac). Both tools allow, users to customize the thalamic cells that are selected from the full ABC Atlas."

"In the ABC Atlas, transcriptomic variation is used to assign cells to a hierarchy of discrete t-types (from broad to narrow categories: class, subclass, supertype, cluster). Thalamic neurons belong to 5 classes, 58 subclasses, 163 supertypes, and 563 clusters (Table S1; Figure S2) (taxonomy version CCN20230722) (Yao et al. 2023), but these numbers will change as the taxonomy evolves. Whereas the cluster level is based on unsupervised clustering, higher levels of the taxonomy are additionally informed by developmental origin via shared transcription factor expression, anatomical location, and classical (pre-scRNAseq) neuron types (DeFelipe et al. 2013) defined by key molecular markers (e.g. neuropeptides, calcium-binding proteins).

Transcriptomic classes typically correspond to major brain structures. Glutamatergic neurons in thalamus proper form a single class (class 18 TH Glut), whereas habenula neurons belong to a distinct class (class 17 MH-LH Glut) (Figure S2B). The other three classes are not unique to the thalamus: class 12 HY GABA is found in the reticular thalamus, zona incerta and ventral lateral geniculate nucleus (CCFv3 annotations; RT, ZI, and LGv, respectively, see Table S1) but also in the hypothalamus; class 19 MB Glut is found in thalamic nucleus SPA but also in the midbrain; class 20 MB GABA is found in thalamic nuclei LGv and SPF and the midbrain. Transcriptomic subclasses span multiple thalamic nuclei(Figure S2C), whereas transcriptomic supertypes often define individual nuclei (e.g. AM), and several nuclei contain multiple supertypes (e.g. ZI) (Figure S2D). Most nuclei contain multiple transcriptomic clusters (Figure 2A, Figure S2E).

We explored correspondence between clusters and thalamic nuclei."

"A comprehensive understanding of how the thalamus routes information to the cortex will require multiple types of data and a community-wide e"ort. Our work contributes to this program by curating and analyzing the correspondence between state-of-the-art spatially resolved, single-cell transcriptomic data, and widely used parcellations of thalamus into discrete nuclei. In addition, we make our data and analyses available in an accessible, documented, and reproducible bioinformatics resource in the cloud. Using the AM nucleus and the somatosensory nuclei as examples, we illustrate how this interactive resource can be used to identify novel patterns of gene expression and how these patterns correspond to prior parcellations of the thalamus.

The THALMANAC resource streamlines researchers' access to thalamic spatial transcriptomic data. We anticipate that scientists will extend our analysis based on the ABC atlas data, or apply similar analyses to additional spatial transcriptomics datasets registered to the CCF coordinate system (Langlieb et al. 2023;Zhang et al. 2023). The tool should also be used as a reference to validate recent studies focused on specific nuclei, such as the spatially segregated projections from PF to distinct dorsal striatum regions (Mandelbaum et al. 2019) and spatially segregated inputs to PO (Casas-Torremocha et al. 2022), the molecular organization of the PVT (Gao et al. 2023), and the core/shell organization of neurons in RT (Li et al. 2020). Our interactive Streamlit app can be used to identify marker genes for post-hoc localization of neurophysiological and neuroanatomical experiments, or to select gene panels for multiplexed mFISH experiments. The THALMANAC is already being used in the e"ort to create and characterize enhancerbased genetic tools (Ben-Simon et al. 2025) to target thalamic cell types and regions.

The THALMANAC shows that at the scale of thalamic nuclei, the molecular organization of the thalamus resembles existing parcellations. "

"However, both discrete cell type composition and gene expression gradients vary at finer spatial scales within and across nuclei. Given the evidence for circuit and functional variation at the same length scales, additional data need to be integrated with molecular characterization to understand thalamic organization. We see two promising paths forward for data integration. First, existing data registered to the CCFv3 3D coordinate space, such as the databases of single-neuron morphological reconstruction (Winnubst et al. 2019) or neurophysiology data (Chen et al. 2024), could be linked to cell types and gene expression described in the THALMANAC. Second, targeted multimodal datasets, such as single-neuron morphological reconstructions combined with posthoc mFISH for transcriptomic cell typing would link cell types and gene expression to projection targets at the single cell level.

Beyond data integration, we also recognize the need for thalamus-focused, community annotation, similar to previous e"orts to annotate the fly connectome (Schlegel et al. 2024). We envision a refined parcellation of the thalamus where scientists with specific domain knowledge would combine molecular information from the THALMANAC with neuroanatomical and functional data to reflect progress in understanding of thalamic organization. Diagram illustrating the three workflows that can be launched from the Code Ocean Collection accompanying this paper. The Code Ocean platform provides built-in support for code versioning (via Git), environment replicability (via Docker), data and analysis provenance (via tracked data assets and linked pipeline runs), cloud compute resources (via AWS EC2 instances). Users can launch an interactive, no-code Streamlit application to explore spatial distributions of transcriptomic cell types and gene expression from their browser, as well as visualize locations of annotated cell types for each thalamic nucleus and highlight a specific nucleus or group of related thalamic nuclei. Tutorial Jupyter notebooks and Python code replicate all analyses and visualizations in a cloud workstation. These visualizations can be customized and extended. Users can download an AnnData-formatted .h5ad"

(Yao et al., 2023)

A high-resolution transcriptomic and spatial atlas of cell types in the whole mouse brain

"The mammalian brain is composed of millions to billions of cells that are organized into numerous cell types with specific spatial distribution patterns and structural and functional properties. An essential step towards understanding brain function is to obtain a parts list, i.e., a catalog of cell types, of the brain. Here, we report a comprehensive and high-resolution transcriptomic and spatial cell type atlas for the whole adult mouse brain. The cell type atlas was created based on the combination of two single-cell-level, whole-brain-scale datasets: a single- cell RNA-sequencing (scRNA-seq) dataset of ~ 7million cells profiled, and a spatially resolved transcriptomic dataset of ~ 4.3million cells using MERFISH. The atlas is hierarchically organized into five nested levels of classification: 7 divisions, 32 classes, 306 subclasses, 1,045 supertypes and 5,200 clusters. We systematically analyzed the neuronal, non-neuronal, and immature neuronal cell types across the brain and identified a high degree of correspondence between transcriptomic identity and spatial specificity for each cell type. The results reveal unique features of cell type organization in different brain regions, in particular, a dichotomy between the dorsal and ventral parts of the brain: the dorsal part contains relatively fewer yet highly divergent neuronal types, whereas the ventral part contains more numerous neuronal types that are more closely related to each other. We also systematically characterized cell-type specific expression of neurotransmitters, neuropeptides, and transcription factors. The study uncovered extraordinary diversity and heterogeneity in neurotransmitter and neuropeptide expression and co-expression patterns in different cell types across the brain, suggesting they mediate a myriad of modes of intercellular communications. Finally, we found that transcription factors are major determinants of cell type classification in the adult mouse brain and identified a combinatorial transcription factor code that defines cell types across all parts of the brain. The whole-mouse-brain transcriptomic and spatial cell type atlas establishes a benchmark reference atlas and a foundational resource for deep and integrative investigations of cell type and circuit function, development, and evolution of the mammalian brain."

"In this study, we created a comprehensive, high-resolution transcriptomic cell type atlas for the whole adult mouse brain based on the combination of two whole-brain-scale datasets: a scRNAseq dataset of ~7 million cells and a MERFISH dataset of a similar scale (~4.3 million cells from the AIBS MERFISH brain). We used ~4.1 million high-quality single-cell transcriptomes after stringent QC to create a transcriptomic cell type taxonomy. We used the MERFISH data, which were generated using marker genes derived from the whole-brain transcriptomic taxonomy, to annotate the spatial location of each subclass and each cluster in the taxonomy. We then built a hierarchically organized transcriptomic and spatial cell type atlas with five nested levels: 7 divisions, 32 classes, 306 subclasses, 1,045 supertypes and 5,200 clusters (Figure 1). The neuronal cell type composition in each major brain structure were systematically analyzed (Figure 2) and distinct features in different brain structures identified (Figure 6). We discovered many sets of neuronal types with varying degrees of similarity with each other, including transitional neuronal types across regions as well as highly distinct neuronal types. We also systematically analyzed all divisions of non-neuronal cell types as well as immature neuronal types present in the adult brain, and identified their unique spatial distribution and spatial interaction patterns (Figure 4). Finally, we systematically characterized cell-type specific neurotransmitters, neuropeptides, and transcription factors, and discovered unique characteristics for each as discussed below. This large-scale study allowed us to delineate several principles regarding cell type organization across the whole mouse brain. It provides a benchmark reference cell type atlas as a resource for the community that will enable many more discoveries in the future.

One of the most striking findings from our study is the high degree of correspondence between transcriptomic identity and spatial specificity. Every subclass (and all supertypes and many clusters within each) has a unique and specific spatial localization pattern within the brain. Furthermore, the relative relatedness between transcriptomic types (as revealed in 2D and 3D UMAPs) is strongly correlated with the spatial relationship between them. Transcriptomically related cell types"

(Gao et al., 2023)

Molecular and spatial profiling of the paraventricular nucleus of the thalamus

"The paraventricular nucleus of the thalamus (PVT) is known to regulate various cognitive and behavioral processes. However, while functional diversity among PVT circuits has often been linked to cellular differences, the molecular identity and spatial distribution of PVT cell types remain unclear. To address this gap, here we used single nucleus RNA sequencing (snRNA-seq) and identified five molecularly distinct PVT neuronal subtypes in the mouse brain. Additionally, multiplex fluorescent in situ hybridization of top marker genes revealed that PVT subtypes are organized by a combination of previously unidentified molecular gradients. Lastly, comparing our dataset with a recently published single-cell sequencing atlas of the thalamus yielded novel insight into the PVT's connectivity with the cortex, including unexpected innervation of auditory and visual areas. This comparison also revealed that our data contains a largely non-overlapping transcriptomic map of multiple midline thalamic nuclei. Collectively, our findings uncover previously unknown features of the molecular diversity and anatomical organization of the PVT and provide a valuable resource for future investigations."

"Although this broad classification into aPVT and pPVT subregions has aided advances in our understanding of the functional organization of the PVT, systematic dissections of the local organization of PVT subnetworks are currently lacking (Barson et al., 2020;McGinty and Otis, 2020;Iglesias and Flagel, 2021). Importantly, consistent with the notion that molecular identity can delineate cell types with different anatomical, functional, and electrophysiological properties (Huang and Zeng, 2013;Huang, 2014;Fuzik et al., 2016;Oh et al., 2014;Luo et al., 2008;Luo et al., 2018), studies have recently identified functional and anatomical differences that tie onto genetically defined subpopulations of PVT neurons (Mátyás et al., 2018;Gao et al., 2020;Kessler et al., 2021;Engelke et al., 2021). From this perspective, transcriptional profiling of the PVT could lead to valuable insights into the functional heterogeneity of this thalamic structure (McGinty and Otis, 2020).

In this study, we employ high throughput single-nucleus RNA sequencing and multiplex fluorescent in situ hybridization (ISH) labeling to identify five PVT subtypes that segregate differentially across the antero-posterior, medio-lateral, and dorso-ventral axes of the PVT. Our data highlights novel features about the molecular organization of the PVT, thereby offering a glimpse into how genetic diversity may tie into the various functions associated with this region of the thalamus. In addition, by performing the reciprocal principal component analysis (RPCA)-based integration and comparative analysis with the ThalamoSeq atlas, we find that PVT subtypes differentially innervate cortical areas and that our dataset contains complementary and previously unexplored mouse thalamic single-nuclei transcriptomes, including mediodorsal thalamus (MD) and intermediodorsal (IMD) thalamus (Phillips et al., 2019)."

"To cross-reference our snRNA-seq dataset of the mouse midline thalamus, we performed integration using RPCA together with the recently published ThalamoSeq RNA-Seq atlas (thalamoseq.janelia. org), which contains single-cell sequenced thalamic cells that project to either motor, auditory, visual, somatosensory, or prefrontal cortices (PFC) (Figure 5, Figure 5-figure supplement 1; Phillips et al., 2019;Hao et al., 2021). Upon clustering analysis and plotting the UMAP of the integrated dataset with all midline-thalamic and BSTpr neurons, we found that our midline-thalamic dataset and the Thal-amoSeq dataset contained mostly non-overlapping transcriptomes (Figure 5-figure supplement 1).

Next, we performed integration of the ThalamoSeq dataset and PVT neurons specifically (Figure 5a-b).

By mapping the integrated dataset based on cortical projection location from the ThalamoSeq cells or their original cluster identity (either PVT1-5 or ThalamoSeq), we could observe the cluster relationship between ThalamoSeq cortical projection location and our

five PVT molecular subtypes (Figure 5c-e). DGE analysis on the integrated dataset revealed that our original PVT subtypes (PVT1-5) could be mapped onto clusters 0, 1, 3, 5, and 4, respectively, of the integrated dataset (Supplementary file 1, Figure 5a and c). Following this, comparing the proportion of cortical-projectors in the integrated dataset showed that the clusters representing PVT1 P and PVT2 edge contained overlap with PFCprojectors, whereas, unexpectedly, the cluster representing PVT4 AM had a small amount of overlap with auditory and visual projectors (Figure 5b). The clusters representing PVT3 AL and PVT5 pan exhibited a small degree of overlap with prefrontal and visual cortical projectors, respectively (Figure 5b)."

(Phillips et al., 2019)

A repeated molecular architecture across thalamic pathways

"The thalamus is the central communication hub of the forebrain and provides the cerebral cortex with inputs from sensory organs, subcortical systems and the cortex itself. Multiple thalamic regions send convergent information to each cortical region, but the organizational logic of thalamic projections has remained elusive. Through comprehensive transcriptional analyses of retrogradely labeled thalamic neurons in adult mice, we identify three major profiles of thalamic pathways. These profiles exist along a continuum that is repeated across all major projection systems, such as those for vision, motor control and cognition. The largest component of gene expression variation in the mouse thalamus is topographically organized, with features conserved in humans. Transcriptional differences between these thalamic neuronal identities are tied to cellular features that are critical for function, such as axonal morphology and membrane properties. Molecular profiling therefore reveals covariation in the properties of thalamic pathways serving all major input modalities and output targets, thus establishing a molecular framework for understanding the thalamus. This work describes comprehensive transcriptomic sequencing from murine thalamic pathways. By integrating this molecular information with anatomical and functional features, this study reveals a repeated architecture across thalamocortical systems."

(Shima et al., 2022)

Distinctiveness and continuity in transcriptome and connectivity in the anterior-posterior axis of the paraventricular nucleus of thalamus

"The paraventricular nucleus of the thalamus (PVT) projects axons to multiple areas and mediates a wide range of behaviors. Heterogeneity of functions and axonal projections in PVT have been reported, but what cell types exist in PVT and how different they are have not been addressed. We applied single-cell RNA sequencing to depict transcriptomic characteristics of mouse PVT neurons. The transcriptome of PVT neurons had a continuous distribution with the largest variance corresponding to the anterior-posterior axis. Although the single-cell transcriptome classified PVT neurons into four types, transcriptomic and histological analyses showed their continuity. Similarly, anterior and posterior subpopulations had nearly non-overlapping axon projection patterns, while another population showed intermediate patterns. In addition, they responded differently to appetite-related neuropeptides, and their chemogenetic activation showed opposing effects in food consumption. Our studies showed contrasts and continuity of PVT neurons underlying their function as a behavior-modulating hub."

(Kapustina et al., 2024)

The cell-type-specific spatial organization of the anterior thalamic nuclei of the mouse brain.

"Understanding the cell-type composition and spatial organization of brain regions is crucial for interpreting brain computation and function. In the thalamus, the anterior thalamic nuclei (ATN) are involved in a wide variety of functions, yet the cell-type composition of the ATN remains unmapped at a single-cell and spatial resolution. Combining single-cell RNA sequencing, spatial transcriptomics, and multiplexed fluorescent in situ hybridization, we identify three discrete excitatory cell-type clusters that correspond to the known nuclei of the ATN and uncover marker genes, molecular pathways, and putative functions of these cell types. We further illustrate graded spatial variation along the dorsomedial-ventrolateral axis for all individual nuclei of the ATN and additionally demonstrate that the anteroventral nucleus exhibits spatially covarying protein products and long-range inputs. Collectively, our study reveals discrete and continuous cell-type organizational principles of the ATN, which will help to guide and interpret experiments on ATN computation and function."

(Zhou et al., 2023)

Brain-wide Correspondence Between Neuronal Epigenomics and Long-Distance Projections

"In any given brain, each neuron contributes uniquely to brain function. Nevertheless, neurons can be grouped into types based on similarities and differences across multiple dimensions, including epigenetic state, gene expression, anatomy, and physiology. Single-cell genomic technologies have been particularly impactful for cell type classification due to their high throughput (millions of cells assayed) and dimensionality (thousands of genes and even more genetic loci) leading to the identification of large numbers of transcriptomic and epigenomic clusters corresponding to possible cell types across the entire mouse brain.

A prominent and distinguishing anatomical feature of many brain neuron types is their long-distance axonal projections. Long-distance projections can be directly related to single neuron gene expression or epigenomes by use of powerful linking technologies, including BARseq 1,2 , Retro-seq 3,4 , and Epi-Retro-Seq 5 . Previous studies have used Retro-Seq and Epi-Retro-Seq to link mouse neocortical [3][4][5] , hypothalamic 6 and thalamic projection cell types 7 to their genetic and epigenetic clusters, revealing complex but predictable relationships. For example, cortical neurons projecting solely to intra-telencephalic (IT) targets fall into different

clusters than those that project to extra-telencephalic (ET) targets. On the other hand, cortical layer 2/3 (L2/3) IT neuron types projecting to different cortical areas typically co-cluster despite having quantifiable and predictable genetic and epigenetic differences across the populations 4,5 . In the face of this complexity, how can single-cell genetic and epigenetic assays be used to inform the structure and function of brain cell types and how can neuronal structure predict genetics, epigenetics, and function? Further, can the principles learned from more limited previous studies be extended to the entire brain? Are there different principles linking projection status to epigenetics for different brain areas?

To address these questions we employed Epi-Retro-Seq to assay 33,034 neurons from 225 source→target combinations across the entire mouse brain."

"The thalamus is a primary hub in sensory and cortical information processing and also projects to subcortical structures. Similar to hypothalamus, thalamus consists of a large number of nuclei that are organized into multiple functional groups. The main, central regions of the thalamus are composed of exclusively excitatory regions (except for a few local GABAergic interneurons in the dorsal lateral geniculate nucleus (LGd)) that are reciprocally connected with cortical areas 20 . Other more ventral and lateral regions of the thalamus (such as LGv and RT) contain GABAergic inhibitory neurons that are either reciprocally connected with thalamic excitatory neurons or project to subcortical structures such as the basal ganglia and brainstem 20 . In contrast to the hypothalamus, the thalamus had a lower degree of cell type complexity as shown by the smaller number of cell clusters identified through gene expression analysis 16 . Despite both the thalamus and hypothalamus showing a high level of heterogeneity in their anatomical nuclei and projections, the differences in their cell type complexity prompted us to investigate whether the relationships between cell types, their projections, and their spatial locations in the thalamus differ from those observed in the hypothalamus, as discussed above.

We analyzed thalamic neurons that project to twelve different targets, including nine cortical areas (PFC, MOp, SSp, ACA, AI, AUDp, RSP, PTLp, and VISp), SC, VTA, and P. To gain a comprehensive understanding of these neurons, we combined Epi-Retro-Seq data with unbiased snmC-seq and single-cell RNA-seq data from the thalamus. "

"Through this integration, we identified a total of 58 thalamic neuronal cell clusters (Fig. 4a), of which 33 clusters were enriched for Epi-Retro-Seq neurons (Fig. 4b). It is worth noting that neurons dissected from different anatomical regions within the thalamus were located in distinct sets of clusters 17 (Fig. 4c), as expected from prior descriptions based on analysis of scRNA-seq data 7 , suggesting that these molecularly defined cell clusters also have a spatial organization.

In Figure 4b, we assessed the degree of enrichment of each projection in each cluster. Similar to what was observed in the hypothalamus, each population of thalamic projection neurons exhibited enrichment in distinct subsets of cell clusters, with each cluster showing enrichment for a specific set of projections, sometimes only one. Notably, the clusters enriched for TH→SC, TH→VTA, TH→P and TH→cortex were mostly mutually exclusive. Regarding cortical projections, TH→PTLp and TH→VISp neurons exhibited enrichment in a largely overlapping set of clusters, but with varying degrees of enrichment. TH→MOp and TH→SSp neurons also shared most of their enriched clusters, which differed from those enriched for TH→PTLp and TH→VISp. These results support the notion of a separation of thalamic cell types between the visual and motor pathways in the thalamus and highlight the heterogeneity of cell types within each pathway. Notably, TH→RSP neurons showed no overlap in enriched clusters with any other cortical projections, and were uniquely enriched in clusters 13, 26 and 47. These clusters were annotated by their gene expression patterns as belonging to the anteroventral (AV) nucleus (clusters 13 and 26), and anterodorsal (AD) nucleus (cluster 47), which is consistent with TH→RSP projections originating from anterior thalamic nuclei 21 . In summary, TH neurons projecting to cortex vs. subcortical targets were enriched in distinct sets of clusters. "

"The enriched cell clusters for cortical projections were further segregated by different thalamic pathways, with multiple enriched cell clusters observed for each pathway or projection. These findings highlight the cell type specificity as well as heterogeneity at the level of TH projections.

Such cell type specificity and heterogeneity of TH projection neurons were also reported in transcriptomic analysis of single TH projection neurons. Phillips et al. conducted a single-cell RNA-seq study on thalamic neurons projecting to the prefrontal, motor, somatosensory, auditory, and visual cortices 7 . We will refer to their dataset as Retro-Seq of these thalamocortical (TC) projections. In their study, clustering analysis of Retro-Seq revealed that neurons of each projection were enriched in a specific set of clusters, indicating specific and shared cell types between different TC projections, as well as cell type heterogeneity within each TC projection.

To compare Retro-Seq and Epi-Retro-Seq data for these TC projections, we mapped the Retro-Seq data onto our 58 integrated TH clusters (Extended Data Fig. 8a, Methods). The t-SNE of Retro-Seq and Epi-Retro-Seq neurons from comparisons of the most closely corresponding projections showed that they occupied similar spaces and were enriched in a common set of clusters (Extended Data Fig. 8a,b). However, in addition to the shared projection-enriched clusters, certain clusters were found to be enriched for each projection only in either Retro-Seq or Epi-Retro-Seq. These differences are most likely due to the use of different injection coordinates for each cortical target (Supplementary Table 4), resulting in overlapping yet different populations of retrogradely labeled TH neurons being analyzed in the two data sets. The degree of alignment between Epi-Retro-Seq and Retro-Seq was quantified by the overlap score and cosine distance (Extended Data Fig. 8c, Methods), which revealed that Retro-Seq neurons of each projection were more similar to Epi-Retro-Seq neurons for the corresponding projections than for any other projections."

(Peng et al., 2021)

Morphological diversity of single neurons in molecularly defined cell types

"Dendritic and axonal morphology reflects the input and output of neurons and is a defining feature of neuronal types1,2, yet our knowledge of its diversity remains limited. Here, to systematically examine complete single-neuron morphologies on a brain-wide scale, we established a pipeline encompassing sparse labelling, whole-brain imaging, reconstruction, registration and analysis. We fully reconstructed 1,741 neurons from cortex, claustrum, thalamus, striatum and other brain regions in mice. We identified 11 major projection neuron types with distinct morphological features and corresponding transcriptomic identities. Extensive projectional diversity was found within each of these major types, on the basis of which some types were clustered into more refined subtypes. This diversity follows a set of generalizable principles that govern long-range axonal projections at different levels, including molecular correspondence, divergent or convergent projection, axon termination pattern, regional specificity, topography, and individual cell variability. Although clear concordance with transcriptomic profiles is evident at the level of major projection type, fine-grained morphological diversity often does not readily correlate with transcriptomic subtypes derived from unsupervised clustering, highlighting the need for single-cell cross-modality studies. Overall, our study demonstrates the crucial need for quantitative description of complete single-cell anatomy in cell-type classification, as single-cell morphological diversity reveals a plethora of ways in which different cell types and their individual members may contribute to the configuration and function of their respective circuits. Sparse labelling and whole-brain imaging are used to reconstruct and classify brain-wide complete morphologies of 1,741 individual neurons in the mouse brain, revealing a dependence on both brain region and transcriptomic profile."

"Dendritic and axonal morphology reflects the input and output of neurons and is a defining feature of neuronal types 1,2 , yet our knowledge of its diversity remains limited. Here, to systematically examine complete single-neuron morphologies on a brain-wide scale, we established a pipeline encompassing sparse labelling, wholebrain imaging, reconstruction, registration and analysis. We fully reconstructed 1,741 neurons from cortex, claustrum, thalamus, striatum and other brain regions in mice. We identified 11 major projection neuron types with distinct morphological features and corresponding transcriptomic identities. Extensive projectional diversity was found within each of these major types, on the basis of which some types were clustered into more refined subtypes. This diversity follows a set of generalizable principles that govern long-range axonal projections at different levels, including molecular correspondence, divergent or convergent projection, axon termination pattern, regional specificity, topography, and individual cell variability. Although clear concordance with transcriptomic profiles is evident at the level of major projection type, fine-grained morphological diversity often does not readily correlate with transcriptomic subtypes derived from unsupervised clustering, highlighting the need for single-cell cross-modality studies. Overall, our study demonstrates the crucial need for quantitative description of complete single-cell anatomy in cell-type classification, as single-cell morphological diversity reveals a plethora of ways in which different cell types and their individual members may contribute to the configuration and function of their respective circuits.

Neurons exhibit extraordinary diversity across molecular, morphological, physiological and connectional features, thus accurate classification and mapping of cell types needs to consider and integrate these distinct yet related cellular properties 1,2 . Single-cell RNA-sequencing (scRNA-seq) has enabled systematic classification at the transcriptomic level [3][4][5] , capturing major cell types with known anatomical and functional properties and revealing many potentially new cell types. Classification of cortical neurons using a combination of transcriptomic, electrophysiological and local morphological properties has also been achieved 6,7 . Brain-wide inter-areal connectivity has been mapped extensively using anterograde and retrograde tracing of projection neuron populations [8][9][10][11] ."

"At the lowest, single-cell level, the degree of similarity or variability between individual neurons within a given type also varies across types. Within-type individual cell variability is high in cortical and claustral neurons, moderate in thalamic matrix neurons, and low in thalamic core neurons and striatal MSNs.

A major question is how morphological and projectional properties compare and correlate with the neurons' molecular identities. We attempted to address this question with two approaches: using validated driver lines to define the subclass-level molecular identities of reconstructed neurons and using Retro-seq to obtain transcriptomic profiles of neurons projecting to specific targets. Both approaches show that subclasses of neurons have highly distinct morphological and projectional properties; however, within these major types, especially for cortical and claustral neurons, many aspects of morphological diversity cannot be accounted for by currently identified transcriptomic subtypes or clusters in the adult stage. Previous studies showed that L2/3 SSp pyramidal neurons projecting to MOp or SSs have distinct intrinsic and network physiological properties 42,43 . Even though they may not belong to distinct transcriptomic subtypes, it will be interesting to examine gene-expression differences that might correspond to the differential connectional and physiological properties for these neurons, as found for primary visual cortical neurons projecting differentially to medial or lateral higher visual areas 44 . Several mechanisms may explain the origin of the morphological diversity, such as molecular instructions that act transiently during development 45 , activity-dependent cell interactions, or stochastic processes. It will be informative to develop methods that enable complete reconstruction of morphology and in-depth gene-expression profiling to be conducted on the same cell, and apply them to single cells in both adult stage and during brain development, so that the developmental correlations of molecular and morphological and connectional features can be identified."

(Roy et al., 2022)

Thalamic subnetworks as units of function

"To appreciate the levels of organization discussed in this Review, we need to explicitly define the three key terms that we use throughout this paper. We refer to the term 'cell type' as a feature that is identified via a singular approach. Within individual thalamic nuclei, cell types have traditionally been defined on the basis of morphology, neurochemistry or physiology. For example, morphology can identify neurons based on differences in soma size and dendritic branching patterns 1,3 , whereas neurochemistry indicates that cell types can be either excitatory or inhibitory (for example, by the expression of enzymes for the synthesis and packaging of glutamate or GABA) 3 . Physiology, on the other hand, can identify neurons using distinct sensory response patterns (for example, in the LGN) 2 . The advent of modern genetic profiling has enabled the identification of 'subpopulations': groups of cells that share a particular pattern of gene expression, morphology, physiology and fine-grained connectivity. In this Review, we define one or more subpopulations serving a particular function as a 'subnetwork' (Fig. 2). Because each neuron or subpopulation often has multiple inputs and outputs, using the same subpopulation in different functional subnetworks is likely to be a common and efficient approach for functional circuit assembly. For instance, subpopulation A may form a subnetwork with subpopulation B for a given function, but forms a subnetwork with subpopulation X to serve a different function. Admittedly, this definition is forward-looking and will require future experiments to continue to validate and refine it.

One example of a putative thalamic subnetwork came from examining neural activity patterns within the TRN 15 , which, up to that point, had been considered a functional monolith 16 . A combination of electrophysiological recordings and connectivity-based optical tagging revealed the existence of subpopulations that share activity patterns depending on the principal thalamic structures to which they project 15 . More recently, state-of-the-art single-cell RNA sequencing (scRNA-seq) has indicated that these 'TRN subnetworks' could be further specified using gene expression gradients 17 subpopulations in several brain regions, such as"

"brainstem, histaminergic afferents from the hypothalamus and noradrenergic afferents from the parabrachial region have heterogeneous terminals in the thalamus 1,2 .

Physiology. Building on these findings, researchers have consistently observed electrophysiological differences in thalamic nuclei.

Whereas thalamic principal cells are thought to have comparable intrinsic properties across sensory nuclei, there is substantial heterogeneity in their action potential parameters [39][40][41][42][43] . Recently, whole-cell recordings from three motor-related thalamic nuclei (central medial (CM), ventral anterior (VA) and ventral lateral (VL) nuclei) showed that electrophysiological membrane and synaptic properties varied along a gradient from CM to VA to VL nuclei 44 . Whereas previous studies revealed electrophysiological heterogeneity of thalamic neurons, this study 44 found that single-cell physiological properties across thalamic nuclei exist along a continuum rather than forming unique, discrete profiles (Fig. 3b). However, the functional implications of this observation have not been revealed. This study also linked these physiological observations to graded transcriptional profiles (Fig. 3c) and morphological differences, suggesting that cellular morphology, terminal arborization, input type and electrophysiological properties are effective approaches for characterizing thalamic cellular diversity. Regarding the fundamental intrinsic feature of thalamic principal neurons 45-47 their two distinct firing modes (tonic versus bursting)-heterogeneity in these modes has been associated with somatodendritic morphological differences 27,48,49 .

Neurotransmitters and receptors. Finally, differences in the localization of neurotransmitters and their receptors [50][51][52][53][54][55][56][57] led to the idea that variation in gene expression profiles may serve as an organizing principle across thalamic nuclei. One group identified four transcription factor genes that showed differential expression within excitatory thalamic nuclei 58 that these candidate genes might reflect distinct subpopulations that are shared across excitatory or inhibitory nuclei, many groups attempted to identify thalamic nucleus-and subpopulation-specific molecular markers ["

"Although studies that have identified subnetworks within the thalamus up to now have used rodents, there is substantial evidence to suggest that heterogeneity among thalamic neurons is conserved from rodents to primates. Here, we briefly discuss several primate studies that support the idea of diverse thalamic subpopulations or show that some cellular properties of primate thalamic neurons are comparable to those in rodents. Neurotransmitter expression patterns are usually conserved across species; for instance, nicotinic binding sites are enriched in the LGN, the MD nucleus and the rhomboid nucleus in mice and monkeys 52 . Similarly, a study examined the expression of genes in the monkey thalamus that were enriched in different mouse thalamic nuclei and found highly consistent patterns 63 . Several genes were expressed in excitatory nuclei in monkeys but not in rodents. A study using high-density oligonucleotide arrays to identify thalamic nuclei-specific gene expression in adult monkeys 120 showed that over 550 genes were selectively expressed in the anterior thalamus or the CM, MD or ventral posterior nucleus. This study identified a marker for excitatory nuclei (coding for transcription factor 7-like 2), a marker for excitatory nuclei but not the CM nucleus or the PF (coding for Purkinje cell protein 4), a CM-PF-specific marker (coding for cerebellin 1 precursor) and an

LGN marker (Spp1). These markers have comparable expression patterns in mice 69 . A study that showed distinct subpopulations labeled by PV or somatostatin (SST) in the mouse TRN showed that these two subpopulations are also found in human TRN samples 87 . Additionally, in primates, SST-expressing neurons exhibit differential distribution in the visual sector of the TRN 121 . As in rodents, the LGN in monkeys has distinct subpopulations based on single-cell electrophysiological properties, spatial localization and inputs from distinct RGC subtypes 122,123 .

Tracing studies have also revealed subpopulations within thalamic nuclei in monkeys 109,124,125 . For example, the PF in monkeys contains projection-specific subpopulations 90 , including neurons that project to striatal regions and neurons that project"